Doctoral thesis

Purity in the tame cohomology

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July 2, 2025

*Funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) TRR 326 *Geometry and Arithmetic of Uniformized Structures*, project number 444845124.

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Abstract

Let (X, Z) be a regular pair of pure codimension r over a base scheme S of characteristic p. Under the assumption of resultion of singularities, we prove purity for the tame cohomology with coefficient in the logarithmic de Rham-Witt sheaves $\nu_m(n)$ for the regular pair (X, Z), i.e. the existence of a natural isomorphism

$$Ri^{!}\nu_{X,m}(n) \cong \nu_{Z,m}(n-r)[-r]$$

for all $m, n \geq 0$, where the logarithmic de Rham-Witt sheaves $\nu_{X,m}(n)$ are the Frobenius-fixed elements in the de Rham-Witt sheaves $W_m\Omega^n_{X/\mathbb{F}_p}$.

1. Introduction

1.1. Purity in the prime to p case

Let X and Z be a regular schemes, and let $i : Z \hookrightarrow X$ be a closed subscheme of pure codimension r. We refer to the pair (X, Z) as a *regular pair* of codimension r. If X and Z are smooth over a base scheme $S = \mathrm{Spec}(k)$, where k is a field and $i : Z \hookrightarrow X$ is a closed immersion, then we call (X, Z) a *smooth pair* over k.

Given a regular pair (Z, X) of codimension r, the theorem of absolute cohomological purity, due to Fujiwara [Fuj02], asserts the following:

Theorem 1.1.1. [Fuj02] Let (X, Z) be a regular pair of codimension r. Then for all integers $j \geq 0$ and $n \in \mathbb{Z}$, and for all integers l invertible on X, there exist canonical isomorphisms

$$H^j_{Z,\text{ét}}(X, \mathbb{Z}/l(n)) \xrightarrow{\sim} H^{j-2r}_{\text{ét}}(Z, \mathbb{Z}/l(n-r)).$$

Here, the twist $\mathbb{Z}/l(n)$ denotes the n-th Tate twist, defined as $\mathbb{Z}/l(n) := \mu_l^{\otimes n}$.

This statement is equivalent to a purity theorem at the level of the derived category, which we now state:

Theorem 1.1.2. Let (X, Z) be a regular pair of codimension r, and let $i : Z \hookrightarrow X$ be the closed immersion. Let l be an integer which is invertible on X. Then there exists a canonical isomorphism in the derived category of étale sheaves:

$$Ri^!\mathbb{Z}/l(n)[2r] \cong i^*\mathbb{Z}/l(n-r). \tag{1.1}$$

The implication of the second statement from the first statement is due to the following argument: Set $\mathcal{F} = \mathbb{Z}/l(n)$. Consider the composition of functors

$$\mathrm{Sh}(X_{\text{ét}}) \xrightarrow{i^!} \mathrm{Sh}(Z_{\text{ét}}) \xrightarrow{\Gamma(Z,-)} \mathcal{A}b,$$

which gives rise to a Grothendieck spectral sequence

$$H^r(Z, R^s i^!\mathcal{F}) \implies H^{r+s}_Z(X, \mathcal{F}).$$

Under the assumptions of the theorem, the cohomology sheaves $R^s i^!\mathcal{F}$ vanish for all $s \neq 2r$, and we have

$$R^{2r} i^!\mathcal{F} \cong i^*\mathcal{F}(-r).$$

The spectral sequence therefore degenerates at the E_2 page, yielding the claimed isomorphism from $H^j_{Z,\text{ét}}(X, \mathbb{Z}/l(n))$ to $H^{j-2r}_{\text{ét}}(Z, \mathbb{Z}/l(n-r))$.

1.2. Semi-purity of the étale cohomology with coefficients in the log de Rham-Witt sheaves

Assume that (X, Z) is a smooth pair over a perfect field k of characteristic p. Let $S := \mathrm{Spec}(k)$. Our goal is to construct a purity isomorphism analogous to (1.1), replacing $\mathbb{Z}/l$ with $\mathbb{Z}/p^m$. However, defining $\mathbb{Z}/p^m(n)$ as $\mu_{p^m}^{\otimes n}$ does not make sense, since the group scheme $\mu_{p^m} \cong \mathrm{Spec}(k[T]/(T^{p^m} - 1))$ is not an étale sheaf.

In [Mil86, Theorem 2.1], Milne presented an alternative, which is the logarithmic de Rham-Witt sheaf $\nu_m(n)$ -whose definition we recall below- and proved a semi-purity theorem for those sheaves. Namely that canonical isomorphisms

$$R^j i^! \nu_{X,m}(n) = \begin{cases} 0 & \text{if } j \neq r, r+1 \\ \nu_{Z,m}(n-r) & \text{if } j = r \end{cases} \tag{1.2}$$

exist.

This omits the term $R^{r+1} i^! \nu_{X,m}(n)$, which may be non-zero in general. Nevertheless, by [Mil86, Corollary 2.2], semi-purity implies that $H^i_Z(X, \nu_m(n)) = 0$ for all $i < r$, that the map

$$H^0(Z, \nu_{Z,m}(n-r)) \to H^r_Z(X, \nu_{X,m}(n))$$

is an isomorphism, and that there is an exact sequence

$$0 \to H^1(Z, \nu_m(n-r)) \to H^{r+1}_Z(X, \nu_m(n)) \to H^0(Z, R^{r+1} i^! \nu_m(n)) \to \cdots$$

Full purity —i.e., the vanishing of $R^{r+1} i^! \nu_{X,m}(n)$— would yield canonical isomorphisms

$$H^j(Z, \nu_m(n-r)) \cong H^{j+r}_Z(X, \nu_m(n))$$

for all $j \in \mathbb{N}$.

Theorem 2.1 of [Mil86] was also independently proven by Gros in [Gro85], using similar methods.

To introduce the logarithmic de Rham-Witt sheaves, we begin with some notation. For simplicity, set $m = 1$ and write $\nu(n)$ for $\nu_1(n)$. A differential form $\omega \in \Omega^n_{X/S}$ is said to be *closed* if $d\omega = 0$. The sheaf of closed forms is denoted by $Z\Omega^n_{X/S}$. The forms in the image of the differential $\Omega^{n-1}_{X/S} \xrightarrow{d} \Omega^n_{X/S}$ are *exact*, and their sheafification is denoted by $B\Omega^n_{X/S}$.

The logarithmic de Rham-Witt sheaves $\nu(n)$ are characterized by the short exact sequence (cf. [Mil76]):

$$0 \to \nu(n) \to Z\Omega^n_{X/S} \xrightarrow{C-1} \Omega^n_{X/S} \to 0, \tag{1.3}$$

where C denotes the Cartier operator (see [Mil76, Lemma 1.1] for its definition and existence). An important property of C is that $C(f^p \eta) = fC(\eta)$ for any closed form η. In the case $n = 0$, this sequence is reduced to

$$0 \to \mathbb{F}_p \to \mathcal{O}_X^p \xrightarrow{F^{-1}-1} \mathcal{O}_X \to 0,$$

where F is the absolute Frobenius $x \mapsto x^p$. Since X is reduced, the map $F : \mathcal{O}_X \to \mathcal{O}_X^p$ is an isomorphism, and the previous sequence becomes the Artin–Schreier sequence

$$0 \to \mathbb{F}_p \to \mathcal{O}_X \xrightarrow{1-F} \mathcal{O}_X \to 0,$$

after replacing $\mathcal{O}_X^p$ with $\mathcal{O}_X$.

Milne used the short exact sequence (1.3) in order to prove semi-purity for $\nu_X(n)$ (1.2) ([Mil86, Theorem 2.1]). However, the term $R^{r+1}i^!\nu_X(n)$ may fail to vanish. Indeed, from the long exact sequence of cohomology associated to (1.3), we get

$$0 \to R^r i^! \nu_X(n) \to R^r i^! Z\Omega^n_{X/k} \xrightarrow{C-1} R^r i^! \Omega^n_{X/k} \to R^{r+1} i^! \nu_X(n) \to 0,$$

This shows that $R^{r+1}i^!\nu_X(n)$ vanishes if and only if the map $C-1$ is surjective on $R^r i^! \Omega^n_{X/k}$. This is not always the case.

Let us assume that $X = \mathrm{Spec}(A)$ is strictly henselian, $Z = V(f)$ is defined by a single equation, so that $r = 1$. Consider the case $n = 0$ (so that $\nu(n) = \mathbb{Z}/p$). By excision, the sheaves $R^{r+1}i^!\mathcal{O}_X(n)$ and $R^{r+1}i^!\mathcal{O}_X^p(n)$ are isomorphic to the quasi coherent sheaves over X (resp. F_*X) associated to $A[f^{-1}]/A$ and $(A[f^{-1}])^p/A^p$ respectively. Note that $(A[f^{-1}])^p/A^p$ is isomorphic to $A[f^{-1}]/A$ as a group.

However, the resulting map $F - 1 : A[f^{-1}]/A \to A[f^{-1}]/A$ may fail to be surjective. For example, consider $X = \mathrm{Spec}(k[T])^{\mathrm{sh}}_{(T)}$ and $f = T$. Then the element $1/f$ does not lie in the image of $F - 1$, since the equation $\gamma^p - \gamma = 1/f$ has no solution in $A[f^{-1}]$.

1.3. Purity of the tame cohomology in positive characteristic

As purity fails in the étale topology, we instead work with the *tame topology*, defined in [HS21]. Roughly speaking, this is the topology whose covers are étale morphisms that are tamely ramified at the boundary. For instance, Artin–Schreier covers that ramify at the boundary are excluded. For more details, see [HS21] or Chapter 3 of this document. For a scheme X over a field k, we denote the tame site by $(X/k)_t$.

We now state the main theorem of this manuscript, along with a sketch of its proof.

Theorem 1.3.1. Let X be a smooth scheme over a perfect field k. Let $i : Z \hookrightarrow X$ be a smooth closed subscheme of pure codimension r. Assume that resolution of singularities and embedded resolution of singularities (as stated in Conjecture B.2.1 and Conjecture B.2.3) hold. Then there is an isomorphism in the derived category of tame sheaves $D((Z/k)_t)$:

$$\nu_{Z,m}(n-r)[-r] \xrightarrow{\sim} Ri^!\nu_{X,m}(n).$$

Remark 1.3.2. The following sketch of the proof uses discretely ringed adic spaces, introduced in Chapter 3. We define $\mathcal{X} := \mathrm{Spa}(X, k)$ and $\mathcal{Z} := \mathrm{Spa}(Z, k)$. The idea of the proof should remain accessible even without a full familiarity in adic spaces.

The adic version of the theorem states that

$$\nu_{\mathcal{Z},m}(n-r)[-r] \xrightarrow{\sim} Ri^!\nu_{\mathcal{X},m}(n)$$

is an isomorphism in $D(\mathcal{Z}_t)$. The two formulations are equivalent by [HS21, §14].

Sketch of the proof. 1. Under the assumption of resolution of singularities, we prove the existence of the short exact sequence (Proposition 4.3.13)

$$0 \to \nu_{\mathcal{X}}(n) \to Z\Omega^{+,n}_{\mathcal{X}/k} \xrightarrow{C-1} \Omega^{+,n}_{\mathcal{X}/k} \to 0.$$

The key input is that $\Omega^{+,n}(\mathrm{Spa}(X,\bar{X})) = \Omega^{n,\log}(X,\bar{X})$ by the main theorem of [Hüb24] whenever $(X,\bar{X})$ is a log smooth log scheme. It is well-known that $\Omega^{n,\log}(X,\bar{X})$ fits into such a short exact sequence (see Proposition 4.3.3).

Note that the analogous sequence for Ω^n is not exact: the preimage of $f \in \mathcal{O}_X(X)$ under $C-1$ lies in $\mathcal{O}_Y(Y)$, where $Y = \mathrm{Spec}_{\mathcal{O}_X}\mathcal{O}_X[T]/(T^p - T - f)$ is an Artin–Schreier cover of X. Such covers are not tamely ramified when f is not invertible on $\bar{X}$.

In Chapter 6, we present an alternative proof of Theorem 4.3.13, without the assumption of resolution of singularities.

2. Prove that $Ri^!\Omega^{+,n}_{\mathcal{X}/k} \cong \Omega^{+,n-r}_{\mathcal{Z}/k}[-r]$ (Corollary 5.2.4) and $Ri^!Z\Omega^{+,n}_{\mathcal{X}/k} \cong Z\Omega^{+,n-r}_{\mathcal{Z}/k}[-r]$ (Corollary 5.2.8).

   To achieve this, we do the following: we check that the tame cohomology of $\Omega^{+,n}$ and $Z\Omega^{+,n}$ is equal to the cohomology of the underlying topological space (Proposition 5.1.1). For $\mathcal{F} = \Omega^{+,n}$ and $\mathcal{F} = Z\Omega^{+,n}$ we prove that

$$H^i(\mathrm{Spa}(X,S),\mathcal{F}) \cong \mathrm{colim}_{\bar{Y}} H^i(\bar{Y},\mathcal{F}|_{\bar{Y}}),$$

   where $\bar{Y}$ are S-compactifications of X.

   Assuming resolution of singularities, we may compute this using a system of $\bar{Y}$'s for which $\Omega^{+,n}_{\mathrm{Spa}(Y,\bar{Y})}|_{\bar{Y}} = \Omega^{n,\log}_{(Y,\bar{Y})}$ holds. We finally use the vanishing of the higher cohomology groups of $\Omega^{n,\log}_{(Y,\bar{Y})}$ when X and S are affine (Theorem 2.4.5) to deduce the vanishing of $R^si^!\Omega^{+,n}$ and $R^si^!Z\Omega^{+,n}$ for $s \neq \mathrm{codim}_X(Z)$.

3. Apply the long exact sequence of $Ri^!$ to obtain the diagram (5.6)

$$\begin{tikzcd}
0 \arrow[r] & \nu_{\mathcal{Z}}(n-r) \arrow[r] \arrow[d, dotted] & Z\Omega^{+,n-r}_{\mathcal{Z}} \arrow[r, "C-1"] \arrow[d, "\cong"] & \Omega^{+,n-r}_{\mathcal{Z}} \arrow[r] \arrow[d, "\cong"] & 0 \\
0 \arrow[r] & R^ri^!\nu_{\mathcal{X}}(n) \arrow[r] & R^ri^!Z\Omega^{+,n}_{\mathcal{X}} \arrow[r, "C-1"] & R^ri^!\Omega^{+,n}_{\mathcal{X}} \arrow[r] & 0,
\end{tikzcd}$$

   and conclude that

$$\nu_{\mathcal{Z}}(n-r)[-r] \to Ri^!\nu_{\mathcal{X}}(n)$$

   is an isomorphism.

4. Use the short exact sequence in Lemma 4.3.11

$$0 \to \nu_{\mathcal{Z},m}(n) \to \nu_{\mathcal{Z},m+1}(n) \to \nu_{\mathcal{Z},1}(n) \to 0,$$

   to deduce by induction the isomorphism

$$\nu_{\mathcal{Z},m}(n-r)[-r] \to Ri^!\nu_{\mathcal{X},m}(n).$$

□

Finally, following the work of Shiho in [Shi07], we generalize Theorem 1.3.1 to regular pairs over basis of characteristic p.

Corollary 1.3.3. (cf. Theorem 5.4.1) Let $i : Z \to X$ be a closed immersion of Noetherian schemes over a Noetherian scheme S of characteristic p. Assume that (X, Z) is a regular pair of pure codimension c. Then in $D((Z/S)_t)$ we have the isomorphism

$$Ri^! \nu_{m,X}(n) \cong \nu_{m,Z}(n-r)[-r]$$

under the assumption of resolution of singularities.

Sketch of the proof. Apply Popescu theorem (Proposition 5.4.2) in order to reduce to the smooth case over $\mathbb{F}_p$, which is solved by Theorem 1.3.1. □

1.4. Avoiding resolution of singularities

We keep the notation of Theorem 1.3.1.

In Theorem 1.3.1, resolution of singularities was applied in order to construct the Cartier operator $C : Z\Omega^{+,n}_{\mathcal{X}} \to \Omega^{+,n}_{\mathcal{X}}$ and to compute $R^s i^! \Omega^n_{\mathcal{X}}$ for $s > 0$. This relied on construction of log smooth compactifications via resolution of singularities.

In contrast, in Chapter 6, we will show that resolution of singularities can be avoided in order to construct the Cartier operator (see Theorem 6.1.2 and Corollary 6.1.3). Furthermore, we were able to compute $R^1 i^! \Omega^n_{\mathcal{X}}$ (Corollary 6.2.8) when the pair (X, Z) is of codimension 1, without the use of resolution of singularities. Note however, that resolution of singularities cannot be avoided altogether, yet: a proof of the vanishing of $R^j i^! \Omega^n_{\mathcal{X}}$, where $j \neq \operatorname{codim}_X Z$, is still out of reach without the use of resolution of singularities.

Theorem 1.4.1. (cf. Corollary 6.1.3) Let X be a smooth scheme over a perfect field k of characteristic $p > 0$, and let $\mathcal{X} = \operatorname{Spa}(X, k)$ be the associated discretely ringed adic space. Then the short exact sequence of sheaves on $\mathcal{X}$

$$0 \to B\Omega^n_{\mathcal{X}} \to Z\Omega^n_{\mathcal{X}} \xrightarrow{C} \Omega^n_{\mathcal{X}} \to 0$$

restricts to a short exact sequence

$$0 \to B\Omega^{+,n}_{\mathcal{X}} \to Z\Omega^{+,n}_{\mathcal{X}} \xrightarrow{C} \Omega^{+,n}_{\mathcal{X}} \to 0.$$

Sketch of the proof.

- Reduce to the case $\mathcal{X} = \operatorname{Spa}(K, K^+)$ for a valued field (K, K^+) (Proposition 6.1.9). In this case, we have

$$\Omega^{+,n}_{\mathcal{X}} = \Omega^n_{K^+}(\log K^\times).$$

The proof is inspired by a similar result of Gabber [KST21, Appendix].

- For an extension $A^{\log} \subseteq B^{\log}$ of integral log rings (Definition 6.1.11), define the *relative inverse Cartier operator*

$$\begin{aligned} C^{-1}_{B^{\log}/A^{\log}} : A \otimes_{F,B} \Omega^{i,\log}_{B/A} &\to Z\Omega^{i,\log}_{B/A}/B\Omega^{i,\log}_{B/A} \\ a \otimes b db_1 \wedge \cdots \wedge db_m \wedge \\ \mathrm{dlog} b_{m+1} \wedge \cdots \wedge \mathrm{dlog} b_i &\mapsto \begin{array}{c} ab^p(b_1 \cdots b_m)^{p-1} db_1 \wedge \cdots \wedge db_m \wedge \\ \mathrm{dlog} b_{m+1} \wedge \cdots \wedge \mathrm{dlog} b_i. \end{array} \end{aligned} \tag{1.4}$$

- Equip K^+ with the canonical log structure $K^+ \setminus \{0\} \to K^+$, denoted $K^{+,\log}$. Then the absolute inverse Cartier operator

$$C^{-1} \colon \Omega^n_{K^+}(\log K^\times) \to Z\Omega^n_{K^+}(\log K^\times)/B\Omega^n_{K^+}(\log K^\times)$$

  coincides with the relative operator $C^{-1}_{K^{+,\log}/K^{p,+,\log}}$, where $K^{p,+}$ is the restriction of the valuation ring K^+ in K^p.

- Express the extension $K^{+,\log} \subset K^{1/p,+,\log}$ as a filtered colimit of *elementary subextensions* $K^{+,\log} \subseteq A^{\log} \subseteq K^{1/p,+,\log}$ (Theorem 6.1.20). These play the role of log smooth extensions. For each such extension, the operator $C^{-1}_{A^{\log}/K^{+,\log}}$ is an isomorphism (Proposition 6.1.19(1)).

- The operator $C^{-1}_{K^{+,\log}/K^{p,+,\log}}$ is then the filtered colimit of the isomorphisms $C^{-1}_{A^{\log}/K^{+,\log}}$ (Proposition 6.1.19(2)), completing the proof. □

We also establish a form of semi-purity in codimension one for tame cohomology:

Theorem 1.4.2. Let (X, Z) be a smooth pair of pure codimension one over a perfect field k of characteristic p. Then we have a canonical isomorphism

$$R^1 i^! \nu_{\mathcal{X},m}(n) \cong \nu_{\mathcal{Z},m}(n-1),$$

where $\mathcal{X} = \mathrm{Spa}(X, k)$ and $\mathcal{Z} = \mathrm{Spa}(Z, k)$.

The main input is the following residue sequence:

Theorem 1.4.3 (cf. Theorem 6.2.1)**.** Under the same assumptions of Theorem 1.4.2, there exists a short exact residue sequence

$$0 \to \Omega^{+,n}_{\mathcal{X}} \to j_*\Omega^{+,n}_{\mathcal{U}} \xrightarrow{\mathrm{res}} i_*\Omega^{+,n-1}_{\mathcal{Z}} \to 0,$$

which satisfies the following local property:
If $Z = V(f)$ for some $f \in \mathcal{O}_X(X)$, the residue map satisfies

$$\mathrm{res}(\omega \wedge d\log f) = \omega|_{\mathcal{Z}} \quad \text{for all } \omega \in \Omega^{+,n-1}_{\mathcal{X}}(\mathcal{X}).$$

Sketch of the proof of Theorem 1.4.2. Theorem 1.4.3 induces an isomorphism $R^1 i^! \Omega^{+,n}_{\mathcal{X}} \cong \Omega^{+,n-1}_{\mathcal{Z}}$. Simlarly, we obtain an isomrphism $R^1 i^! Z\Omega^{+,n}_{\mathcal{X}} \cong Z\Omega^{+,n-1}_{\mathcal{Z}}$. The rest of the proof is similar to Theorem 1.3.1. □

Guide to the reader Chapter 2 provides an introduction to logarithmic schemes. Chapter 3 introduces discretely ringed adic spaces and tame cohomology. Chapter 4 is devoted to logarithmic differentials on such adic spaces, where we define $\Omega^{+,n}$ and develop its basic properties. Chapter 5 combines the techniques developed so far to prove Theorem 1.3.1. In Chapter 6, we explore valuative methods to prove purity without assuming resolution of singularities. Appendix A collects generalities on logarithmic de Rham-Witt sheaves, and Appendix B summarizes the Nagata compactification and conjectural statements about resolution of singularities.

Acknowledgment I thank Katharina Hübner for her ideas and the endless discussions about every detail of this thesis. I also thank Piotr Achinger and his group, whom I visited in Warsaw, and who helped me clarify some of the log geometric aspects of this work. I also thank Alberto Merici and Morten Lüders for sharing and explaining their respective work on the matter.

2. Logarithmic schemes and their differentials

Logarithmic differentials appear when discussing purity for p-torsion sheaves in characteristic p. Logarithmic schemes enable a more natural definition of these differentials.

2.1. Logarithmic schemes

In this manuscript, a monoid is assumed to be commutative and to have a unit element.

Definition 2.1.1. Let X be a scheme. A *prelog structure* on X is a morphism of sheaves of monoids

$$\alpha_X : \mathcal{M}_X \to \mathcal{O}_X$$

on the small étale site $X_{\text{ét}}$, where $\mathcal{O}_X$ is considered as a multiplicative monoid. We say that α_X is a *log structure* if

$$\alpha_X^{-1}(\mathcal{O}_X^\times) \xrightarrow{\alpha_X} \mathcal{O}_X^\times$$

is an isomorphism. A log scheme is a pair $(X, \mathcal{M}_X)$ where X is a scheme and $\alpha_X : \mathcal{M}_X \to \mathcal{O}_X$ is a log structure. Here the morphism α_X is a part of the structure but we generally choose to omit it from the notation.

A morphism $(X, \mathcal{M}_X) \to (Y, \mathcal{M}_Y)$ of log schemes is a pair $(f, f^\flat)$ where $f : X \to Y$ is a morphism of schemes and $f^\flat : f^{-1}\mathcal{M}_Y \to \mathcal{M}_X$ is a morphism of monoids which respects the log structure.

To each prelog structure $\alpha_X : \mathcal{M}_X \to \mathcal{O}_X$ we define the associated log structure as the pushout $\mathcal{M}_X^{\log} := \mathcal{M}_X \bigoplus_{\alpha_X^{-1}(\mathcal{O}_X^\times)} \mathcal{O}_X^\times$ together with the natural morphism $\tilde{\alpha}_X : \mathcal{M}_X^{\log} \to \mathcal{O}_X$. The log structure $alpha_X$ is called the *logification* of the prelog structure α_X.

Example 2.1.2. Let X be a scheme and $U \subseteq X$ an open subscheme. Then we define a log structure, called the *compactifying log structure*

$$\alpha_{U/X} : \mathcal{M}_{U/X} \to \mathcal{O}_X$$

as follows. The sheaf $\mathcal{M}_{U/X}$ is the subsheaf of $\mathcal{O}_X$ whose sections are those elements $f \in \mathcal{O}_X(V)$, such that $f|_{U \cap V}$ is invertible in $\mathcal{O}_X(V \cap U)$ and $\alpha_{U/X}$ is the natural inclusion.

A monoid P is said to be *integral* if $P \to P^{\text{gp}}$ is injective.

A log scheme is called *quasi-coherent* (resp. *coherent*) if it is étale locally induced by a constant (resp. finitely generated and constant) prelog structure. Let $\{X_i \to X\}_i$ be an étale covering of X where $\mathcal{M}_X|_{X_i}$ is the logification of a constant prelog structure $\{P_i\}_i$. We say that $\{P_i\}_i$ is a *chart* of $\mathcal{M}_X$.

A coherent log scheme is *fine* if it is charted by integral monoids.

A coherent log scheme is said to be saturated if it has a chart consisting of monoids P which are *saturated*, i.e. they satisfy:

$$\forall a \in P^{\mathrm{gp}} \quad \big[a^n \in P \text{ for some } n \geq 1 \implies a \in P\big].$$

A morphism of logarithmic schemes $Y \to X$ is *strict*, if the log structure of Y is the pullback of the log structure of X, i.e. $\mathcal{M}_Y = f^*\mathcal{M}_X$ which is the logification of the prelog structure $f^{-1}\mathcal{M}_X \to f^{-1}\mathcal{M}_X \to \mathcal{M}_Y$.

Log schemes that are fine and saturated are called fs. They form a subcategory of log schemes. In this paper, we will mostly consider the category of fs log schemes.

Notation 2.1.3. Let X be a log scheme. Denote the underlying scheme by X°. Define

$$X_{\mathrm{tr}} := \{x \in X^\circ | \mathcal{M}_{X,x} = \mathcal{O}_{X,x}^\times\}.$$

Proposition 2.1.4 (Universal property of compactifying log structures)**.** [Ogu18, Propostion 1.6.2.] Let X be a log scheme with compactifying log structure associated to an open morphism $U \subseteq X$. Then for every log scheme Y, the natural map

$$\mathrm{Hom}(X, Y) = \{f \in Hom(U, Y^\circ) | f(U) \subseteq Y_{\mathrm{tr}}\}$$

Corollary 2.1.5. Let X be a log scheme with log strucutre associated to an open morphism $U \subseteq X$. Let $Y \to X^\circ$ be a morphism of schemes. Endow Y with the log structure associated to the preimage of U in Y. Then the morphisms of log schemes $Y \to X$ is strict.

2.2. Logarithmic differentials

Let $f : X \to S$ be a morphism of prelog schemes. Denote by $\alpha : \mathcal{M}_X \to \mathcal{O}_X$ the structure morphism of X. Let $\mathcal{E}$ be an $\mathcal{O}_X$-module. An *S-derivation* from $(\mathcal{O}_X, \mathcal{M}_X)$ to $\mathcal{E}$ is a pair (d, dlog) of maps, where $d : \mathcal{O}_X \to \mathcal{E}$ is an S-derivation and $\mathrm{dlog} : \mathcal{M}_X \to \mathcal{E}$ is a morphism of monoids satisfying:

- $d(\alpha(m)) = \alpha(m)\mathrm{dlog}(m)$ for each section $m \in \mathcal{M}_X$,
- $\mathrm{dlog}(f^\flat(m)) = 0$ for each section $m \in \mathcal{M}_S$.

Note that if $f : X \to Y$ is a morphism of log schemes induced by a morphism of prelog schemes $X^{\mathrm{pre}} \to Y^{\mathrm{pre}}$, then the derivations are the same. In particular, if $X \to Y$ has a global chart $Q \to P$, we may replace the log structures on X and Y by the constant sheaves P and Q, respectively, without affecting the set of derivations.

Lemma 2.2.1. [Ogu18, Theorem IV.1.2.4] Let $f : X \to Y$ be a morphism of prelog schemes. The functor associating to an $\mathcal{O}_X$-module $\mathcal{E}$ the set of S-derivations is represented by the $\mathcal{O}_X$-module of *logarithmic Kähler differentials* $\Omega^{\log}_{X/S}$, together with the *universal S-derivation* (d, dlog).

The module $\Omega^{\log}_{X/S}$ is the quotient of

$$\Omega_{X/S} \bigoplus (\mathcal{O}_X \otimes_{\mathbb{Z}} \mathcal{M}_X^{\mathrm{gp}})$$

by the submodule generated by all elements of the form:

- $d(\alpha_X(a)) - \alpha_X(a) \otimes a$ where a is a section of $\mathcal{M}_X$
- $0 + 1 \otimes f^\flat(b)$ where b is a section of $\mathcal{M}_Y$.

Elements of the form $0 + 1 \otimes a$ are denoted by $d \log a$ for a section a of $\mathcal{M}_X$. The first equivalence relation should tautologically correspond to

$$\mathrm{dlog}(x) = dx/x.$$

Remark 2.2.2. Let X be a scheme over a base scheme S. Let $D \subset X$ be a normal crossing divisor. Endow X with the compactifying log structure for $X \backslash D$. Then we sometimes write $\Omega_{X/S}(\log D)$ instead of $\Omega^{\log}_{X/S}$.

Example 2.2.3. Let $f : X \to Y$ be a strict morphism. Then $\Omega^{\log}_{X/Y} \cong \Omega_{X/Y}$.

Lemma 2.2.4. [Fundamental exact sequence I] [Ogu18, Proposition IV.2.3.1] Let $X \xrightarrow{f} Y$ and $Y \to S$ be morphisms of log schemes. Then there is a natural exact sequence of $\mathcal{O}_X$-modules

$$f^*\Omega^{\log}_{Y/S} \to \Omega^{\log}_{X/S} \to \Omega^{\log}_{X/Y} \to 0.$$

Furthermore, if $X \xrightarrow{f} Y$ is log smooth, then the map on the LHS is injective [Ogu18, Theorem IV.3.2.3].

Lemma 2.2.5. [Fundamental exact sequence II][Ogu18, Proposition IV.2.3.2] Let X be a log scheme over some base S. Let $i : Z \to X$ be a strict closed immersion defined by an ideal $\mathcal{I} \subset \mathcal{O}_X$. Then we have an exact sequence

$$\mathcal{I}/\mathcal{I}^2 \xrightarrow{d} i^*\Omega^{\log}_{X/S} \to \Omega^{\log}_{Z/S} \to 0$$

of $\mathcal{O}_Z$-modules.

We now introduce the notion of coherent ideals in the context of monoids.

Definition 2.2.6. An ideal $\mathcal{K}$ of a sheaf of monoids $\mathcal{M}_X$ is a subsheaf satisfying for each étale morphism $U \to X$:

$$\forall \alpha \in \mathcal{K}(U), \beta \in \mathcal{M}_X(U), \alpha\beta \in \mathcal{K}(U).$$

Assume that $\mathcal{M}_X$ defines a coherent log structure on X. We then say that $\mathcal{K}$ is *coherent* if there exists a chart $\{P_i\}_i$ of $\mathcal{M}_X$ and an ideal K_i of each P_i, such that $\mathcal{K}$ is locally generated by K_i as an ideal of $\mathcal{M}_X$.

Lemma 2.2.7. Let X be a log scheme and $\mathcal{I}$ is an ideal generated by a coherent ideal $\mathcal{K} \subset \mathcal{M}_X$. Let $i : Z \to X$ be a strict closed immersion defined by the ideal $\mathcal{I} \subset \mathcal{O}_X$. Then $i^*\Omega^{\log}_{X/S}$ and $\Omega^{\log}_{Z/S}$ are isomorphic.

Proof. [Ogu18, Corollary IV.2.3.3] Let β be an element in $\mathcal{K}$. Then $d\alpha(\beta) = \alpha(\beta)d\beta$, which maps to zero via $\Omega^{\log}_{X/S} \to i^*\Omega_{X/S}$. Hence the morphism

$$\mathcal{I}/\mathcal{I}^2 \to i^*\Omega^{\log}_{X/S}$$

is trivial. We conclude that $i^*(\Omega^{\log}_{X/S}) \to \Omega^{\log}_{Z/S}$ is an isomorphism by Lemma 2.2.5. □

2.3. Normal Crossing Divisors

Let $D = D_1 \cup \cdots \cup D_r$ be an snc divisor on a scheme X. We consider the compactifying log structure on X which is associated to the open subscheme $X \backslash D$. As D_i are locally cut out by $f_i \in \mathcal{O}_X$, this log structure is coherent and locally defined by the prelog structure:

$$\bigoplus_{i=1}^{r} \mathbb{N} \to \mathcal{O}_X,\ (n_1, \ldots, n_r) \mapsto (f_1^{n_1} \cdots f_r^{n_r}).$$

In this section we are interested in properties of such log structures and their differentials, as well as the log structure on $E = D_1$ induced by the log structure of X.

Example 2.3.1. A DF-structure is a set of morphisms of $\mathcal{O}_X$-modules

$$\{\gamma_i : \mathcal{L}_i \to \mathcal{O}_X | i = 1, \ldots, n \text{ where } \mathcal{L}_i \text{ is an invertible sheaf}\}.$$

The shortcut DF stands for Deligne and Faltings, who developed this structure.

To each DF-structure we can associate a log structure on X. This is the logification of the prelog structure $\mathcal{P} \to \mathcal{O}_X$ defined as follows:
Sections of $\mathcal{P}$ are pairs (a, I), where $I = (I_1, \ldots, I_n)$ is an n-tuple of integers and a generates $\mathcal{L}^I = \mathcal{L}_1^{I_1} \otimes \cdots \otimes \mathcal{L}_n^{I_n}$ as an $\mathcal{O}_X$-module.

Multiplication in $\mathcal{P}$ is defined as

$$\alpha \cdot (a, I) = (\alpha \cdot a, I), \quad \alpha \in \mathcal{O}_X, (a, I) \in \mathcal{P}.$$

The prelog structure is defined by sending a pair $(a = \sum a_1 \otimes \cdots \otimes a_n, I) \in \mathcal{P}$ to $\gamma(a) = \prod_i \gamma_i(a_i) \in \mathcal{O}_X$.

Lemma 2.3.2. Let X be a normal scheme. Let $i : D = D_1 \cup \cdots \cup D_r \to X$ be a strict normal crossing divisor. Let $\mathcal{M}_X$ be the log structure defined by $X \backslash D$. Let $\mathcal{M}_D$ be its pullback via i. Then $\mathcal{M}_D$ is the log structure generated by the DF-structure associated to the pullback of the $\mathcal{O}_X$-ideal sheaves $\mathcal{O}(-D_1), \cdots, \mathcal{O}(-D_r)$.

Proof. The claim follows directly from [Ogu18, example III.1.7.2]: the log structure associated to an snc divisor is the same as the one generated by the DF-structure given by the ideals $\mathcal{O}(-D_1), \cdots, \mathcal{O}(-D_r)$. □

Example 2.3.3. [Ogu18, Proposition III.1.7.5] Let D be an integral effective divisor on a normal scheme X over a base S. Let $\mathcal{I}_D := \mathcal{O}(-D)$ be the ideal sheaf defining D. We can consider the log structure on D making $D \to X$ a strict morphism. This is defined by the DF-structure associated to the invertible $\mathcal{O}_D = \mathcal{O}_X/\mathcal{I}_D$-sheaf $\mathcal{I}_D/\mathcal{I}_D^2$, together with the zero map $\mathcal{I}_D/\mathcal{I}_D^2 \to \mathcal{O}_D$.

From another perspective, this log structure is locally defined by the prelog structure

$$\mathbb{N} \to \mathcal{O}_X \to \mathcal{O}_D, n \mapsto f^n \mapsto 0,$$

where $f = 0$ is a local equation defining D. Hence the sheaf of log differentials $\Omega^{\log}_{D/S}$ is locally isomorphic to

$$\Omega_{D/S} \oplus (\mathbb{N}^{\mathrm{gp}} \otimes_{\mathbb{Z}} \mathcal{O}_D) \cong \Omega_{D/S} \oplus \mathcal{O}_D,$$

since the equivalence relation

$$f\mathrm{dlog}(f) = df$$

as stated in Lemma 2.2.1 is trivial as $f = 0$ in $\mathcal{O}_D$.

The local prelog structure does however not glue to a global prelog structure. Instead we use the log structure associated to the pre-log structure

$$\mathbb{N} \oplus \mathcal{O}_X^\times \to \mathcal{O}_X \to \mathcal{O}_D, \quad n \oplus h \mapsto \begin{cases} h \bmod \mathcal{I}_D, & \text{if } n = 0 \\ 0 & \text{otherwise.} \end{cases}$$

For opens U_1 and U_2 in X, for which D is defined by $V(f_1)$ and $V(f_2)$, respectively, the glueing morphism is then

$$\mathbb{N} \oplus \mathcal{O}^\times_{U_1 \cap U_2} \to \mathbb{N} \oplus \mathcal{O}^\times_{U_1 \cap U_2} : \ n \oplus h \mapsto n \oplus h(f_1/f_2)^n.$$

This defines a glueing morphism on the logarithmic differentials corresponding to the $\mathcal{O}_D$-linear morphism

$$\Omega_{D/S} \oplus \mathcal{O}_D|_{U_1 \cap U_2} \to \Omega_{D/S} \oplus \mathcal{O}_D|_{U_1 \cap U_2} : \ \big(da \oplus h \mapsto da + h\mathrm{dlog}(f_1/f_2)\big) \oplus h.$$

To better understand the previous calculation, we may consider log differentials of D locally as sums of elements of the form

$$bda + h\mathrm{dlog}(f^n) = bda + nh\mathrm{dlog}(f),$$

where f is a section of $\mathcal{O}_X$ generating $\mathcal{O}(-D)$. Therefore

$$bda + h\mathrm{dlog}(f_1) = bda + h\mathrm{dlog}(f_1/f_2) + h\mathrm{dlog}(f_2).$$

Let X be a scheme over a base S. Let $D = D_1 \cup \cdots \cup D_r$ be an snc divisor. Then the pullback log structure on $E := D_1$ is generated by the DF-structure associated to $\mathcal{I}_{D_i \cap E/E} = \iota_E^* \mathcal{I}_{D_i}$ for $i \neq 1$ and $\mathcal{I}_E/\mathcal{I}_E^2$ for $j = 1$.

Alternatively, the log structure of E is locally defined by the map

$$\alpha : \mathbb{N} \oplus i_{2,*}\mathbb{N} \oplus \cdots \oplus i_{r,*}\mathbb{N} \to \mathcal{O}_E, \ n + n_1 + \cdots + n_r \mapsto \begin{cases} f_2^{n_2} \cdots f_r^{n_r} & \text{if } n = 0, \\ 0 & \text{if } n \geq 1, \end{cases}$$

where f_i are the local generators of $\mathcal{O}(-D_i)$.

The gluing morphism sends a section

$$(h, n, n_2, \cdots, n_r) \in \big(\mathcal{O}_E^\times \bigoplus_{\alpha^{-1}(\mathcal{O}_E^\times)} \big[\mathbb{N} \oplus i_{2,*}\mathbb{N} \oplus \cdots \oplus i_{r,*}\mathbb{N}\big]\big)(U_1 \cap U_1^{'})$$

to

$$(h(f_1/f_1')^n \cdots (f_r/f_r')^{n_r}, n, n_2, \cdots, n_r) \in \big(\mathcal{O}_E^\times \bigoplus_{\alpha^{-1}(\mathcal{O}_E^\times)} \big[\mathbb{N} \oplus i_{2,*}\mathbb{N} \oplus \cdots \oplus i_{r,*}\mathbb{N}\big]\big)(U_1 \cap U_1^{'}).$$

Proposition 2.3.4. In the situation above, denote by D^1 the scheme $D_2 \cap \cdots \cap D_r$. Recall that $E := D_1$. Then the sheaf $\Omega^{\log}_{E/S}$ is locally isomorphic to

$$\Omega_{E/S}(\log(D^1 \cap E)) \oplus \mathcal{O}_E$$

on opens $U \subset E$, which $\mathcal{I}_E/\mathcal{I}_E^2$ is principal. The gluing morphism is

$$\alpha \oplus h \mapsto \big[\alpha + h\mathrm{dlog}(f_1/f_2)\big] \oplus h,$$

on any pair of open subsets $U_i \subset E$ of E for which $\mathcal{I}_E/\mathcal{I}_E^2$ is freely generated by a section $[f_i] = f_i \mod \mathcal{I}_E \in \mathcal{O}_E \cong \mathcal{O}_X/\mathcal{I}_E$.

Proof. The local description follows from the local description of E, together with Lemma 2.2.1.

Let $V_i \subset X$ be open subsets such that $U_i = E \cap V_i$. A section

$$\alpha \oplus h \in \Omega_{E/S}(\log(D^1 \cap E)) \bigoplus \mathcal{O}_E(U_i)$$

is the restriction of

$$\alpha' \oplus h\mathrm{dlog}(f_i) \in \Omega_{X/S}(\log D)(V_i).$$

for an adequate choice of $\alpha' \in \Omega_{X/S}(\log(D^1))(V_i)$. Over $V_1 \cap V_2$, we have the equality

$$\alpha' \oplus h\mathrm{dlog}(f_1) = \alpha' + h\mathrm{dlog}(\frac{f_1}{f_2}) \oplus h\mathrm{dlog}(f_2),$$

which translates to the claimed gluing isomorphism after restricting to E. □

We define the sheaf of n-differentials

$$\Omega^{n,\log}_{X/S} := \bigwedge^n \Omega^{\log}_{X/S}.$$

Corollary 2.3.5. We have a short exact sequence

$$0 \to \Omega_{E/S}(\log(D^1 \cap E)) \to \Omega^{\log}_{E/S} \to \mathcal{O}_E \to 0.$$

More generally, we have

$$0 \to \Omega^n_{E/S}(\log(D^1 \cap E)) \to \Omega^{n,\log}_{E/S} \to \Omega^{n-1}_{E/S}(\log(D^1 \cap E)) \to 0. \tag{2.1}$$

Proof. We directly prove 2.1.

The map on the left hand side is the natural inclusion.

On the other hand we have Zariski-locally the isomorphism

$$\Omega^{\log}_{E/S} = \Omega_{E/S}(\log(D^1 \cap E)) \oplus \mathcal{O}_E.$$

Therefore we can describe $\Omega^{n,\log}_{E/S}$ Zariski-locally as

$$\bigwedge^n (\Omega_{E/S}(\log(D^1 \cap E)) \oplus \mathcal{O}_E) = \left[\Omega^{n-1}_{E/S}(\log(D^1 \cap E)) \wedge \mathcal{O}_E\right] \bigoplus \Omega^n_{E/S}(\log(D^1 \cap E).$$

Using this description, we define

$$\begin{array}{ccc} \bigwedge^n \left(\Omega_{E/S}(\log(D^1 \cap E)) \oplus \mathcal{O}_E\right) & \to & \Omega^{n-1}_{E/S}(\log(D^1 \cap E)) \\ \eta \wedge \gamma + \eta' & \mapsto & \eta. \end{array} \tag{2.2}$$

This is obviously surjective with kernel $\Omega^n_{E/S}(\log(D^1 \cap E))$. It is left to prove that this glues to a global map.

Let U_1 and U_2 be two open subsets of X for which $E = V(f_1)$ and $E = V(f_2)$ respectively, for sections f_1, f_2 in $\mathcal{O}_X$. The gluing morphism sends $\eta \wedge \gamma + \eta'$ to

$$\eta \wedge \gamma + \left(\eta \wedge \gamma \mathrm{dlog}(\frac{f_1}{f_2}) + \eta'\right).$$

Both these elements are sent to η in (2.2). □

Lemma 2.3.6. [Ogu18, Corollary IV.1.2.8] Let S be a scheme, and let X be a smooth scheme over S endowed with the log structure defined by an snc divisor $D = \bigoplus_{i=0}^n D_i$. Then the sheaf $\Omega_X(\log(D))$ is a coherent $\mathcal{O}_X$-module.

Proposition 2.3.7. (cf. [EV92, Properties 2.3] for $S = \mathrm{Spec}(k), k = k^{\mathrm{alg}}$) Let X be a smooth scheme over S. We endow it with the log structure defined by an snc divisor $D = D_1 \cup \cdots \cup D_n$. Let $a > 0$ be an integer. Define $D^1 := D - D_1$. We have the following short exact sequences

$$\begin{aligned} &0 \to \Omega_{X/S} \to \Omega_{X/S}(\log(D)) \to \bigoplus_{i=1}^n \mathcal{O}_{D_i} \to 0, \\ &0 \to \Omega^a_{X/S}(\log(D - D_1)) \to \Omega^a_{X/S}(\log(D)) \to \Omega^{a-1}_{D_1/S}(\log(D^1)|_{D_1}) \to 0, \\ &0 \to \Omega^a_{X/S}(\log(D))(-D_1) \to \Omega^a_{X/S}(\log(D^1)) \to \Omega^a_{D_1/S}(\log(D^1|_{D_1}) \to 0. \end{aligned}$$

Remark 2.3.8. We wrote $\mathcal{O}_{D_i}$ and $\Omega^a_{D_i/S}$ instead of $i_*\mathcal{O}_{D_i}$ and $i_*\Omega^a_{D_i/S}$, where $i : D_i \to X$ is the given closed immersion. This is done to simplify the notation.

Proof. We follow the proof in the reference. The condition that S is the spectrum of an algebraically closed field, as in loc. cit., may be dropped.

For the first sequence define the map locally as

$$\alpha : \Omega_{X/S}(\log(D)) \to \bigoplus_{i=1}^{n} \mathcal{O}_{D_i}$$

$$a\alpha(dg) + \alpha(\sum_{i=1}^{n} a_i \mathrm{dlog}(f_i)) = \sum_{i=1}^{n} a_i|_{D_i},$$

where the f_i's locally define the divisors D_i. Note that this is well defined and glues to a global map: given other equations f_i' defining D_i, we have

$$\alpha(a_i \mathrm{dlog} f_i) = \alpha(a_i \mathrm{dlog} f_i' + a_i \mathrm{dlog}(f_i/f_i')) = a_i \in \mathcal{O}_{D_i}.$$

The morphism α is obviously surjective and $\ker(\alpha)$ is $\Omega_{X/S}$.

Let η be a section of $\Omega^a_{X/S}(\log D)$. Then η is locally a sum of elements of the form

$$\eta_1 + \mathrm{dlog} f_1 \wedge \eta_2,$$

where η_1 is a section of $\Omega^a_{X/S}(\log(D - D_1))$ and η_2 is a section of $\Omega^{a-1}_{X/S}(\log(D - D_1))$. We define the map

$$\phi_1 : \Omega^a_{X/S}(\log D) \to \Omega^{a-1}_{D_1/S}(\log(D - D_1)|_{D_1}), \quad \eta_1 + \mathrm{dlog} f_1 \wedge \eta_2 \mapsto \eta_2|_{D_1}.$$

The map ϕ_1 is well defined and independent from the representative f_1. Indeed, let f_1' be another local generator of $\mathcal{O}(-D_1)$. Then

$$\eta_1 + \mathrm{dlog}(f_1) \wedge \eta_2 = \big(\eta_1 + \mathrm{dlog}(\frac{f_1'}{f_1}) \wedge \eta_2\big) + \log(f_1') \wedge \eta_2.$$

The map ϕ_1 is surjective and the elements of the kernel are locally of the form $\eta = \eta_1$. Hence $\ker(\phi_1) = \Omega^a_{X/S}(\log(D - D_1))$.

Consider the third sequence. The morphism

$$\phi_2 : \Omega^a_{X/S}(\log(D - D_1)) \to \Omega^a_{D_1/S}(\log(D - D_1)|_{D_1})$$

is just the restriction, which is surjective. Since X/S is smooth, $\Omega^a_{X/S}(\log(D - D_1))$ has locally a basis generated by wedge products of

$$df_1, \mathrm{dlog} f_2, \ldots \mathrm{dlog} f_n, df_{n+1}, \ldots, df_m,$$

with $D_i = V(f_i), i = 1, \ldots n$ and f_i are sections of $\mathcal{O}_X$. The $\mathcal{O}_{D_1}$-module $\Omega_{D_1/S}(\log(D - D_1)|_{D_1})$ has a basis generated by wedge products of classes of

$$\mathrm{dlog} f_2, \ldots, \mathrm{dlog} f_n, df_{n+1}, \ldots, df_m \qquad \mathrm{mod}\ f_1$$

The morphism ϕ_2 sends basis elements not involving df_1 to basis elements, and those which contain df_1 to zero. The kernel is then generated by elements of the form $f_1\eta$ where η is a wedge product of

$$\mathrm{dlog} f_1, \mathrm{dlog} f_2, \ldots, \mathrm{dlog} f_n, df_{n+1}, \ldots, df_m.$$

Elements of the form $f_1\eta$ constitute a basis of $\Omega_{X/S}(\log(D))(-D_1)$. □

Corollary 2.3.9. For the same setup we have the short exact sequence of closed forms

$$0 \to Z\Omega^a_{X/S}(\log(D-D_1)) \to Z\Omega^a_{X/S}(\log(D)) \to Z\Omega^{a-1}_{D_1/S}(\log(D-D_1)|_{D_1}) \to 0.$$

Proof. The morphisms in the second exact sequence in Proposition 2.3.7 commutes with d. We have therefore the morphism of short exact sequences

$$\begin{array}{ccccccccc} 0 & \to & \Omega^a_{X/S}(\log(D-D_1)) & \to & \Omega^a_{X/S}(\log(D)) & \to & \Omega^{a-1}_{D_1/S}(\log(D-D_1)|_{D_1}) & \to & 0 \\ & & \downarrow d & & \downarrow d & & \downarrow d & & \\ 0 & \to & \Omega^{a+1}_{X/S}(\log(D-D_1)) & \to & \Omega^{a+1}_{X/S}(\log(D)) & \to & \Omega^a_{D_1/S}(\log(D-D_1)|_{D_1}) & \to & 0. \end{array}$$

After applying the snake lemma, we get the sequence

$$\begin{aligned} 0 \to & Z\Omega^a_{X/S}(\log(D-D_1)) \to Z\Omega^a_{X/S}(\log(D)) \to Z\Omega^{a-1}_{D_1/S}(\log(D-D_1)|_{D_1}) \\ & \to B\Omega^{a+1}_{X/S}(\log(D-D_1)) \to \dots. \end{aligned}$$

It is left to check that the boundary sequence induced by the snake lemma is the zero map. Since the claim is étale local and $X \to S$ is log smooth, we may assume that S is affine, $X \cong \mathbb{A}^m_S$ with coordinates $T_1, \dots, T_m$ and $D_i = V(T_i)$ for each $i = 1, \dots, n$. We write $\delta_i := \mathrm{dlog} T_i$ for $i = 1, \dots, n$ and $\delta_i := dT_i$ for $i = n+1, \dots, m$.

Let $\tilde{\eta}$ be a section of $Z\Omega^{a-1}_{D_1/S}(\log(D-D_1)|_{D_1})$ and let $\eta \wedge \mathrm{dlog} T_1 \in \Omega^a_{X/S}(\log(D))$ be a preimage. We may assume that η is a sum of elements of the form

$$g_{i_1,\dots,i_{a-1}} \delta_{i_1} \wedge \dots \wedge \delta_{i_{a-1}}, \quad i_1, \dots, i_{a-1} \neq 1,\ T_1 \not| g_{i_1,\dots,i_{a-1}},$$

where

$$g_{i_1,\dots,i_{a-1}} \in \mathcal{O}_X(X) = \mathcal{O}_S(S)[T_1, \dots, T_{a-1}].$$

We may also assume that all monomials of the polynomials $g_{i_1,\dots,i_{a-1}}$ are not divisible by T_1, since such monomials are trivial when restricting to D_1.

Since $d\tilde{\eta} = 0$, the differential $d(\eta \wedge \mathrm{dlog} T_1) = d\eta \wedge \mathrm{dlog} T_1$ lies in $\Omega^{a+1}_{X/S}(\log(D-D_1))$. Hence $d\eta$ is necessarily divisible by T_1. But all monomials of $g_{i_1,\dots,i_{a-1}}$ are not divisible by T_1, implying that $d\eta = 0$. □

2.4. Acyclicity of Blow-ups

Let k be a field. Let X be a log smooth scheme over k, defined by a simple normal crossing divisor $D \subset X$. Let $\bar{X} \to X$ be a blowup with smooth center Z and exceptional divisor E. Endow $\bar{X}$ with the log structure corresponding to $E \cup \bar{D}$, where $\bar{D}$ is the strict transform of D. We want to compare the sheaf cohomology of $\Omega^{n,\log}_{\bar{X}}$ and $\Omega^{n,\log}_X$ in good cases. We first need some lemmas describing the sheaf cohomology groups of logarithmic differentials of projective spaces.

Lemma 2.4.1. [Stacks, Lemma 0FMH] Let A be a ring. Let $n \geq 1$ be an integer. There exists a short exact sequence

$$0 \to \Omega_{\mathbb{P}^n_A/A} \to \mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus n+1} \to \mathcal{O}_{\mathbb{P}^n_A} \to 0. \tag{2.3}$$

This is called the *Euler sequence*. More generally we have the sequence

$$0 \to \Omega^j_{\mathbb{P}^n_A/A} \to \bigwedge^j (\mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus n+1}) \cong \mathcal{O}_{\mathbb{P}^n_A}(-j)^{\oplus \binom{n+1}{j}} \to \Omega^{j-1}_{\mathbb{P}^n_A/A} \to 0.$$

Let us describe the morphisms in (2.3). The sheaf $\Omega_{\mathbb{P}^n_A/A}$ is locally generated by elements of the form

$$\sum_{i=0}^n f_i dX_i \quad \text{such that } f_i \in \mathcal{O}_{\mathbb{P}^n_A}(-1),$$

where X_i are the homogeneous coordinates of $\mathbb{P}^n_A$. The morphism on the left hand side is

$$\Omega_{\mathbb{P}^n_A/A} \to \mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus n+1} =: \bigoplus_{i=0}^n \mathcal{O}_{\mathbb{P}^n_A} e_i, \quad \sum f_i dX_i \mapsto \sum f_i e_i. \tag{2.4}$$

The right hand side is

$$\mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus n+1} \to \mathcal{O}_{\mathbb{P}^n_A}, \quad \sum_{i=0}^n f_i e_i \mapsto \sum_{i=0}^n f_i T_i. \tag{2.5}$$

Using this, we deduce the following facts on cohomology.

Lemma 2.4.2. [Stacks, TAG 0FUK] For any ring A we have

$$H^i(\mathbb{P}^n_A, \Omega^j_{\mathbb{P}^n_A/A}) \cong \begin{cases} A & \text{if } 0 \leq i = j \leq n \\ 0 & \text{otherwise} \end{cases}$$
$$H^i(\mathbb{P}^n_A, \Omega^j_{\mathbb{P}^n_A/A}(l)) \cong 0 \text{ for all } l \geq 1.$$

Remark 2.4.3. The sheaf cohomology groups $H^i(\mathbb{P}^n_A, \Omega^j_{\mathbb{P}^n_A/A})$ are isomorphic to the Čech cohomology groups $\check{H}^i(\{U_l \to \mathbb{P}^n_A\}, \Omega^j_{\mathbb{P}^n_A/A})$, where $\{U_l \to \mathbb{P}^n_A\}$ is the standard affine open cover defined by $U_l := \{X_l \neq 0\} \subseteq \mathbb{P}^n_A$.

The generators of $\check{\mathrm{H}}^i(\{U_l \to \mathbb{P}^n_A\}, \Omega^j_{\mathbb{P}^n_A/A})$ were described in detail in [Stacks, TAG 0FMI]. A generator of $\check{\mathrm{H}}^j(\{U_l \to \mathbb{P}^n_A\}, \Omega^j_{\mathbb{P}^n_A/A})$ is represented by

$$(\sum_{k=0}^j (-1)^k \mathrm{dlog}(X_{i_0}) \wedge \ldots \widehat{\mathrm{dlog} X_{i_k}} \wedge \mathrm{dlog}(X_{i_j}))_{i_0,\ldots,i_j} \in \check{C}^j(\{U_i \to \mathbb{P}^n_A\}, \Omega^j_{\mathbb{P}^n_A/A}).$$

This can be checked inductively using the short exact sequences from Lemma 2.4.1 as follows.

For $j=1$, the map $\check{\mathrm{H}}^0(\mathbb{P}^n_A, \mathcal{O}_{\mathbb{P}^n_A}) \to \check{\mathrm{H}}^1(\{U_l \to \mathbb{P}^n_A\}, \Omega^1_{\mathbb{P}^n_A/A})$ sends $(1)_i$ to

$$(dX_m/X_m - dX_n/X_n)_{mn} = (\mathrm{dlog}X_m - \mathrm{dlog}X_n)_{mn}.$$

For $j \geq 1$, we consider the general form of the Euler sequence and use the fact that the cohomology of $\mathcal{O}_{\mathbb{P}^n_A}(-1)$ vanishes. Then the image of

$$\left(\sum_{k=0}^{j}(-1)^k \mathrm{dlog}X_{i_0} \wedge \ldots \widehat{\mathrm{dlog}X_{i_k}} \wedge \cdots \wedge \mathrm{dlog}X_{i_j}\right)_{i_0,\ldots,i_j} \in \check{C}^j(\{U_i \to \mathbb{P}^n_A\}, \Omega^j_{\mathbb{P}^n_A/A})$$

under the boundary morphism $H^j(\mathbb{P}^n_A, \Omega^j_{\mathbb{P}^n_A}) \to H^{j+1}(\mathbb{P}^n_A, \Omega^{j+1}_{\mathbb{P}^n_A})$ is

$$(e_{i_0}/X_{i_0} \wedge \cdots \wedge e_{i_j}/X_{i_j})_{i_0,\ldots,i_j} \in \check{C}^j(\{U_i \to \mathbb{P}^n_A\}, \bigwedge^{j+1}(\mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus n+1})).$$

This maps to the element

$$\left(\sum_{k=0}^{j+1}(-1)^k e_{i_0}/X_{i_0} \wedge \cdots \wedge \widehat{e_{i_k}/X_{i_k}} \cdots \wedge e_{i_{j+1}}/X_{i_{j+1}}\right)_{i_0,\ldots,i_{j+1}}$$

in $\check{C}^{j+1}(\{U_i \to \mathbb{P}^n_A\}, \bigwedge^{j+1}(\mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus n+1}))$, which corresponds to

$$\left(\sum_{k=0}^{j+1}(-1)^k \mathrm{dlog}(X_{i_0}) \wedge \cdots \wedge \widehat{\mathrm{dlog}X_{i_k}} \wedge \cdots \wedge \mathrm{dlog}X_{i_{j+1}}\right)_{i_0,\ldots,i_{j+1}}$$

in $\check{C}^{j+1}(\{U_i \to \mathbb{P}^n_A\}, \Omega^{j+1}_{\mathbb{P}^n_A/A})$.

Lemma 2.4.4. Let E be the projective space $\mathbb{P}^n_R$ over a regular ring R. Let $D = V(X_0) \subset E$ be a hyperplane divisor. Then $\Omega^n_{E/R}(\log(D)) \otimes_{\mathcal{O}_E} \mathcal{O}_E(l)$ has trivial cohomology groups for each $l \geq 0$, $n \geq 1$.

Proof. We may assume that R is a local ring. Indeed, any cohomology group $H^i(E, \mathcal{F})$ for a coherent sheaf $\mathcal{F}$ corresponds to the global sections of the coherent sheaf $R^i\pi_*\mathcal{F}$, where π is the projection $\mathbb{P}^n_R \to \mathrm{Spec}(R)$. Note that in this case $\mathrm{Pic}(\mathbb{P}^n_R) = \mathbb{Z}$ and all invertible sheaves are of the form $\mathcal{O}(l)$, $l \in \mathbb{Z}$ (see [Stacks][Tag 0BXJ], note that regular local rings are UFD).

Proof for $l = 0, n = 1$:

We have a short exact sequence of residues as in Proposition 2.3.7

$$0 \to \Omega_{E/R} \to \Omega_{E/R}(\log(D)) \to \mathcal{O}_D \to 0. \tag{2.6}$$

The cohomology group $H^i(E, \Omega_{E/R})$ and $H^i(D, \mathcal{O}_D)$ are trivial for $i \neq 1$ and $i \neq 0$ respectively by Lemma 2.4.2.

Now we have the long exact sequence of cohomology associated to the sequence (2.6)

$$\begin{aligned}\ldots &\to H^0(E,\mathcal{O}_D) \to H^1(E,\Omega_{E/R})\\ &\to H^1(E,\Omega_{E/R}(\log(D))) \to H^1(E,\mathcal{O}_D) = 0 \to \cdots .\end{aligned}$$

We obtain that $H^j(E,\Omega_E(\log(D))) = 0$ for all $j \geq 2$. It is left to check that the map $H^0(E,\mathcal{O}_D) \to H^1(E,\Omega_{E/R})$ is surjective. We claim that the morphism

$$H^0(D,\mathcal{O}_D) \to H^1(E,\Omega_{E/R})$$

arising from the short exact sequence is an isomorphism. For this we may instead consider Čech cohomology groups corresponding to the standard affine open covering $\{U_i = D(X_i)\}$ of $\mathbb{P}^n_R \cong E$. Now by the following diagram chase we determine the image of $1 \in R \cong \check{\mathrm{H}}^0(D,\mathcal{O}_D)$:

$$\begin{tikzcd}
 & & & \check{\mathrm{H}}^0(D,\mathcal{O}_D) \arrow[d] \arrow[dddll, bend left] & \\
 & \prod_{i=0}^n \Gamma(U_i,\Omega_{E/R}) \arrow[r] \arrow[d] & \prod_{i=0}^n \Gamma(U_i,\Omega_{E/R}(\log(D))) \arrow[r] \arrow[d] & \prod_{i=1}^n R[\frac{X_j}{X_i}|j=1\ldots n]_0 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Z^1(E,\Omega_{E/R}) \arrow[r] \arrow[d] & Z^1(E,\Omega_{E/R}(\log(D))) \arrow[r] & Z^1(D,\mathcal{O}_D) & \\
 & \check{\mathrm{H}}^1(E,\Omega_{E/R}) & & &
\end{tikzcd} \tag{2.7}$$

$$\begin{tikzcd}
 & & & 1 \arrow[d, mapsto] & \\
 & 0 \arrow[r, mapsto, dotted] \arrow[d, mapsto, dotted] & (\mathrm{dlog}(X_0/X_i))_i \arrow[r, mapsto] \arrow[d, mapsto] & (1)_i \arrow[r, mapsto, dotted] \arrow[d, mapsto, dotted] & 0, \\
0 \arrow[r, mapsto] & (\mathrm{dlog}(X_j) - \mathrm{dlog}(X_i))_{ij} \arrow[r, mapsto] \arrow[d, mapsto] & (\mathrm{dlog}(X_j) - \mathrm{dlog}(X_i))_{ij} \arrow[r, mapsto, dotted] & 0 & \\
 & [(\mathrm{dlog}(X_j) - \mathrm{dlog}(X_i))_{ij}] & & &
\end{tikzcd}$$

where $Z^1(E,-)$ is the kernels of the boundary morphisms

$$\prod_{i,j=0}^n \Gamma(U_{ij},-) \to \prod_{i,j,k=0}^n \Gamma(U_{ijk},-)$$

of the Čech complex. Each of the elements

$$\mathrm{dlog}(X_0/X_i) \in \Gamma(U_i,\Omega_{E/R}(\log(D)))$$

maps to

$$1 \in \Gamma(D \cap U_i,\mathcal{O}_D) = R[X_j|j=1,\ldots,n][X_i^{-1}]_0, \quad \text{for } i > 0,$$

by the first residue sequence in Proposition 2.3.7. For $i = 0$, $\mathrm{dlog}(X_0/X_i) = 0$, but $U_0 \cap D_1 = \emptyset$, hence

$$\Gamma(D \cap U_0, \mathcal{O}_D) \cong 0$$

does not contribute to the Čech cohomology group $\check{\mathrm{H}}^0(D, \mathcal{O}_D)$.

On the other hand, by Remark 2.4.3, we know that $\check{\mathrm{H}}^1(E, \Omega_{E/R})$ is generated by the class $(\mathrm{dlog}(X_j) - \mathrm{dlog}(X_i))_{ij}$.

Proof for $l = 0, n > 1$: By the second residue sequence in Proposition 2.3.7, we have a short exact sequence

$$0 \to \Omega^n_{E/R} \to \Omega^n_{E/R}(\log D) \to \Omega^{n-1}_{D/R} \to 0. \tag{2.8}$$

Note that D is also a projective space isomorphic to $\mathbb{P}^{n-1}_R$. Hence

$$H^{j-1}(D, \Omega^{n-1}_{D/R}) \cong H^j(E, \Omega^n_{E/R}) \cong \begin{cases} R & \text{if } j = n \\ 0 & \text{otherwise.} \end{cases}$$

We now follow the steps of (2.7) to prove that the boundary morphism

$$H^{n-1}(D, \Omega^{n-1}_{D/R}) \to H^n(E, \Omega^n_{E/R}),$$

associated to the short exact sequence (2.8), is an isomorphism. Note that by Remark 2.4.3 a generator of

$$H^{n-1}(D, \Omega^{n-1}_{D/R}) = \check{\mathrm{H}}^{n-1}(\{U_i \to \mathbb{P}^{n-1}_R\}, \Omega^{n-1}_{\mathbb{P}^{n-1}/R})$$

is

$$(\sum_{k=0}^{n-1} (-1)^k \mathrm{dlog}(X_{i_0}) \wedge \cdots \wedge \widehat{\mathrm{dlog} X_{i_k}} \wedge \cdots \wedge \mathrm{dlog}(X_{i_{n-1}}))_{i_0,\ldots,i_{n-1}}. \tag{2.9}$$

Note that i_j lies in $\{1, \ldots n-1\}$. A preimage of (2.9) in

$$\prod_{i_0,\ldots,i_{n-1}} \Gamma(U_{i_0,\ldots,i_{n-1}}, \Omega^n_{E/R}(\log(D)))$$

by the second residue sequence in Proposition 2.3.7 is

$$(\sum_{k=0}^{n-1} (-1)^k \mathrm{dlog}(\frac{X_0}{X_{i_k}}) \wedge \mathrm{dlog}(X_{i_0}) \wedge \cdots \wedge \widehat{\mathrm{dlog} X_{i_k}} \wedge \cdots \wedge \mathrm{dlog}(X_{i_{n-1}}))_{i_0,\ldots,i_{n-1}},$$

as $\{\frac{X_0}{X_{i_k}} = 0\} = D \cap U_{i_0,\ldots,i_{n-1}}$ for any k and each $U_{i_0,\ldots,i_{n-1}}$. Here we allow $i_j = 0$. This may be rewritten as

$$\begin{aligned} &\big(\sum_{k=0}^{n-1} (-1)^k \mathrm{dlog} X_0 \wedge \mathrm{dlog} X_{i_0} \wedge \cdots \wedge \widehat{\mathrm{dlog} X_{i_k}} \wedge \cdots \wedge \mathrm{dlog} X_{i_{n-1}} \\ &- \sum_{k=0}^{n-1} \mathrm{dlog} X_{i_0} \wedge \cdots \wedge \mathrm{dlog} X_{i_k} \wedge \cdots \wedge \mathrm{dlog} X_{i_{n-1}}\big)_{i_0,\ldots,i_{n-1}}. \end{aligned}$$

Finally, by straightforward computations this maps to the element

$$(\sum_{k=0}^{n}(-1)^k \mathrm{dlog} X_{i_0} \wedge \cdots \wedge \widehat{\mathrm{dlog} X_{i_k}} \wedge \cdots \wedge \mathrm{dlog} X_{i_n})_{i_0,\ldots,i_n}$$

which lies in

$$Z^n(X, \Omega_{E/R}) = \ker\left(\prod_{i_0,\ldots,i_n} \Gamma(U_{i_0,\ldots,i_n}, \Omega_{E/R}) \to \prod_{i_0,\ldots,i_{n+1}} \Gamma(U_{i_0,\ldots,i_{n+1}}, \Omega_{E/R})\right),$$

and whose class is a generator of

$$H^n(X, \Omega^n_{E/R}) = \check{\mathrm{H}}^n(\{U_i \to \mathbb{P}^n_R\}, \Omega^n_{\mathbb{P}^n/R})$$

as an R-module by Remark 2.4.3.

Proof for $l \geq 1$: In this case the proof is simpler. We tensor the residue sequence with $\mathcal{O}_E(l)$ to obtain

$$0 \to \Omega^n_{E/R}(l) \to \Omega^n_{E/R}(\log D)(l) \to \Omega^{n-1}_{D/R}(l) \to 0.$$

We have seen in Lemma 2.4.2 that $\Omega^n_{E/R}(l)$ and $\Omega^{n-1}_{D/R}(l)$ have no higher cohomology groups since E/R and D/R are projective spaces. Note that tensoring $\Omega^{n-1}_{D/R}$ with $\mathcal{O}_E(l)$ corresponds to tensoring with $\mathcal{O}_D(l)$, thereby concluding that $\Omega^n_{E/R}(\log D)(l)$ has no higher cohomology groups. □

Theorem 2.4.5. Let k be a field of positive characteristic. Let $X = \mathrm{Spec}(A)$ be a connected affine log regular log scheme over k whose log structure is defined by a normal crossing divisor $D \subset X$. Let $f : \bar{X} \to X = \mathrm{Spec}(A)$ be a blowup in a regular center $Z \subset D$, such that the exceptional divisor E intersects the strict transform $\bar{D}$ of the divisor D transversally. Then

$$H^i(\bar{X}, \Omega^n_{\bar{X}/k}(\log(E + \bar{D}))) = 0 \quad \text{for} \quad i > 0, n \geq 0.$$

Proof. We prove the theorem by induction on n. For $n = 0$, $\Omega^n_{\bar{X}/k}(\log D) = \mathcal{O}_{\bar{X}}$. This case is proven by Chatzistamatiou and Rülling in [CR15, Theorem 1.1] . Let $n \geq 1$.

Note that $R^i f_* \Omega^{n,\log}_{\bar{X}}$ is a coherent sheaf, as $\Omega^n_{\bar{X},\log}$ is also coherent and f is proper [Stacks, TAG 02O4]. Hence $R^i f_* \Omega^{n,\log}_{\bar{X}}$ is the coherent sheaf corresponding to the A-module $H^i(\bar{X}, \Omega^{n,\log}_{\bar{X}})$.

Reduction Step 1: We first reduce to the case when D is a regular irreducible divisor containing Z:

Write $D = D_1 \cup \cdots \cup D_n$, where D_i are the irreducible components of D. Write $\bar{D} = \bar{D}_1 \cup \cdots \cup \bar{D}_n$, where each $\bar{D}_i$ is the strict transform of D_i. Since Z is smooth and connected, we may assume that it is contained in a single D_i, say D_1. We also define $\tilde{D} := \bar{D} \cap E$.

The second residue sequence in Proposition 2.3.7 states that

$$0 \to \Omega^n_{\bar{X}/S}(\log(E+\bar{D}-\bar{D}_i)) \to \Omega^n_{\bar{X}/S}(\log(E+\bar{D})) \to \Omega^{n-1}_{\bar{D}_1/S}(\log(E+\bar{D}-\bar{D}_i)|_{\bar{D}_i}) \to 0 \tag{2.10}$$

is exact. By induction on the number of divisors $D_i \neq D_1$, we reduce to the case where $D = D_1$ as follows:

$Z \cap D_i$ is contained in $D_1 \cap D_i$, which allows us to use the induction step on n by applying theorem 2.4.5 to $\Omega^{n-1}_{\bar{D}_i/S}(\log(E+\bar{D}-\bar{D}_i)|_{\bar{D}_i})$. Thus by the long exact sequence of cohomology we obtain

$$\begin{aligned}
&\cdots \to H^1\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}-\bar{D}_i))\big) \to H^1\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}))\big) \to \\
&H^1\big(\bar{D}_i, \Omega^{n-1}_{\bar{D}_i/S}(E+\bar{D}-\bar{D}_i)|_{\bar{D}_i}\big) \cong 0 \to \cdots \to H^j\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}-\bar{D}_i))\big) \\
&\to H^j\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}))\big) \to H^j\big(\bar{D}_i, \Omega^{n-1}_{\bar{D}_i/S}(E+\bar{D}-\bar{D}_i)|_{\bar{D}_i}\big) \cong 0 \to \dots .
\end{aligned}$$

Hence $H^1\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}-\bar{D}_i))\big) \to H^1\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}))\big)$ is surjective and $H^j\big(\bar{X}, \Omega^n_{\bar{X}/S}(\log(E+\bar{D}-\bar{D}_i))\big) \to H^j\big(\bar{X}, \Omega^n_{\bar{X}/S}(E+\bar{D})\big)$ is an isomorphism for $j \geq 2$. Note here that it was essential not to remove the divisor D_1. Otherwise, one can not apply the induction step of the theorem on D_i.

Reduction Step 2: Zariski-locally, Z is a complete intersection in X, defined by $(f_1, \dots, f_c)$, where $c = \operatorname{codim}_X(Z)$. We may furthermore assume that $D = V(f_1)$:

Indeed, let $x \in X$ be a closed point. Then $\mathcal{I}_Z/\mathfrak{m}_x\mathcal{I}_Z$ has a basis defined by $(\bar{f}_1, \dots, \bar{f}_c)$. Let α be a generator of $\mathcal{I}_{D,x}$ as an $\mathcal{O}_{X,x}$-module. Then $\bar{\alpha}$ lies in $\mathcal{I}_Z/\mathfrak{m}_x\mathcal{I}_Z$ and is nonzero. In particular we may replace one of the basis elements, say $\bar{f}_1$ by $\bar{\alpha}$. By Nakayama, $(\bar{f}_1, \dots, \bar{f}_n)$ lift to a set of minimal generators $(f_1, \dots, f_c)$ of $\mathcal{I}_{Z,x}$. We may then find an affine open U of X containing x, for which $(f_1, \dots, f_c)$ generate $\mathcal{I}_Z$ since $\mathcal{I}_Z$ is coherent.

We can now describe the inclusion $\tilde{D} = E \cap \bar{D} \to E$. We have

$$\tilde{D} = \operatorname{Proj}(\bigoplus_{n=1}^{\infty} (I_Z/f_1)^n/(I_Z/f_1)^{n+1}) \cong \operatorname{Proj}(A/I_Z[X_2, \dots X_c]) \cong \mathbb{P}^{c-2}_Z$$

and

$$E = \operatorname{Proj}(\bigoplus_{i=1}^{\infty} I_Z^n/I_Z^{n+1}) \cong \operatorname{Proj}(A/I_Z[X_1, \dots X_c]) \cong \mathbb{P}^{c-1}_Z,$$

where $\tilde{D}$ corresponds to $V(X_1)$ in $\mathbb{P}^{c-1}_Z \cong E$.

Reduction Step 3: We reduce the problem to computing sheaf cohomology groups of the projective space $E = \mathbb{P}^{c-1}_Z$:

To prove the claim we only need to check that $(R^i f_* \Omega^{n,\log}_{\bar{X}})_x$ is zero at every point $x \in \operatorname{Spec}(A)$.

The scheme $V := X \backslash Z$ is isomorphic to $\bar{X} \backslash E$. We can then cover V by affine opens V_i for which $H^i(V_i, \Omega^{n,\log}_{\bar{X}}) = 0$ as $\Omega^{n,\log}_{\bar{X}}$ is coherent. Hence $(R^i f_* \Omega^{n,\log}_{\bar{X}})_x$ is zero at every point $x \in V$.

It remains to check that $(R^i f_* \Omega^{n,\log}_{\bar{X}})_x$ is zero at each point $x \in Z$. This group is isomorphic to $H^i(\bar{X}, \Omega^{n,\log}_{\bar{X}}) \otimes_A A_{\mathfrak{p}}$, where $\mathfrak{p}$ is the prime ideal of A associated to x. The theorem on formal functions [Stacks][TAG 02OC] states that

$$H^i(\bar{X}, \Omega^{n,\log}_{\bar{X}}) \otimes_A \hat{A} \cong \mathrm{colim}_l H^i(E, \Omega^{n,\log}_{\bar{X}} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{O}_{\bar{X}}/\mathcal{I}_E^l),$$

where $\hat{A}$ is the completion of A with respect to the ideal I_E cutting out E. Since $A_{\mathfrak{p}} \to \hat{A}_{\mathfrak{p}} \cong \hat{A} \otimes_A A_{\mathfrak{p}}$ is faithfully flat [Stacks, TAG 00MO], $H^i(\bar{X}, \Omega^{n,\log}_{\bar{X}}) \otimes_A A_{\mathfrak{p}}$ is trivial if the same holds for $H^i(\bar{X}, \Omega^{n,\log}_{\bar{X}}) \otimes_A \hat{A}_{\mathfrak{p}} = H^i(\bar{X}, \Omega^{n,\log}_{\bar{X}}) \otimes_A \hat{A} \otimes_A A_{\mathfrak{p}}$. It then suffices to check that $H^i(E, \Omega^{n,\log}_{\bar{X}} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{O}_{\bar{X}}/\mathcal{I}_E^l)$ is trivial. We prove this by induction on l.

Computations:
Case $l = 1$: Let $\iota_E : E \to \bar{X}$ be the natural inclusion. We investigate

$$\Omega^{n,\log}_{\bar{X}} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{O}_E = \iota_{E,*} \Omega^{n,\log}_{\bar{X}}.$$

The blowup centre Z is smooth, hence E is irreducible (we already showed that $E \cong \mathbb{P}^{c-1}_Z$)). Therefore E is defined by cutting out an ideal in $\mathcal{O}_{\bar{X}}$ generated by the coherent monoidal ideal $\mathcal{I}_E \cap \mathcal{M}_{\bar{X}}$ in $\mathcal{M}_{\bar{X}}$ which is locally the logification of the monoid $\iota_{E,*}\mathbb{N}$. By Lemma 2.2.7

$$\Omega^{\log}_{\bar{X}/S} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{O}_E \cong \Omega^{\log}_{E/S},$$

where we endow E with the log structure inherited from $\bar{X}$. By Corollary 2.3.5 we have a short exact sequence

$$0 \to \Omega^n_{E/S}(\log \tilde{D}) \to \Omega^{n,\log}_{E/S} \to \Omega^{n-1}_{E/S}(\log \tilde{D}) \to 0. \tag{2.11}$$

Hence, it suffices to check that the higher cohomology groups of $\Omega^n_{E/S}(\log \tilde{D})$ are trivial for each $n \geq 0$. Since Z is smooth, locally on X, $\Omega_{Z/S}$ is free [Stacks, TAG 02G1]. After localization of X, the fundamental exact sequence takes the form

$$0 \to \Omega_{Z/S} \otimes \mathcal{O}_E \cong \mathcal{O}_E^{\oplus m} \to \Omega_{E/S}(\log \tilde{D}) \to \Omega_{E/Z}(\log \tilde{D}) \to 0.$$

where $m = \dim Z$. Note that the map on the LHS is injective since $E \to Z$ is log smooth. We obtain by Lemma 2.4.7 below a filtration

$$0 = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_n \subseteq F_{n+1} := \Omega^n_{E/S}(\log \tilde{D})$$

such that

$$F_{i+1}/F_i \cong \bigwedge^{n-i} \mathcal{O}_E^{\oplus m} \otimes \Omega^i_{E/Z}(\log \bar{D}).$$

The module F_{i+1}/F_i is then a direct sum of copies of $\Omega^i_{E/Z}(\log \bar{D})$, whose higher cohomology vanishes for every $i \geq 1$ by Lemma 2.4.4. For $i = 0$, $\Omega^0_{E/Z}(\log \tilde{D}) = \mathcal{O}_E$. Its cohomology groups vanish by

Case $l > 1$: We prove this part using induction on l. We have the short exact sequence

$$0 \to \Omega^{n,\log}_{\bar{X}/S} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{I}^l_E/\mathcal{I}^{l+1}_E \to \Omega^{n,\log}_{\bar{X}/S} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{O}_{\bar{X}}/\mathcal{I}^{l+1}_E \to \Omega^{n,\log}_{\bar{X}/S} \otimes_{\mathcal{O}_{\bar{X}}} \mathcal{O}_{\bar{X}}/\mathcal{I}^{l}_E \to 0.$$

Note that $\mathcal{I}^l_E/\mathcal{I}^{l+1}_E \cong \mathcal{O}_E(l)$. Therefore we have to prove that

$$\Omega^{n,\log}_{E/S} \otimes_{\mathcal{O}_E} \mathcal{O}_E(l) =: \Omega^{n,\log}_{E/S}(l)$$

has no higher cohomology. Tensoring sequence (2.11) with $\mathcal{O}_E(l)$ reduces to proving that $\Omega^i_{E/Z}(\log\tilde{D})(l)$ has no higher cohomology, which is the case by Lemma 2.4.4. □

Remark 2.4.6. I was notified by A. Merici that a proof of Theorem 2.4.5 has been given in [KM23, Theorem 6.1] and is also a special case of Theorem 5.2. of [Sai23].

Technicalities about wedge products

The next lemma is essential for studying the cohomology of wedge products of locally free sheaves. It is often used in the literature, however with no proof that I found.

Lemma 2.4.7. [two step filtration, folklore] Let $(X, \mathcal{O}_X)$ be a locally ringed space. Let

$$0 \to \mathcal{U} \to \mathcal{V} \xrightarrow{g} \mathcal{W} \to 0$$

be a short exact sequence of locally free sheaves. Then there is for every $k \in \mathbb{N}$ a filtration

$$0 = \mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_k \subseteq \mathcal{F}_k \subseteq \mathcal{F}_{k+1} = \bigwedge^k \mathcal{V}$$

such that

$$\mathcal{F}_{i+1}/\mathcal{F}_i \cong \bigwedge^{k-i} \mathcal{U} \otimes_{\mathcal{O}_X} \bigwedge^{i} \mathcal{W}.$$

Proof. Define $\mathcal{F}_i$ as the subsheaf of $\bigwedge^k \mathcal{V}$ generated locally by

$$u_1 \wedge \cdots \wedge u_{k+1-i} \wedge v_1 \wedge \cdots \wedge v_{i-1}$$

where each u_j is a section of $\mathcal{U}$ and each v_j is a section of $\mathcal{V}$. The isomorphism

$$\mathcal{F}_{i+1}/\mathcal{F}_i \xrightarrow{\phi} \bigwedge^{k-i} \mathcal{U} \otimes_{\mathcal{O}_X} \bigwedge^{i} \mathcal{W},$$

is defined by

$$u_1 \wedge \cdots \wedge u_{k-i} \wedge v_1 \wedge \cdots \wedge v_i \mapsto u_1 \wedge \cdots \wedge u_{k-i} \otimes g(v_1) \wedge \cdots \wedge g(v_i).$$

We now prove that this is an isomorphism. Arguing locally , we need to prove that the claim holds when $\mathcal{U}$, $\mathcal{V}$ and $\mathcal{W}$ are free modules over a local ring A. By Nakayama's lemma $\mathcal{V}$ has a basis $v_1, \ldots, v_m$ where $m = \operatorname{rank}(\mathcal{V})$, such that $v_1, \ldots v_l$ is a basis of

$\mathcal{U}$ where $l = \operatorname{rank}(\mathcal{U})$. Therefore $\mathcal{F}_{i+1}$ has a basis consisting of $v_{i_1} \wedge \cdots \wedge v_{i_k}$ where $i_1 < \cdots < i_k$ and $i_1, \ldots i_{k-i} \leq l$. The latter has a basis of $\mathcal{F}_i$ of the same form as a subbasis. Hence $\mathcal{F}_{i+1}/\mathcal{F}_i$ has a basis generated by classes of $v_{i_1} \wedge \cdots \wedge v_{i_k}$ such that $i_1 < \cdots < i_{k-i} \leq l < i_{k-i+1} < \cdots < i_k$. On the other hand

$$\bigwedge^{k-i} \mathcal{U} \otimes_A \bigwedge^{i} \mathcal{W}$$

has a basis generated by $v_{i_1} \wedge \cdots \wedge v_{i_{k-i}} \otimes g(v_{i_{k-i+1}}) \wedge \cdots \wedge g(v_{i_k})$ where $i_1 < \cdots < i_{k-i} \leq l < i_{k-i+1} < \cdots < i_k$. Since ϕ sends the basis elements of $\mathcal{F}_{i+1}/\mathcal{F}_i$ to the respective basis elements of $\bigwedge^{k-i} \mathcal{U} \bigotimes_A \bigwedge^{i} \mathcal{W}$, we obtain an isomorphism. $\square$

We get the evident corollaries

Corollary 2.4.8. In the case of the previous corollary, if $\mathcal{U}$ is of rank 1 then

$$0 \to \mathcal{U} \otimes \bigwedge^{k-1} \mathcal{W} \to \bigwedge^{k} \mathcal{V} \to \bigwedge^{k} \mathcal{W} \to 0,$$

is exact. If $\mathcal{W}$ is of rank 1 then

$$0 \to \bigwedge^{k} \mathcal{U} \to \bigwedge^{k} \mathcal{V} \to \bigwedge^{k-1} \mathcal{U} \otimes \mathcal{W} \to 0$$

is exact.

3. Discretely ringed adic spaces

3.1. Generalities about adic spaces

Discretely ringed adic spaces

We will use adic spaces as defined in [Hub13, §1.1.]. An excellent overview on the subject may also be found in Wedhorn's book [Wed19]. We are only concerned with adic spaces which are locally of the form $\mathrm{Spa}(A, A^+)$, where A has the discrete topology. We call such adic spaces *discretely ringed*.

Definition 3.1.1. Let $X \to S$ be a morphism of schemes. We define the set $\mathrm{Spa}(X, S)$ as the set of triples $P = (x, v, \epsilon)$ where x is a point of X, v an equivalence class of valuations on $k(x)$, and $\epsilon : \mathrm{Spec}(\mathcal{O}_v) \to S$ is a morphism fitting into the following diagram

$$\begin{tikzcd} \mathrm{Spec}(k(x)) \arrow[r, "x"] \arrow[d] & X \arrow[d] \\ \mathrm{Spec}(\mathcal{O}_v) \arrow[r, "\epsilon"] & S. \end{tikzcd}$$

For a given discretely ringed Huber pair (A, A^+)

$$\mathrm{Spa}(\mathrm{Spec}(A), \mathrm{Spec}(A+)) \cong \mathrm{Spa}(A, A^+),$$

by [Tem11, Lemma 3.1.2].

We endow $\mathrm{Spa}(X, S)$ with the topology induced by open subsets of the form $\mathrm{Spa}(A, A+)$ fitting into the diagram

$$\begin{tikzcd} \mathrm{Spec}(A) \arrow[r, hook] \arrow[d] & X \arrow[d] \\ \mathrm{Spec}(A^+) \arrow[r] & S \end{tikzcd}$$

where $\mathrm{Spec}(A) \hookrightarrow X$ is an open immersion and $\mathrm{Spec}(A^+) \to S$ is of finite type and (A, A^+) is a Huber pair. This also endows $\mathrm{Spa}(X, S)$ with the structure of an adic space. To elaborate, $\mathrm{Spa}(X, S)$ is the adic space $\mathcal{X}$ associated to the triple

$$(\mathrm{Spa}(X, S), \mathcal{O}_{\mathcal{X}}, (v)_{(x,v,\epsilon)})$$

where $\mathcal{O}_{\mathcal{X}}(\mathrm{Spa}(A, A^+)) := A$. The assignment

$$\mathrm{Spa} : \{\text{morphism of schemes}\} \to \{\text{Discretely ringed adic spaces}\}$$

is functorial [Hüb21, Lemma 2.1].

Tame and strongly étale topology

We assign to an adic space $\mathcal{X}$ the tame étale site $\mathcal{X}_t$ and the strongly étale site $\mathcal{X}_{\text{sét}}$ as defined in [Hüb21]. We recall their definitions.

Definition 3.1.2. Let L/K be a finite separable extension of valued fields, whose valuation rings are $\mathcal{O}_L$ and $\mathcal{O}_K$. Let p be the residue characteristic of L and K. We say that L/K is *unramified* if the extension $\mathcal{O}_L$ is the localization of an étale $\mathcal{O}_K$-algebra. It is said to be *ramified* if it is not unramified.

The extension L/K is called *tamely ramified* if $L^{\text{sh}}/K^{\text{sh}}$ is a prime-to-p extension, where L^{sh} and K^{sh} are the strict henselizations of L and K respectively.

An étale morphism $f : \mathcal{Y} \to \mathcal{X}$ of adic space is *tame* (resp. *strongly*) étale if for every $x \in \mathcal{Y}$, the extension of valued fields $k(x)/k(f(x))$ is tamely ramified (resp. unramified).

The tame étale and strongly étale coverings are étale coverings $\{\mathcal{X}_i \to \mathcal{X}\}$ whose morphisms are tame or strongly étale, respectively. These coverings induce the tame étale and strongly étale sites, denoted by $\mathcal{X}_{\text{t}}$ and $\mathcal{X}_{\text{sét}}$, respectively.

We then have the chain of morphisms of sites

$$\mathcal{X}_{\text{ét}} \xrightarrow{\phi} \mathcal{X}_{\text{t}} \xrightarrow{\psi} \mathcal{X}_{\text{sét}}.$$

Assume that $\mathcal{X}$ is defined over $\mathbb{F}_p$. Given a p-torsion sheaf $\mathcal{F}$ on the tame site, we have

$$H^i(\mathcal{X}_{\text{t}}, \mathcal{F}) \cong H^i(\mathcal{X}_{\text{sét}}, \psi_*\mathcal{F}), \quad \forall i \geq 0,$$

by [Hüb21, Proposition 8.5]. Likewise, for a torsion sheaf $\mathcal{G}$ on the étale site with torsion prime to p, we have

$$H^i(\mathcal{X}_{\text{ét}}, \mathcal{G}) \cong H^i(\mathcal{X}_{\text{t}}, \phi_*\mathcal{G}), \quad \forall i \geq 0,$$

by [Hüb21, Proposition 8.2].

Given a closed adic subspace $i : \mathcal{Z} \to \mathcal{X}$ we denote by $j : \mathcal{U} = \mathcal{X} \backslash \mathcal{Z} \to \mathcal{X}$ the natural open immersion. For $\tau \in \{\text{sét}, \text{t}\}$, define

$$\begin{aligned} i^! : \quad \{\text{abelian sheaves over } \mathcal{X}_\tau\} &\to \{\text{abelian sheaves on } \mathcal{X}_\tau \text{ with support on } \mathcal{Z}\} \\ \mathcal{F} &\mapsto \ker(\mathcal{F} \to j_* j^* \mathcal{F}). \end{aligned}$$

Furthermore one defines the cohomology with support

$$H^n_{\mathcal{Z},\tau}(\mathcal{X}, -) = R^n(\Gamma(\mathcal{X}, i^!(-)).$$

This fits into the long exact sequence of excision:

$$\cdots \to H^{n-1}_\tau(\mathcal{U}, \mathcal{F}) \to H^n_{\mathcal{Z},\tau}(\mathcal{X}, \mathcal{F}) \to H^n_\tau(\mathcal{X}, \mathcal{F}) \to H^n_\tau(\mathcal{U}, \mathcal{F}) \to \cdots \tag{3.1}$$

Remark 3.1.3. Using Propositions 8.2 and 8.5 of [Hüb21] and the five lemma, one obtains analogous isomorphisms for cohomology with support

$$H^i_{\mathcal{Z}}(\mathcal{X}_{\text{t}}, \mathcal{F}) \cong H^i_{\mathcal{Z}}(\mathcal{X}_{\text{sét}}, \psi_*\mathcal{F}), \quad \forall i \geq 0$$

when $\mathcal{F}$ is a p-torsion sheaf on $\mathcal{X}_{\text{t}}$. Similarly, for torsion sheaves $\mathcal{G}$ on $\mathcal{X}_{\text{ét}}$ with torsion prime to p, one has

$$H^i_{\mathcal{Z}}(\mathcal{X}_{\text{ét}}, \mathcal{G}) \cong H^i_{\mathcal{Z}}(\mathcal{X}_{\text{t}}, \phi_*\mathcal{G}), \quad \forall i \geq 0.$$

3.2. Comparison with Riemann-Zariski spaces

Horizontal specializations

Let $\mathcal{X}$ be a discretely ringed adic space. We say that a point $y \in \mathcal{X}$ specializes to $x \in \mathcal{X}$ if every open neighborhood of x contains y. We recall the definition of horizontal (or primary) specializations.

Definition 3.2.1 (horizontal specialization)**.** Let $\mathcal{X} = \mathrm{Spa}(A, A^+)$ be a discretely ringed affinoid adic space. A point of $\mathcal{X}$ is a valuation $v : A \to \Gamma_v \cup \{0\}$ satisfying $v(a) \leq 1$ for all $a \in A^+$. Consider a convex subgroup $H \subset \Gamma_v$ which bounds A, i.e. for all $a \in A$, there exists $h \in H$ s.t. $v(a) \leq h$. One defines

$$v|H : A \to H \cup \{0\} : a \mapsto \begin{cases} v(a) & \text{if } v(a) \in H, \\ 0 & \text{otherwise.} \end{cases}$$

This is a specialization of v. Such specializations are called *horizontal* (or primary).

A specialization $y \succ x$ in a discretely ringed adic space $\mathcal{Y}$ is horizontal if there exists an affinoid open set containing x for which $y \succ x$ is horizontal.

A point $x \in \mathcal{Y}$ is called *Riemann-Zariski* if it has no nontrivial horizontal specialization.

The set of Riemann-Zariski points is called the Riemann-Zariski space. For a morphism $X \to S$, we define the topological space

$$\mathrm{RZ}(X, S) = \{\text{Riemann-Zariski points of } \mathrm{Spa}(X, S)\},$$

together with the underlying topology of $\mathrm{Spa}(X, S)$

In [Tem11], Temkin defines the Riemann-Zariski space as the topological space

$$\mathrm{RZ}(X, S) = \lim_{X \to \bar{Y} \to S} \bar{Y}, \tag{3.2}$$

where $\bar{Y}$ ranges over all X-*modifications* $\bar{Y}$ of S: these are factorizations $X \xrightarrow{f} \bar{Y} \xrightarrow{g} S$ of $X \to S$, such that f is schematically dominant and g is proper. We often write $\bar{Y}$ instead of $X \xrightarrow{f} \bar{Y} \xrightarrow{g} S$. He then shows that the two descriptions agree if $X \to S$ is separated [Tem11, Corollary 3.4.7 and Theorem 3.5.1].

Definition 3.2.2. A morphism $\mathcal{X} \to \mathcal{Y}$ of discretely ringed adic spaces is said to be Riemann-Zariski if it sends Riemann-Zariski points to Riemann-Zariski points.

Let $\mathcal{X}$ be a discretely ringed adic space. We define the Riemann-Zariski open site

$$\mathcal{X}_{\mathrm{RZ}}$$

for which coverings are open coverings which consist of Riemann-Zariski open immersions.

Lemma 3.2.3. There is an equivalence between Riemann-Zariski open subsets in $\mathrm{Spa}(X,S)$ and open subsets in $\mathrm{RZ}(X,S)$, defined by

$$\begin{aligned} \{\text{Riemann-Zariski open immersions to } \mathrm{Spa}(X,S)\} &\to \mathrm{Op}\big(\mathrm{RZ}(X,S)\big) \\ \mathcal{U} &\mapsto \mathcal{U}_{\mathrm{RZ}} = \{\text{RZ-points of } \mathcal{X} \text{ which are contained in } \mathcal{U}\}. \end{aligned} \tag{3.3}$$

Proof. **Injectivity:** Let $\mathcal{U}$ be an RZ-open in $\mathrm{Spa}(X,S)$. Then

$$\mathcal{U} = \{x \in \mathrm{Spa}(X,S), \quad x \succ y \quad \text{for some RZ-point } y \text{ of } \mathcal{X} \text{ s.t.} \quad y \in \mathcal{U}\}.$$

Indeed, if $x \in \mathrm{Spa}(X,S)$ lies in the RHS, then $x \in \mathcal{U}$. If x on the other hand is in $\mathcal{U}$, then there exists a unique RZ-point x_{RZ} in $\mathcal{U}$ which is a horizontal specialization of x.

Hence $\mathcal{U}$ is uniquely defined by its Riemann-Zariski points.

Surjectivity: We described $\mathrm{RZ}(X,S)$ as a subspace of $\mathrm{Spa}(X,S)$, which inherits its topology. Hence every open of $RZ(X,S)$ is of the form $\mathcal{U} \cap \mathrm{RZ}(X,S)$. □

Definition 3.2.4. By Lemma 3.2.3, the functor (3.3) defines an isomorphism of topoi that we denote by

$$\gamma : \mathrm{RZ}(X,S) \to \mathrm{Spa}(X,S)_{\mathrm{RZ}}.$$

Comparison between the open and Riemann-Zariski open sites

Temkin [Tem11] defines an isomorphism between the spaces $\mathrm{RZ}(X,S)$ of Riemann-Zariski points in $\mathrm{Spa}(X,S)$, which he denotes by $\mathrm{Val}_S(X)$, and the space $\lim_{X\to\bar{Y}\to S}\bar{Y}$, where $\bar{Y}$ ranges over all X-*modifications* of S, denoted by $\mathrm{RZ}_S(X)$. We prove a similar lemma, inspired by [Hüb25].

Lemma 3.2.5. Let $X \to S$ be a separated morphism of qcqs schemes. We have a morphism of sites

$$\Psi : \mathrm{Spa}(X,S)_{\mathrm{RZ}} \to \lim_{\bar{Y}} \bar{Y}, \tag{3.4}$$

where $\bar{Y}$ ranges over all modifications $X \to \bar{Y} \to S$.

Proof. For each X-modification $\bar{Y}$ of S, one defines the morphism of sites

$$\Psi_{\bar{Y}} : \mathrm{Spa}(X,S)_{\mathrm{RZ}} \to \bar{Y}, \tag{3.5}$$

associated to the functor

$$\begin{aligned} \mathrm{Op}(\bar{Y}) \to &\{\text{Riemann-Zariski open immersions to } \mathrm{Spa}(X,S)\} \\ \bar{U} \subset \bar{Y} \mapsto &\mathrm{Spa}(\bar{U} \times_S X, \bar{U}) \subset \mathrm{Spa}(X,S). \end{aligned} \tag{3.6}$$

The morphisms $\Psi_{\bar{Y}}$ are compatible for all modifications $\bar{Y}$. Hence they induce the morphism Ψ. □

Lemma 3.2.6. Assume that $X \to S$ is a separated morphism of qcqs schemes. The morphism (3.4)

$$\Psi : \mathrm{Spa}(X, S)_{\mathrm{RZ}} \to \lim_{\bar{Y}} \bar{Y},$$

is a homeomorphism.

Proof. Using [Tem11, Theorem 1.1.3.], we may assume that $X \to S$ is affine, and therefore by [Tem11, Corollary 3.4.7], $\mathrm{RZ}(X, S) \to \lim_{\bar{Y} \to S} \bar{Y}$ is a homeomorphism.

We can then factor Ψ as a chain of isomorphisms

$$\mathrm{Spa}(X, S)_{\mathrm{RZ}} \to \mathrm{RZ}(X, S) \to \lim_{\bar{Y}} \bar{Y}. \qquad \square$$

Lastly, we need to check that the site $\mathrm{Spa}(X, S)_{\mathrm{RZ}}$ can be used to compute sheaf cohomology groups of $\mathrm{Spa}(X, S)$.

Definition 3.2.7. The inclusion

$$\text{RZ-open subsets in } \mathrm{Spa}(X, S) \to \mathrm{Op}(\mathrm{Spa}(X, S))$$

induces a morphism of sites

$$\tilde{\pi} : \mathrm{Spa}(X, S) \to \mathrm{Spa}(X, S)_{\mathrm{RZ}}$$

For a sheaf $\mathcal{F}$ in $\mathrm{Spa}(X, S)$ write $\mathcal{F}_{\mathrm{RZ}}$ for $\tilde{\pi}_* \mathcal{F}$.

Notation 3.2.8. Let $\bar{Y}$ be an X-modification of S. Let $\mathcal{F}$ be a sheaf on the topological space $\mathrm{Spa}(X, S) = \mathrm{Spa}(X, \bar{Y})$. We define the notation

$$\mathcal{F}|_{\bar{Y}} := \Psi_{Y,*} \mathcal{F}_{\mathrm{RZ}}.$$

Consider the following morphisms

$$\mathrm{RZ}(X, S) \xrightarrow{\iota} \mathrm{Spa}(X, S) \xrightarrow{\pi} \mathrm{RZ}(X, S).$$

defined by Temkin [Tem11]. The first morphism is the inclusion, which is a topological embedding. The second morphism sends $x \in \mathrm{Spa}(X, S)$ to its RZ-point x_{RZ}. Note that the functor

$$\tilde{\pi} : \mathrm{Spa}(X, S) \to \mathrm{Spa}(X, S)_{\mathrm{RZ}}$$

is the composition of $\mathrm{Spa}(X, S) \xrightarrow{\pi} \mathrm{RZ}(X, S) \xrightarrow{\gamma} \mathrm{Spa}(X, S)_{\mathrm{RZ}}$. The next proposition is featured in greater generality in [Hüb25, Corollary 5.7]

Proposition 3.2.9. We have a natural isomorphism of cohomology groups

$$H^i(\mathrm{Spa}(X, S), \mathcal{F}) \cong H^i(\mathrm{Spa}(X, S)_{\mathrm{RZ}}, \mathcal{F}_{\mathrm{RZ}}).$$

Proof. It suffices to check that $R\tilde{\pi}_*\mathcal{F}$ is concentrated in degree zero. Since $\tilde{\pi} = \gamma \circ \pi$, and γ is an equivalence, it suffices to check that $R\pi_*\mathcal{F}$ is concentrated in degree zero. To prove that we check that π_* is equal to ι^*, which is exact.

The open subsets of $\mathrm{RZ}(X, S)$ are of the form $\mathcal{U}_{\mathrm{RZ}}$ for some RZ-open subset of $\mathrm{Spa}(X, S)$. By definition $\iota^*\mathcal{F}$ is the sheaffification of

$$\mathcal{U}_{\mathrm{RZ}} \mapsto \mathrm{colim}_{\mathcal{V}}\mathcal{F}(\mathcal{V}),$$

where the colimit is taken over $\mathcal{V}$ open in $\mathrm{Spa}(X, S)$, such that $\mathcal{U}_{\mathrm{RZ}} \to \mathrm{RZ}(X, S)$ factors through $\mathcal{V} \cap \mathrm{RZ}(X, S)$. The RZ-open $\mathcal{U}$ is initial for such open morphisms, hence

$$\mathrm{colim}_{\mathcal{V}}\mathcal{F}(\mathcal{V}) = \mathcal{F}(\mathcal{U}) = \pi_*\mathcal{F}(\mathcal{U}_{\mathrm{RZ}}).$$

It follows that $\iota^*\mathcal{F} = \pi_*\mathcal{F}$, which concludes the proof. □

Combining Lemma 3.2.6 and Proposition 3.2.9 we obtain the corollary.

Corollary 3.2.10. Let $X \to S$ be a separated morphism of qcqs schemes. We have a natural isomorphism

$$H^i(\mathrm{Spa}(X, S), \mathcal{F}) \cong \mathrm{colim}_{\bar{Y}} H^i(\bar{Y}, \mathcal{F}|_{\bar{Y}})$$

where the colimit is taken over all X-modifications $\bar{Y}$ of S.

Proof. Apply [Stacks, Lemma 09YP] to Lemma 3.2.6. Then apply Proposition 3.2.9. □

3.3. Vertical specialization and local Huber pairs

Another type of specializations in adic spaces is called *vertical specialization*. We will learn later that all specializations may be decomposed into vertical and horizontal specialization.
Fix a Huber pair (A, A^+).

Definition 3.3.1. Let $v \in \mathrm{Spa}(A, A^+)$ be a valuation

$$v : A \to \Gamma_v.$$

Let $H \subset \Gamma_v$ be a convex subgroup. Then

$$v/H : A \to \Gamma_v/H \cup \{0\}, a \mapsto \begin{cases} |a|_v \mod H & \text{if } |a|_v \neq 0 \\ 0 & \text{if } |a|_v = 0. \end{cases}$$

A specialization of the form $v/H \geq v$ is called *vertical* (or less commonly *secondary*).

Let $\mathcal{X}$ be an adic space. One says that that $w \succ v$ is a *vertical* if there exists an open $\mathrm{Spa}(A, A^+) \subseteq$ containing v (and therefore w) such that $w \succ v$ is vertical in $\mathrm{Spa}(A, A^+)$.

Alternatively we may describe vertical specialization as follows

Lemma 3.3.2. [HS21, Lemma 10.3] Let v and w be two points of $\mathrm{Spa}(A, A^+)$ associated to the valuation rings $k^+(v)$ and $k^+(w)$. Then w is a vertical specilization of v if and only if

- The supports of v and w are equal, i.e.

$$\mathrm{supp}(v) := \{a \in A; |a|_v = 0\} = \{a \in A; |a|_w\} =: \mathrm{supp}(w).$$

  We write $k(v) := \mathrm{Frac}(A/\mathrm{supp}(v)) = k(w) := \mathrm{Frac}(A/\mathrm{supp}(w))$.

- We have the inclusion $k^+(w) \subseteq k^+(v)$ of subrings of $k(v) = k(w)$. In this case $k^+(v)$ is the localization of $k^+(w)$ at some prime ideal [HS21, Proposition 10.5].

Remark 3.3.3. Note that the definitions of horizontal and vertical specialization in $\mathrm{Spa}(A, A^+)$ only depend on the ring A.

We may furthermore decompose every valuation into a horizontal and vertical component as follows.

Lemma 3.3.4. Let (A, A^+) be a Huber pair. Let $v \succ w$ be a specialization in $\mathrm{Spa}(A, A^+)$, then

- w is a horizontal specialization of a vertical specialization of v [Wed19, Proposition 4.21].

- w is a vertical specialization of a *generalized horizontal specialization*: these are a combination of horizontal specilizations, and speclization of trivial valuations

$$tr_{\mathfrak{p}} \succ tr_{\mathfrak{q}}; \quad \text{for} \quad \mathfrak{p} \succ \mathfrak{q}.$$

Let us now define local Huber pairs: Let $\mathcal{X}$ be an adic space and $x \in \mathcal{X}$. The *localization* of $\mathcal{X}$ is the adic space

$$\mathcal{X}_x := \lim_{x \in \mathcal{U} \subseteq \mathcal{X}} \mathcal{U}.$$

If $\mathcal{X}$ is a discretely ringed affinoid $\mathrm{Spa}(A, A^+)$, then

$$\mathrm{Spa}(A, A^+)_x = \mathrm{Spa}(B, B^+),$$

where $B = A_{\mathrm{supp}(x)}$ and B^+ is the preimage of $k^+(x) \subset k(x) = A_{\mathrm{supp}(x)}/\mathfrak{m}_{\mathrm{supp}(x)}$ in $A_{\mathrm{supp}(x)} = B$. In particular, $\mathcal{X}$ is an affinoid adic space.

More generally, a Huber pair (A, A^+) is called *local* if A is local with maximal ideal $\mathfrak{m}$ and A^+ is the preimage of a valuation ring $\mathcal{O} \subseteq A/\mathfrak{m}$ in A. In particular, A^+ is a local ring with a valuation $|\cdot|$ and a maximal ideal

$$\mathfrak{m}^+ = \{a \in A|, |a| < 1\}.$$

Note also that $A^+ = \{a \in A|, |a| \leq 1\}$ and $\mathfrak{m} = \{a \in A, |a| = 0\}$.

The localization of an adic space is local. For more details on local Huber pairs, see [HS21, Section 10].

We now consider a discretely ringed adic spaces $\mathcal{X} = \mathrm{Spa}(X, S)$ such that both X and S are finite-dimensional, qcqs and $X \to S$ is separated. Let $x \in \mathcal{X}$ be a point and let (A, A^+) the local Huber pair associated to x. Replace furthermore S by a compactification $\bar{X}$ of X

As $A^+/\mathfrak{m}$ is isomorphic to the valuation ring $k^+(x)$, the prime ideals of A^+ containing $\mathfrak{m}$ are totally ordered. By Lemma 3.3.2, there is a bijection between vertical generalizations of x and prime ideals of A^+ containing $\mathfrak{m}$.

Proposition 3.3.5. Up to replacing $\bar{X}$ with an X-admissible blowup, the natural morphism

$$\mathrm{Spec}(A^+) \to \bar{X}$$

is injective.

Proof. Since $A^+_{\mathfrak{m}} = A$ is local, Prime ideals of $\mathrm{Spec}(A^+)$ are either contained in $\mathrm{Spec}(A)$ or contain $\mathfrak{m}$. The ideals contained in $\mathrm{Spec}(A)$ map injectively into $\mathrm{Spec}(X) \subseteq \mathrm{Spec}(\bar{X})$. It is therefore enough to check injectivity on prime ideals of $\mathrm{Spec}(A^+)$ containing $\mathfrak{m}$. Such ideals correspond to prime ideals in $\mathrm{Spec}(k^+(x))$. It then suffices to prove the injectivity of the morphism

$$\mathrm{Spec}(k^+(x)) \to \bar{X},$$

after replacing $\bar{X}$ with an X-admissible blowup.

Let $\mathfrak{p}$ and $\mathfrak{p}'$ be different prime ideals of $\mathrm{Spec}(k^+(x))$. Let y and z be the points of $\mathrm{Spa}(A, A^+)$ associated to the valuation rings $k^+(x)_{\mathfrak{p}}$ and $k^+(x)_{\mathfrak{p}'}$ respctively. The images of $\mathfrak{p}$ and $\mathfrak{p}'$ in $\bar{X}$ are therefore the images of the maximal ideals of $k^+(y)$ and $k^+(z)$ in $\bar{X}$, respectively.

We claim that $y_{\mathrm{RZ}} \neq z_{\mathrm{RZ}}$. Indeed, the Riemann-Zariski points associated to y, resp. z, have the valuation ring of the form $k^+(y)/\mathfrak{q}$, resp. $k^+(z)/\mathfrak{q}'$, for prime ideals $\mathfrak{q}, \mathfrak{q}' \in \mathrm{Spec}(A)$ [HS21, Lemma 10.3]. The maximal ideal of $k^+(y_{\mathrm{RZ}})$, resp. $k^+(z_{\mathrm{RZ}})$ are mapped to $\mathfrak{p}$, resp. $\mathfrak{p}'$, in $\mathrm{Spec}(A^+)$.

By Lemma 3.2.6, the space of Riemann-Zariski points is homeomorphic to the topological space $\lim\limits_{X \subseteq \bar{Y} \to \bar{X}} \bar{Y}$, where the limit is taken over all X-admissible blowups of $\bar{X}$. It associates y_{RZ} and z_{RZ} to the image of $\mathfrak{p}$ and $\mathfrak{p}'$ in X-admissible blow-ups of $\bar{X}$. Since $y_{\mathrm{RZ}} \neq z_{\mathrm{RZ}}$, there exists an X-admissible blowup of $\bar{X}$ which separates $\mathfrak{p}$ and $\mathfrak{p}'$. $\square$

Remark 3.3.6. Proposition 3.3.5 implies that the ring A^+ is finite-dimensional. Note that A^+ is generally not Noetherian, except in the cases where $A^+/\mathfrak{m}$ is a DVR or $A = A^+$.

3.4. Unramified sheaves

In this section we will define unramified sheaves. These will help relate between certain sheaves which are defined on logarithmic schemes and sheaves on discretely ringed adic spaces. For more details see section 6 in [Hüb24].

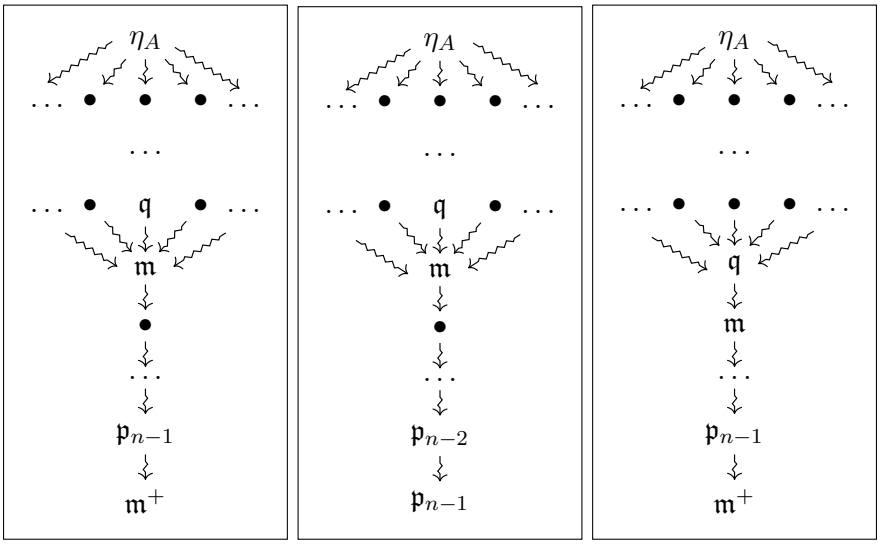

Figure 3.1.: SpecA^+, a vertical generalization $\mathrm{Spec}(A^+_{\mathfrak{p}_{n-1}})$ and a horizontal generalization

The log site

We fix, as usual, a perfect field k of positive characteristic p.

Consider a morphism of schemes $X \to S$ over k, where X is smooth over k. A *log smooth presentation* of $\mathcal{X} = \mathrm{Spa}(X, S)$ is a pair $(X, \bar{X})$ with X open in $\bar{X}$, such that $\mathcal{X} = \mathrm{Spa}(X, \bar{X})$ and $\bar{X}$ is log smooth over k when endowed with the compactifying log structure associated to $X \subset \bar{X}$. In particular, a discretely ringed adic space with a log smooth presentation is also smooth over $\mathrm{Spa}(k, k)$.

We define the site $(X, S)_{\log}$, whose objects are log smooth schemes $\bar{Y}$ such that we have the following commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ \bar{Y} & \longrightarrow & S \end{array}$$

where $Y \to \bar{Y}$ is an open immersion, $\bar{Y}$ carries the compactifying log structure associated to Y and $\mathrm{Spa}(Y, \bar{Y}) \to \mathrm{Spa}(X, S)$ defines an open immersion. In particular $Y \to X$ is an open immersion of schemes and $\bar{Y} \to S$ is the normalization in Y of a scheme of finite type over S.

Morphisms in $(X, S)_{\log}$ are morphisms of log schemes. Coverings in $(X, S)_{\log}$ are defined to be surjective families $(\bar{Y}_i \to \bar{Y})$ of (strict) open immersions.

Example 3.4.1. The first example of interest are logarithmic differentials. Consider the presheaf $\Omega^{n,\log}$ defined by

$$\bar{Y} \mapsto \Omega^{n,\log}_{\bar{Y}}(\bar{Y}).$$

This is a sheaf on $(X, S)_{\log}$ since $\Omega^{n,\log}_{\bar{Y}}$ is a Zariski sheaf on $\bar{Y}$.

Unramified sheaves

Definition 3.4.2. [Hüb24, Definition 6.1] An *isomorphism in codimension one* in $(X, S)_{\log}$ is a morphism $\bar{Y}' \to \bar{Y}$ of log schemes such that $\bar{Y}'_{\mathrm{tr}} \cong \bar{Y}_{\mathrm{tr}} \times_{\bar{Y}} \bar{Y}'$ and there exists an open subscheme $U \subset \bar{Y}$ containing all points of codimension ≤ 1 in $\bar{Y}$ such that $U \times_{\bar{Y}} \bar{Y}'$ is isomorphic to U.

We write $\bar{Y} \sim_1 \bar{Y}'$.

We now can define what unramified sheaves are.

Definition 3.4.3. [Hüb24, Definition 6.2] A sheaf $\mathcal{F}$ on $(X, S)_{\log}$ is *unramified* if for every open immersion $\bar{Y}' \to \bar{Y}$ in $(X, S)_{\log}$ with dense image

$$\mathcal{F}(\bar{Y}) \to \mathcal{F}(\bar{Y}')$$

is injective, and for each isomorphism in condimension one $\bar{Y} \sim_1 \bar{Y}'$

$$\mathcal{F}(\bar{Y}) \to \mathcal{F}(\bar{Y}')$$

is an isomorphism.

Example 3.4.4. [Hüb24, Lemma 6.10]

- The presheaf $\bar{\mathcal{O}} : \bar{Y} \mapsto \mathcal{O}_{\bar{Y}}(\bar{Y})$ is unramified.
- The sheaf of log differentials $\Omega^{n,\log}$ is unramified.
- The sheaf of exact differentials $Z\Omega^{n,\log} = \ker(\Omega^{n,\log} \xrightarrow{d} \Omega^{n+1,\log})$ is unramified.
- The sheaf $B\Omega^{n,\log} = d(\Omega^{n-1,\log})$ is unramified. This will be proven later in Corollary 4.3.5 using the Cartier operator.

More generally, any locally free $\bar{\mathcal{O}}$ module in $(X, S)_{\log}$ is unramified.

Lemma 3.4.5. Consider the exact sequence of sheaves in $(X, S)_{\log}$

$$0 \to \mathcal{F} \to \mathcal{G} \xrightarrow{f} \mathcal{H}.$$

If $\mathcal{G}$ and $\mathcal{H}$ are unramified, then the same is true for $\mathcal{F}$.

Proof.

- **Sheaf condition:** $\mathcal{F}$ is a sheaf on $(X, S)_{\log}$ by definition.
- **Injectivity:** This condition holds because $\mathcal{F}$ is a subsheaf of the unramified sheaf $\mathcal{G}$: Consider an open immersion $\bar{Y}' \to \bar{Y}$ with dense image. We have the diagram

$$\begin{tikzcd} \mathcal{F}(\bar{Y}) \arrow[r] \arrow[d] & \mathcal{F}(\bar{Y}') \arrow[d] \\ \mathcal{G}(\bar{Y}) \arrow[r] & \mathcal{G}(\bar{Y}') \end{tikzcd}$$

  The lower horizontal and left vertical arrow are injections. Hence the upper horizontal arrow is also an injection.

- **Bijection for isomorphisms in codimension one:** Let $\bar{Y} \sim_1 \bar{Y}'$ be an isomorphism in codimension one. We have the following diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{F}(\bar{Y}') \arrow[r] \arrow[d] & \mathcal{G}(\bar{Y}') \arrow[r] \arrow[d, "\cong"] & \mathcal{H}(\bar{Y}') \arrow[d, "\cong"] \\ 0 \arrow[r] & \mathcal{F}(\bar{Y}) \arrow[r] & \mathcal{G}(\bar{Y}) \arrow[r] & \mathcal{H}(\bar{Y}). \end{tikzcd}$$

  Since $\mathcal{G}(\bar{Y}') \to \mathcal{G}(\bar{Y})$ and $\mathcal{F}(\bar{Y}') \to \mathcal{F}(\bar{Y})$ are isomorphisms we may apply the 5-lemma to obtain that $\mathcal{F}(\bar{Y}') \to \mathcal{F}(\bar{Y})$ is also an isomorphism. □

Induced sheaves on adic spaces

An unramified sheaf $\mathcal{F}$ in $(X, S)_{\log}$ induces a presheaf on adic spaces denoted $\mathcal{F}_{\lim}$. Before stating the definition of $\mathcal{F}_{\lim}$ we need the following lemma.

Lemma 3.4.6. [Hüb24, Lemma 6.3] Let $\mathcal{F}$ be an unramified sheaf on $(X, S)_{\log}$. If $\bar{Y}' \to \bar{Y}$ induces an isomorphism $\mathrm{Spa}(Y', \bar{Y}') \to \mathrm{Spa}(Y, \bar{Y})$, then the map $\mathcal{F}(\bar{Y}) \to \mathcal{F}(\bar{Y}')$ is an isomorphism.

This enables us to associate a presheaf on $\mathrm{Spa}(X, S)$ to each unramified sheaf.

Definition 3.4.7. Let $\mathcal{F}$ be an unramified sheaf on $(X, S)_{\log}$. We define the presheaf $\mathcal{F}_{\lim}$ on $\mathrm{Spa}(X, S)$ as

$$\mathcal{F}_{\lim}(\mathcal{U}) = \lim_{\bar{Y} \in (X,S)_{\log}/\mathcal{U}} \mathcal{F}(\bar{Y}),$$

where $(X, S)_{\log}/\mathcal{U}$ denotes all objects $\bar{Y}$ in $(X, S)_{\log}$, such that $\mathrm{Spa}(\bar{Y}_{\mathrm{tr}}, \bar{Y}) \to \mathrm{Spa}(X, S)$ factors through $\mathcal{U}$.

Actually, $\mathcal{F}_{\lim}$ is not only a presheaf.

Proposition 3.4.8. [Hüb24, Proposition 8.9] The presheaf $\mathcal{F}_{\lim}$ is a sheaf on the topological space $\mathrm{Spa}(X, S)$.

Remark 3.4.9. Note that if $\mathcal{U} = \mathrm{Spa}(Y_{\mathrm{tr}}, \bar{Y})$ is asociated to a log smooth log scheme $\bar{Y}$, then $(X, S)_{\log}/\mathcal{U}$ has a final object, namley $\bar{Y}$. Hence $\mathcal{F}_{\lim}(\mathcal{U}) = \mathcal{F}(\bar{Y})$

The operation $(-)_{\lim}$ taking unramified sheaves to sheaves on $\mathrm{Spa}(X, S)$ is functorial and left exact.

We now can focus on our examples of interest

Example 3.4.10. Let $\mathcal{X} = \mathrm{Spa}(X, S)$ be a discretely ringed adic space. Consider the sheaf $\Omega^{n,\log}_{\lim}$. We have a differential

$$d : \Omega^{n,\log}_{\lim} \to \Omega^{n+1,\log}_{\lim},$$

defined by taking the limit on all differentials. By left exactness of the lim functor on unramified sheaves we have an isomorphism

$$(Z\Omega^{n,\log})_{\lim} \cong \ker(\Omega^{n,\log}_{\lim} \xrightarrow{d} \Omega^{n+1,\log}_{\lim})$$

4. Differentials over adic spaces

4.1. Rangers and definition of Ω^+

As usual, let $\mathcal{X} = \mathrm{Spa}(X, S)$ be a discretely ringed adic space. The sheaf Ω^+ was introduced in [Hüb24] and is defined by

$$\Omega^+(\mathcal{U}) := \{\eta \in \Omega(\mathcal{U}) \mid |\eta|_x \leq 1 \text{ for all } x \in \mathcal{U}\}$$

for a suitable definition of the seminorm

$$|\cdot|_x : \Omega_{A_{\mathrm{supp}(x)}} \to \mathcal{R}_{\Gamma_x},$$

known as the *Kähler seminorm* (see below). This construction originates in Temkin's paper [Tem16], where he defines a similar Kähler seminorm on the module of differentials $\Omega_{K/k}$ for an extension of real-valued fields K/k as

$$|\eta|_\Omega := \inf_{\eta = \sum f_i dg_i} \max_i \{|f_i|_K |g_i|_K\}.$$

This definition, however, does not directly extend to local Huber pairs (A, A^+), as the value group Γ_A may lack infima. To address this, Hübner introduced the concept of *rangers* in [Hüb24, §5], which were further applied to the study of adic curves in recent work by Hübner and Temkin [HT24, p. 2.1.5].

Definition 4.1.1. Let Γ be an ordered abelian group (or more generally, an ordered set). A *ranger* is a pair (Γ', γ'), where Γ' is an ordered group (or ordered set) containing Γ, and $\gamma' \in \Gamma'$. Two rangers (Γ_1, γ_1) and (Γ_2, γ_2) are *equivalent* if there exists a ranger (Γ', γ') such that $\Gamma' \subseteq \Gamma_1 \cap \Gamma_2$, and γ' maps to γ_1 in Γ_1 and to γ_2 in Γ_2.

The set of equivalence classes of rangers is denoted $\mathcal{R}_\Gamma$. It is a totally ordered set.

Rangers are used to define the Kähler seminorm on modules of differentials over local Huber pairs.

Definition 4.1.2. [Hüb24, §7] Let (A, A^+) be a local Huber pair over a perfect field k of positive characteristic. The *Kähler seminorm*

$$|\cdot| : \Omega_{A/k} \to \mathcal{R}_\Gamma$$

is defined by

$$|\omega| := \inf_{\omega = \sum f_i dg_i} \max_i \left\{|f_i|_A |g_i|_A\right\},$$

where Γ is value group associated to (A, A^+).

The usual Kähler differentials $\Omega^n_{X/k}$ define a coherent $\mathcal{O}_{\mathcal{X}}$-module

$$\Omega^n_{\mathcal{X}/k} : \mathrm{Spa}(A, A^+) \mapsto \Omega^n_{A/k},$$

Using rangers. we can also define the sheaf $\Omega^{+,n}_{\mathcal{X}/k}$.

Definition 4.1.3. Let $\mathcal{X}$ be a discretely ringed adic space over a field k of positive characteristic. For any open subset $\mathcal{U} \subseteq \mathcal{X}$, define

$$\Omega^+_{\mathcal{X}/k}(\mathcal{U}) := \left\{\eta \in \Omega_{\mathcal{X}/k}(\mathcal{U}) \;\middle|\; |\eta|_v| \leq 1 \text{ for all } v \in \mathcal{U}\right\}.$$

Then set

$$\Omega^{+,n}_{\mathcal{X}/k} := \bigwedge\nolimits^n_{\mathcal{O}^+_{\mathcal{X}}} \Omega^+_{\mathcal{X}/k}.$$

We often omit the subscript and simply write $\Omega^{+,n}$ when the context is clear.

The next property enables us to use $\Omega^{+,n}$ to compute tame cohomology.

Proposition 4.1.4. [Hüb24, Proposition 7.9] The presheaf $\Omega^{+,n}$ is a sheaf in the tame and strongly étale topology.

We also define a valuation on Ω^n_A with values in $\mathcal{R}_\Gamma$, generalizing Definition 4.1.2.

Definition 4.1.5. Let (A, A^+) be a local Huber pair over a field k of positive characteristic. Let Γ be the value group associated to (A, A^+). Let n be an integer. The *Kähler seminorm* on $\Omega^n_{A/k}$ is the map

$$|\cdot| : \Omega^n_A \to \mathcal{R}_\Gamma$$

defined by

$$|\omega| := \inf_{\omega = \sum f_i dg_{1,i} \wedge \cdots \wedge dg_{n,i}} \max_i \left\{|f_i \cdot g_{1,i} \cdots g_{n,i}|\right\}.$$

Proposition 4.1.6. For every integer n, we have the equality

$$\Omega^{+,n}_{(A,A^+)} = \left\{\omega \in \Omega^n_A; |\omega| \leq 1\right\} \tag{4.1}$$

Proof. Let ω be a differential in $\Omega^{+,n}_{(A,A^+)}$. Then ω is a sum of wedge products of elements of the form fdg satisfying $|fg| \leq 1$ [Hüb24, Lemma 7.3.](cf. Lemma 4.2.1). Therefore $\omega = \sum_i f_i dg_{1,i} \wedge \cdots \wedge dg_{n,i}$ such that $|f_i g_{1,i} \ldots g_{n,i}| \leq 1$. Hence ω is in the RHS of (4.1).

Consider $\omega \in \Omega^n_A$ lying in the RHS of (4.1). According to [Hüb24, Lemma 7.2], one may express ω as a sum $\sum_i f_i dg_{1,i} \wedge \cdots \wedge dg_{n,i}$ such that $|f_i g_{1,i} \ldots g_{n,i}| \leq 1$. Thus the proof of Proposition 4.1.6 is reduced to the following claim.

Claim 4.1.7. Let $f, g_1, \ldots, g_n$ be elements in A such that $|fg_1 \ldots g_n| \leq 1$, then $\eta := fdg_1 \wedge \cdots \wedge g_n \in \Omega^{+,n}_{(A,A^+)}$.

We prove Claim 4.1.7 inductively. When $n = 1$ this is true by definition. Assume that the claim is true for $n - 1$.

Case 1: $|g_1| = \cdots = |g_n| = 0$. Then $f dg_1 \wedge \cdots \wedge d_{n-1} \in \Omega^{+,n-1}_{(A,A^+)}$ and $dg_n \in \Omega^+_{(A,A^+)}$. Hence $\eta \in \Omega^{+,n}_{(A,A^+)}$.

Case 2: $|fg_1 \dots g_{n-1}| = 0$ and $|g_n| \neq 0$. Then express η as the wedge product

$$(fg_n dg_1 \wedge \cdots \wedge dg_{n-1}) \wedge (dg_n/g_n).$$

Then $fg_n dg_1 \wedge \cdots \wedge dg_{n-1} \in \Omega^{+,n-1}_{(A,A^+)}$ due to the induction step and $dg_n/g_n \in \Omega^+_{(A,A^+)}$ by definition.

Case 3: We may now assume, without loss of generality, that $|f|, |g_1|, \dots, |g_n| \neq 0$. Then

$$\eta = (fg_1 \dots g_n)(dg_1/g_1) \wedge \cdots \wedge (dg_n/g_n),$$

lies in $\Omega^{+,n}_{(A,A^+)}$. □

4.2. Logarithmic differentials on adic spaces

For every morphism of Huber pairs $(B, B^+) \to (A, A^+)$ we can define the associated logarithmic differential

$$\Omega^{\log}(\mathrm{Spa}(A, A^+)) = \Omega^{\log}_{(A,A^+)/(B,B^+)} = \Omega^{\log}_{(A^+\cap A^\times \to A)/(B^+\cap B^\times \to B)}.$$

In this manuscript we will mostly consider $(B, B^+) = (k, k)$ for a perfect field k of characteristic $p > 0$.

We also define $\Omega^{n,\log} := \bigwedge^n \Omega^{\log}$. This is however not a sheaf, as shown in [Hüb24, Example 4.4.].

Lemma 4.2.1. [Hüb24, Lemma 7.3.] Let k be a perfect field of characteristic p. Let (A, A^+) be a local Huber pair. We have

$$\Omega^{n,\log}_{(A,A^+)} \cong \Omega^{n,+}_{(A,A^+)}.$$

This leads to the proposition:

Proposition 4.2.2. [Hüb24, Proposition 7.11] Let $\mathcal{X}$ be a discretely ringed adic space over a perfect field k of characteristic p. Then $\Omega^{+,n}$ is the sheafification of $\Omega^{n,\log}$.

Remark 4.2.3. Under the same assumptions, we have the following inclusion

$$\Omega^{+,n}_{\mathcal{X}/k} \subseteq \Omega^{n,\log}_{\mathcal{X}/k,\lim} \subseteq \Omega^n_{\mathcal{X}/k},$$

as $\Omega^{n,\log}_{\mathcal{X}/k,\lim}$ is a sheaf that contains $\Omega^{n,\log}_{\mathcal{X}/k}$.

If $\mathcal{X} = \mathrm{Spa}(X, \bar{X})$ is associated to a log smooth pair $(X, \bar{X})$, then

$$\Omega^{+,n}_{\mathcal{X}/k}(\mathrm{Spa}(X, \bar{X})) \to \Omega^{n,\log}_{\mathcal{X}/k,\lim}(\mathrm{Spa}(X, \bar{X}))$$

is an isomorphism. This is [Hüb24, Theorem 8.12].

Conjecture 4.2.4. Given a smooth discretely ringed adic space $\mathcal{X} = \mathrm{Spa}(X, S)$ over k, we conjecture that

$$\Omega^{n,\log}_{\mathcal{X},\lim} \cong \Omega^{n,+}_{\mathcal{X}}$$

for every $n \in \mathbb{N}$.

Proposition 4.2.5. Under the assumption of resolution of singularities (namely Conjecture B.2.1 and Conjecture B.2.2) over k) Conjecture 4.2.4 holds.

Proof. Under the given assumptions, every open in $\mathcal{X}$ is covered by adic spaces of the form $\mathcal{U} = \mathrm{Spa}(U, \bar{U})$, for which the pair $(U, \bar{U})$ is a log smooth presentation of $\mathcal{U}$. Hence over this covering $\Omega^{+,n}$ and $\Omega^{n,\log}_{\lim}$ agree as subsheaves of Ω^n. The result follows. □

4.3. The Cartier Operator I

The Cartier operator for logarithmic differentials

Let k be a perfect field of positive characteristic. We write $S = \mathrm{Spec}(k)$.

Let $\mathcal{X} = \mathrm{Spa}(X, T)$ be a smooth discretely ringed adic space over k. Assume that $\mathcal{X} = \mathrm{Spa}(X, \bar{X})$ where $\bar{X}$ is a log smooth log scheme with the compactifying log structure associated to the open subscheme X. Denote the open immersion $X \to \bar{X}$ by j.

We start by introducing some notation. Consider the absolue Frobenius endomorphism

$$F_{\mathrm{abs}} : X \to X$$

induced by sending a section α in $\mathcal{O}_X$ to α^p. We denote by $X^{(p)}$ the pullback

$$\begin{tikzcd} X^{(p)} \arrow[r, "W"] \arrow[d, "\pi^{(p)}"] & X \arrow[d, "\pi"] \\ S \arrow[r, "F_{\mathrm{abs}}"] & S \end{tikzcd}$$

The morphism W induces an isomorphism on $X^{(p)} \to X$. The structure morphism $\pi^{(p)} : X^{(p)} \to S$ is associated to the morphism $\mathcal{O}_S \to \mathcal{O}_{X^{(p)}} (\cong \mathcal{O}_X) : f \mapsto f^{\frac{1}{p}}$. On the level of topological spaces, the morphism W is just the identity.

The absolute Frobenius $F_{\mathrm{abs}} : X \to X$ induces the relative Frobenius $F : X \to X^{(p)}$.

Now note that $\Omega_{X^{(p)}/S} = F_* \Omega_{X/S}$. Similarly, if $D \subseteq X$ is a normal crossing divisor, then $\Omega_{X^{(p)}/S}(\log D^{(p)}) = F_* \Omega_{X/S}(\log D)$.

Let us first recall a theorem of Cartier, see [Kat70, Theorem 7.2].

Proposition 4.3.1 (Inverse Cartier operator)**.** Assume that X is smooth over the prefect field k.

There exists for each $n \in \mathbb{N}$ an isomorphism

$$\bar{C}^{-1} : \Omega^n_{X^{(p)}/S} \to F_*\big(Z\Omega^n_{X/S} / B\Omega^n_{X/S}\big)$$

of $\mathcal{O}_{X^{(p)}}$-modules, which satisfies

- $\bar{C}^{-1}(1) = 1$ when $n = 0$.
- $\bar{C}^{-1}(df) = [f^{p-1}df]$ when $n = 1$.
- $\bar{C}^{-1}(\omega \wedge \eta) = \bar{C}^{-1}(\omega) \wedge \bar{C}^{-1}(\eta)$.

If on the other hand $\bar{X}$ has the structure of a log smooth log scheme then the same holds for the log differentials:

$$\bar{C}^{-1} : \Omega^{n,\log}_{\bar{X}^{(p)}/S} \to F_*(Z\Omega^{n,\log}_{\bar{X}/S}/B\Omega^{n,\log}_{\bar{X}/S}).$$

This isomorphism is compatible with

$$j_*\bar{C}^{-1} : j_*\Omega^n_{X^{(p)}/S} \to j_*F_*(Z\Omega^n_{X/S}/B\Omega^n_{X/S}),$$

where $j_* : X \to \bar{X}$ is the open immersion associated to the trivializing subscheme X of X.

We apply the fact that $\Omega^{n,\log}_{X^{(p)}/S}$ is isomorphic to $F_*\Omega^{n,\log}_{X/S}$ as an $\mathcal{O}_{X^{(p)}}$-module, and hence as a group, to directly obtain the following proposition.

Proposition 4.3.2. There exists a unique family of additive maps $C : Z\Omega^{n,\log}_{\bar{X}/k} \to \Omega^{n,\log}_{\bar{X}/k}$ with the properties

- $C(1) = 1$ in $\mathcal{O}_{\bar{X}} = \Omega^{0,\log}_{\bar{X}/k}$,
- $C(f^p\omega) = fC(\omega)$ for each $f \in \Omega_X, \omega \in Z\Omega^{r,\log}_{X/k}$,
- $C(\omega \wedge \omega') = C(\omega) \wedge C(\omega')$ for closed ω and ω' in $Z\Omega^{r,\log}_{\bar{X}/k}$,
- $C(\omega) = 0$ if and only if $\omega = d\eta$, where η is a logarithmic differnetial,
- $C(f^{p-1}df) = df$.

Furthermore, this is compatible with the pushforward of the Cartier operator

$$j_*C : j_*Z\Omega_{X/k} \to j_*\Omega_{X/k}.$$

Proof. This follows directly from the previous proposition. The $\mathcal{O}_{X(p)}$-linearity translates to the second condition. The fact that $\bar{C}^{-1}$ is an isomorphism corresponds to the fourth condition. The uniqueness of $\bar{C}^{-1}$ is equivalent to that of C. □

The next proposition is a motivation for the use of logarithmic differentials.

Proposition 4.3.3. [Sat07, Lemma 2.4.6] Let X be a log smooth log scheme over a perfect field k. The sequence

$$0 \to j_*\nu_{X_{\mathrm{tr}}}(r) \to Z\Omega^{r,\log}_X/k \xrightarrow{C-1} \Omega^{r,\log}_{X/k} \to 0$$

is exact in $\mathrm{Sh}(X_{\text{ét}})$. The sheaf $j_*\nu_{X_{\mathrm{tr}}}(r)$ is the pushforward of $\nu_{X_{\mathrm{tr}}}$ via $j : X_{\mathrm{tr}} \to X$.

Cartier operator on adic spaces

In this section we study the properties of the Cartier operator on adic spaces. Consider a discretely ringed adic space $\mathcal{X}$ over a perfect field k. Recall that we have the inclusions

$$\Omega^{+,n}_{\mathcal{X}} \subseteq \Omega^{n,\log}_{\mathcal{X},\lim} \subseteq \Omega^{n}_{\mathcal{X}}.$$

To prove some of the needed properties of the Cartier operator we sometimes need to work with $\Omega^{+}_{\mathcal{X}}$ and sometimes with $\Omega^{n,\log}_{\mathcal{X},\lim}$. The existence of a Cartier operator may first be checked on the site $(X,S)_{\log}$.

Proposition 4.3.4. The Cartier operator defined in Proposition 4.3.2 induces a surjective morphism

$$C : Z\Omega^{n,\log} \to \Omega^{n,\log}$$

of sheaves on $(X,S)_{\log}$. Its kernel is $B\Omega^{n,\log}$.

Proof. This is a direct consequence of Proposition 4.3.2. □

Corollary 4.3.5. The sheaf $B\Omega^{n,\log}$ is unramified.

Proof. This follows from the fact that the $B\Omega^{n,\log}$ is the kernel of the surjective morphism C between unramified sheaves and from Lemma 3.4.5. □

Applying the functor $(\cdot)_{\lim}$ yields the following Proposition.

Proposition 4.3.6. There exists an exact sequence of sheaves on $\mathrm{Spa}(X,S)$

$$0 \to B\Omega^{n,\log}_{\mathcal{X}/k,\lim} \to Z\Omega^{n,\log}_{\mathcal{X}/k,\lim} \xrightarrow{C} \Omega^{n,\log}_{\mathcal{X}/k,\lim}.$$

Corollary 4.3.7. Under the assumption of Conjecture 4.2.4 we have a short exact sequence of sheaves on $\mathcal{X} := \mathrm{Spa}(X,S)$

$$0 \to B\Omega^{+,n}_{\mathcal{X}/k} \to Z\Omega^{+,n}_{\mathcal{X}/k} \xrightarrow{C} \Omega^{+,n}_{\mathcal{X}/k} \to 0.$$

Proof. Since we assumed that $\Omega^{+,n}_{\mathcal{X}/k} \cong \Omega^{n,\log}_{\mathcal{X}/k,\lim}$ by Conjecture 4.2.4, the exactness of the LHS follows from Proposition 4.3.6.

The exactness of the RHS is checked on the level of stalks. Let $x \in \mathcal{X}$ and (A, A^+) be the local Huber pairs associated to x. Let $\eta \in \Omega^{+,n}_{\mathcal{X}/k,x} \cong \Omega^{+,n}_{(A,A^+)/k}$. We can express η as a sum

$$\eta = \sum_i f_i dg_{i,1} \wedge \cdots \wedge dg_{i,n},$$

such that $f_i, g_{i,1}, \ldots, g_{i,n}$ lie in A and $|f_i \cdot g_{i,1} \ldots g_{i,n}|_x \leq 1$. Then

$$\gamma := \sum_i f_i^p (g_{i,1} \cdots g_{i,n})^{p-1} dg_{i,1} \wedge \cdots \wedge dg_{i,n},$$

is a preimage of η under $Z\Omega^n_{A/k} \xrightarrow{C} \Omega^n_{A/k}$. Since

$$|(f_i \cdot g_{i,1} \dots g_{i,n})^p|_x = |f_i \cdot g_{i,1} \dots g_{i,n}|_x^p \leq 1,$$

we deduce that γ lies in $Z\Omega^{+,n}_{(A,A^+)/k} \cong \Omega^{+,n}_{\mathcal{X}/k,x}$.

In conclusion, the Cartier operator $Z\Omega^{+,n}_{\mathcal{X}/k} \xrightarrow{C} \Omega^{+,n}_{\mathcal{X}/k}$ is surjective. □

Definition 4.3.8. The log de Rham-Witt sheaf $\nu_{\mathcal{X}}(n)$ is defined as the étale additive subsheaf of $\Omega^n_{\mathcal{X}}$ whose sections are étale locally generated by wedge products over $\mathbb{F}_p$ of the form

$$df/f, \quad f \in \mathcal{O}^\times_{\mathcal{X}}.$$

Alternatively, $\nu_{\mathcal{X}}(n)$ is the kernel of $C - 1 : Z\Omega^n_{\mathcal{X}/k} \to \Omega^n_{\mathcal{X}/k}$.

The latter definition is stated just after Remark 1.2. in [Mil76]. The fact that it is equivalent to the first definition is stated in [Mil86, Lemma 1.6].

In [Mil86], one also defines the sheaves of $\mathbb{Z}/p^m\mathbb{Z}$-modules $\nu_m(n)$ on a given scheme X over a perfect field k of characteristic p. These are the subsheaves of the de Rham Witt sheaves $W_m\Omega^n_{X/k}$ generated by $\mathrm{dlog}\underline{f_1} \wedge \dots \wedge \mathrm{dlog}\underline{f_n}$ (see the paragraph after Lemma 1.6 in [Mil86] or Appendix A).

Definition 4.3.9. Over a discretely ringed adic space $\mathcal{X}$ over k we define the sheaves

$$\nu_{m,\mathcal{X}}(n) : \mathrm{Op}(\mathrm{Spa}(X,S)) \to \mathbb{Z}/p^m\mathbb{Z} - \text{modules} : \mathrm{Spa}(U,T) \mapsto \nu_{m,\mathcal{X}}(n)(U).$$

These are étale sheaves and hence by restriction sheaves in the tame topology. We drop the subscript $\mathcal{X}$ when the context is clear. We may also drop the index m if $m = 1$.

Remark 4.3.10. The sheaves $\nu_r(n)$ are Zariski-locally generated sub-$\mathbb{Z}/p^r\mathbb{Z}$-sheaves of the de Rham-Witt sheaves $W_r\Omega^n_{\mathcal{X}}$ which are generated by $\mathrm{dlog}\underline{f_1} \wedge \dots \wedge \mathrm{dlog}\underline{f_n}$. This is [Mor15, Theorem 1.2] (cf. Appendix A) .

The next lemma is needed in order to generalize the purity theorem to $\mathbb{Z}/p^m\mathbb{Z}$-modules for any m.

Lemma 4.3.11. We have a short exact sequence of sheaves on the topological space of $\mathcal{X}$

$$0 \to \nu_m(n) \to \nu_{m+m'}(n) \to \nu_{m'}(n) \to 0.$$

Proof. This is a direct consequence of the same result on schemes proven in [Mil86, Lemma 1.7]. □

Proposition 4.3.12. Assume Conjecture 4.2.4. Then the morphism

$$Z\Omega^{+,n}_{\mathcal{X}'} \xrightarrow{C-1} \Omega^{+,n}_{\mathcal{X}'}$$

is surjective in the tame étale and strong étale topology over $\mathcal{X}$.

Proof. Let us check the surjectivity of $C-1$. This is checked on stalks. Assume we have a discretely ringed local Huber pair (A, A^+). Denote by $|\cdot| : A \to \Gamma$ the valuation associated to (A, A^+). Consider an element

$$\eta = h\mathrm{dlog} f_1 \wedge \cdots \wedge \mathrm{dlog} f_n \in \Omega^{+,n}_{(A,A^+)},$$

where h lies in A^+.

A preimage of this in $Z\Omega^n_A$ under $C-1$ is of the form

$$\eta' = \gamma^p \mathrm{dlog} f_1 \wedge \cdots \wedge \mathrm{dlog} f_n,$$

with $\gamma \in A^+$ satisfying $\gamma^p - \gamma = h$. Such a γ exists since

$$\mathrm{Spec} A^+[T]/(T^p - T - h)$$

is étale and hence induces a finite strongly étale morphism of adic spaces

$$\mathrm{Spa}(A[T]/(T^p - T - h), A^+[T]/(T^p - T - h)) \to \mathrm{Spa}(A, A^+).$$

It is left to check that η' lies in $\Omega^+_{(A,A^+)}$. To prove this, we need to check that $|\gamma^p| \leq 1$. Assume that $|\gamma| > 1$. Then $|\gamma^p| = |\gamma|^p > |\gamma|$. Hence $1 < |\gamma|^p = |\gamma^p - \gamma| = |h| \leq 1$, which is a contradiction.

It follows that $|\gamma^p| \leq 1$. Therefore η' lies in $Z\Omega^{+,n}_{(A,A^+)}$ □

Theorem 4.3.13. The sheaf $\nu(n)$ is the kernel of

$$Z\Omega^{n,\log}_{\mathcal{X},\lim} \xrightarrow{C-1} \Omega^{n,\log}_{\mathcal{X},\lim}.$$

In particular, if Conjecture 4.2.4 holds, then we have a short exact sequence

$$0 \to \nu(n) \to Z\Omega^{+,n}_{\mathcal{X}} \xrightarrow{C-1} \Omega^{+,n}_{\mathcal{X}} \to 0,$$

which further implies that $\nu(n)$ is a sheaf in the tame topology.

Proof. Note that $\nu(n)$ is the kernel of $Z\Omega^n_{\mathcal{X}} \xrightarrow{C-1} \Omega^n_{\mathcal{X}}$. Furhtermore $\nu(n)$ is the subsheaf of $Z\Omega^n_{\mathcal{X}}$ generated by sums of wedge products of logarithmic differentials, i.e. elements of the form $\mathrm{dlog} f = df/f$, for f a section in $\mathcal{O}_X$. These elements have valuation ≤ 1 for any point $x \in \mathcal{X}$ since $|df|_x \leq |f|_x$ by definition.

But we already know that $Z\Omega^{+,n}_{\mathcal{X}} \subseteq Z\Omega^{n,\log}_{\mathcal{X},\lim}$ with equality under the assumption of Conjecture 4.2.4. □

Remark 4.3.14.

- An alternative less conceptual proof of Theorem 4.3.13 uses Proposition 4.3.3 together with assumption of resolution of singularities.
- We will present in Chapter 6 an alternative proof of the existence of the Cartier operator $C : Z\Omega^{+,n}_{\mathcal{X}} \to \Omega^{+,n}_{\mathcal{X}}$ without the need of resolution of singularities or Conjecture 4.2.4.

5. Gysin morphism and purity in positive characteristic

5.1. Cohomology of $\Omega^{+,n}$

In this section we will check that the tame étale and the strongly étale cohomology of $\Omega^{+,n}_{\mathcal{X}/k}$ agree with the Riemann-Zariski open cohomology. For simplicity we write $\Omega^{+,n}$ instead of $\Omega^{+,n}_{\mathcal{X}/k}$. We similarly drop the subscript on $\Omega^{n,\log}$.

Proposition 5.1.1. Let $\mathcal{X}$ be a discretely ringed adic space. Then

$$H^i_{\text{sét}}(\mathcal{X}, \Omega^{+,n}) \cong H^i(\mathcal{X}, \Omega^{+,n}).$$

In order to prove this claim, we first need to study the sheaf $\Omega^{+,n}$ over the strongly étale site associated to a discrete local Huber pair (A, A^+).

Lemma 5.1.2. Let (A, A^+) be a discretely ringed local Huber pair. Then $\Omega^{n,+}(Y) = \Omega^{n,\log}(Y)$ for any Cartesian strongly étale map $Y \to \mathrm{Spa}(A, A^+)$.

Proof. By [Hüb21, Corollary 12.2], every strongly étale covering of $\mathrm{Spa}(A, A^+)$ has a refinement by *Cartesian* morphisms: these are strongly étale coverings

$$\mathrm{Spa}(B, B^+) \to \mathrm{Spa}(A, A^+)$$

for which

$$\begin{tikzcd} \mathrm{Spec}(B) \arrow[r] \arrow[d] & \mathrm{Spec}(B^+) \arrow[d] \\ \mathrm{Spec}(A) \arrow[r] & \mathrm{Spec}(A^+) \end{tikzcd} \tag{5.1}$$

is Cartesian. By [Hüb21, Proposition 11.10] the morphism $A^+ \to B^+$ is étale. We conclude that $\mathrm{Spa}(A, A^+)_{\mathrm{Cart,sét}}$ and $\mathrm{Spec}(A^+)_{\text{ét}}$ are equivalent, where $\mathrm{Spa}(A, A^+)_{\mathrm{Cart,sét}}$ is the site of Cartesian strongly étale maps over $\mathrm{Spa}(A, A^+)$.

We have inclusions of presheaves on $\mathrm{Spa}(A, A^+)_{\text{sét}}$

$$\Omega^{n,\log} \subset \Omega^{+,n} \subset \Omega^n.$$

But $\Omega^{+,n}$ is the sheafification of $\Omega^{n,\log}$ in $\mathrm{Spa}(A, A^+)_{\text{sét}}$ and therefore in $\mathrm{Spa}(A, A^+)_{\mathrm{Cart,sét}}$. The presheaf

$$\mathrm{Spec}(A^+)_{\text{ét}} \ni Y \mapsto \Omega^{n,\log}(\mathrm{Spa}(Y \otimes_{A^+} A, Y))$$

is a quasi-coherent sheaf in the étale topology (see [Stacks, Section 03DR]). This follows from the following two fact

- By [Ogu18, Corollary IV.1.2.8], $\Omega^n_Y(\log Y \otimes_{A^+} A)$ is a sheaf in the Zariski topology of Y.
- The log structure on Y is induced by the log structure on A. Thus the morphism of log schemes $(Y \otimes_{A^+} A, Y) \to (\mathrm{Spec} A, \mathrm{Spec} A^+)$ is strict, which implies that $\Omega^{n,\log}(\mathrm{Spa}(Y \otimes_{A^+} A, Y))$ the pullback of $\Omega^{n,\log}_{(A,A^+)}$.

Hence $\Omega^{n,\log}$ is already a sheaf on $\mathrm{Spa}(A, A^+)_{\mathrm{Cart,sét}}$. The claim follows. □

Proof of Proposition 5.1.1. To prove the claim we need to check that

$$H^i_{\text{sét}}(\mathcal{X}, \Omega^{+,n}) = 0$$

when $\mathcal{X} = \mathrm{Spa}(A, A^+)$ is a discretely ringed local adic space. We follow the steps of [Hüb21, Proposition 12.3]. As in loc. cit. we only need to prove that the Čech complex for $\Omega^{+,n}$ associated to *Cartesian* strongly étale coverings

$$\mathrm{Spa}(B, B^+) \to \mathrm{Spa}(A, A^+)$$

is trivial. Over such coverings $\Omega^{+,n} = \Omega^{n,\log}$.

Taking integral closures commutes with étale base change. Therefore $B^+ \otimes_{A^+} \cdots \otimes_{A^+} B^+$ is integrally closed in $B \otimes_A \cdots \otimes_A B$.

$$\mathrm{Spa}(B, B^+) \times_{\mathrm{Spa}(A,A^+)} \cdots \times_{\mathrm{Spa}(A,A^+)} \mathrm{Spa}(B, B^+) \cong \mathrm{Spa}(B \otimes_A \cdots \otimes_A B, B^+ \otimes_{A^+} \cdots \otimes_{A^+} B^+).$$

Furthermore, we may interpret $\Omega^{n,\log}$ as a quasi-coherent sheaf over $\mathrm{Spec}(A^+)$ [Ogu18, Corollary IV.1.2.8]). Therefore the Čech complex in question is equal to the Čech complex of the quasi-coherent sheaf $\Omega^{n,\log}$, which is exact since $A^+ \to B^+$ is faithfully flat. □

We furthermore compare the strongly étale and tame étale cohomology of $\Omega^{+,n}$.

Proposition 5.1.3. Let $\mathcal{X}$ be a discretely ringed adic space over a valued field $\mathrm{Spa}(k, k^+)$. Then

$$H^i_{\text{sét}}(\mathcal{X}, \Omega^{+,n}) = H^i_t(\mathcal{X}, \Omega^{+,n}).$$

Proof. Under the assumption that $\mathcal{X}$ is defined over a field of characteristic p, the claim follows from [Hüb21, Proposition 8.5], as $\Omega^{+,n}_{\mathcal{X}}$ is p-torsion. □

Lastly, we aim to prove that $H^i(\mathrm{Spa}(A, A^+), \Omega^{+,n})$ is trivial for log smooth Huber pairs (A, A^+) with discrete topology.

Theorem 5.1.4. Let $\mathcal{X} = \mathrm{Spa}(A, A^+)$ be a discretely ringed affinoid over a perfect field k of positive characteristic p. Assume that (A, A^+) is associated to a log smooth log scheme. Assume Conjectures B.2.1 and B.2.3. Then

$$H^i_t(\mathrm{Spa}(A, A^+), \Omega^{+,n}) = H^i_{\text{sét}}(\mathrm{Spa}(A, A^+), \Omega^{+,n}) = H^i(\mathrm{Spa}(A, A^+), \Omega^{+,n}) = 0,$$

for all $i \geq 1$.

Proof. We proved in Proposition 5.1.1 and Proposition 5.1.3 that the tame, strongly étale and open cohomology groups of $\Omega^{+,n}$ agree. We have proved in Corollary 3.2.10 that the latter is isomorphic to $H^i_{\mathrm{RZ}}(\mathrm{Spa}(A, A^+), \Omega^{+,n})$, which in counterpart is isomorphic to $\mathrm{colim}_{\bar{Y}} H^i(\bar{Y}, \Omega^{+,n}|_{\bar{Y}})$, where $\bar{Y}$ are A-modifications of A^+. As A is of finite type over A^+, we may assume that these are compactifications. Under an adequate assumption of resolution of singularities (i.e. Conjectures B.2.1 and B.2.3) we may assume that $\bar{Y}$ is log smooth over k with the log structure associated to $\bar{Y}\backslash\mathrm{Spec}(A)$. Furthermore, as discussed in Remark B.2.4, we may dominate these modifications by chains of blow ups at smooth centers, intersecting the boundary transversally. Hence $\Omega^+|_{\bar{Y}} = \Omega^{n,\log}_{\bar{Y}}$ by [Hüb24, Theorem 8.12]. Using Theorem 2.4.5, we obtain $H^i(\bar{Y}, \Omega^+|_{\bar{Y}}) = H^i(\bar{Y}, \Omega^{n,\log}_{\bar{Y}}) = 0$. This is proven by induction on the length of the chain of blow-ups:

Let $\pi : \bar{Y}_2 \to \bar{Y}_1$ be a morphism in the chain of blow ups. Denote its smooth center by Z. Note that $\mathrm{Spec}(A)$ is open in $\bar{Y}_i$ and that $\bar{Y}_i\backslash\mathrm{Spec}(A)$ is an snc divisor for $i = 1, 2$. Assume that we know that $H^i(\bar{Y}_1, \Omega^{n,\log}) = 0$. Cover $\bar{Y}_1$ by affine opens U_i. Set $V_i := \bar{Y}_2 \times_{\bar{Y}_1} U_i$. By Theorem 2.4.5, we know that $H^i(V_i, \Omega^{n,\log}) = 0$. We apply [Stacks, Proposition 02O5], to conclude that $R^i\pi_*\Omega^{+,n} = 0$ and hence

$$H^i(\bar{Y}_2, \Omega^{+,n}) = 0. \qquad \square$$

5.2. The Gysin morphism on $\Omega^{+,n}$

In this section we define the Gysin isomorphism for the sheaf of $\mathcal{O}^+$-modules $\Omega^{+,n}$. From now on the functor $i^!$ is taken in the tame topology, unless we explicitly say otherwise.

Theorem 5.2.1. Let k be a perfect field of characteristic p. Assume resolution of singularities and embedded resolution of singularities with boundary over k. Let (X, Z) be a smooth pair over k of pure codimension 1. Set $\mathcal{X} := \mathrm{Spa}(X, k)$ and $\mathcal{Z} := \mathrm{Spa}(Z, k)$. Then the sheaf $R^n i^! \Omega^{+,n}_{\mathcal{X}}$ is trivial for $n \neq 1$.

Proof. Triviality of the sheaves $R^n i^! \Omega^{+,n}_{\mathcal{X}}$ for $n \geq 2$ can be checked on stalks. Let x be a point in $\mathcal{Z} \subset \mathcal{X}$. Then when $n \geq 2$, the group $(R^n i^! \Omega^+_{\mathcal{X}})_x$ is isomorphic to the group

$$\begin{aligned} \operatorname*{colim}_{x \in \mathcal{U} \xrightarrow{\text{tame}} \mathcal{X}} H^n_{\mathcal{Z}\cap\mathcal{U},t}(\mathcal{U}, \Omega^{+,n}) &\cong \operatorname*{colim}_{x \in \mathcal{U} \xrightarrow{\text{tame}} \mathcal{X}} H^{n-1}_t(\mathcal{U}\backslash\mathcal{Z}, \Omega^{+,n}) \\ &\cong \operatorname*{colim}_{x \in \mathcal{U} \xrightarrow{\text{tame}} \mathcal{X}} H^{n-1}(\mathcal{U}\backslash\mathcal{Z}, \Omega^{+,n}) \end{aligned}$$

where the colimit is taken over affinoids $\mathcal{U}$ containing x and which are tame étale over $\mathcal{X}$.

By assuming resolution of singularities (Conjecture B.2.1) as well as embedded resolution of singularities with boundary (Conjecture B.2.3), the category of tame étale neighborhoods $\{x \in \mathcal{U} \to \mathcal{X}\}$ is dominated by adic spaces of the form $\mathrm{Spa}(U, \bar{U})$, such that $(U\backslash Z, \bar{U})$ is a pair associated to a log smooth log scheme.

Since Z has codimension 1 in X, we may assume that both $U \setminus Z$ and $\bar{U}$ are affine, after replacing $\bar{U}$ by an affine open subscheme. Then, by Theorem 5.1.4, we conclude

that

$$H^{n-1}(\mathcal{U}\backslash\mathcal{Z},\Omega^{+,n}) = H^{n-1}(\mathrm{Spa}(U\backslash Z,\bar{U}),\Omega^{+,n}) = 0 \text{ for } n \geq 2.$$

It remains to check that $i^!\Omega^{+,n}_{\mathcal{X}} = 0$. By definition,

$$i^!\Omega^{+,n}_{\mathcal{X}}(\mathcal{U}\cap\mathcal{Z}) = \ker\left(\Omega^{+,n}_{\mathcal{X}}(\mathcal{U}) \to \Omega^{+,n}_{\mathcal{X}}(\mathcal{U}\backslash\mathcal{Z})\right)$$

for every open $\mathcal{U}\subset\mathcal{X}$.

By [Hüb24, Lemma 8.8], $i^!\Omega^{n,\log}_{\mathcal{X},\lim}(\mathcal{U}\cap\mathcal{Z})$ is trivial. Since under our assumptions the equality $\Omega^{n,\log}_{\mathcal{X},\lim} = \Omega^{+,n}$ holds, we conclude that $i^!\Omega^{+,n}_{\mathcal{X}}(\mathcal{U}\cap\mathcal{Z}) = 0$. □

Theorem 5.2.2. Let k be a perfect field of characteristic p. Let (X,Z) be a smooth pair over k of pure codimension 1. Assume that Conjectures B.2.1 and B.2.3 hold. Define $\mathcal{X} := \mathrm{Spa}(X,k)$ and $\mathcal{Z} := \mathrm{Spa}(Z,k)$. Then we have an isomorphism of $\mathcal{O}^+_{\mathcal{Z}}$-modules:

$$R^1 i^!\Omega^{+,n}_{\mathcal{X}} \to \Omega^{+,n-1}_{\mathcal{Z}}.$$

Proof. Define $\mathcal{U} := \mathcal{X}\backslash\mathcal{Z}$ and let $j:\mathcal{U}\to\mathcal{X}$ be the natural inclusion. By the excision long exact sequence we know that $R^1 i^!\Omega^{+,n}_{\mathcal{X}}$ is isomorphic to the cokernel of $j_*\Omega^{+,n}_{\mathcal{U}} \to \Omega^{+,n}_{\mathcal{X}}$.

By resolution of singularities, we may assume that $\mathcal{U} = \mathrm{Spa}(U,\bar{U})$ where $\bar{U}$ is smooth, and both $D := \bar{U}\setminus U$ and $\bar{U}\setminus(U\setminus Z) = D\cup\bar{Z}$ are simple normal crossings divisors. Here, $\bar{Z}\subseteq\bar{U}$ denotes the schematic image of $Z\cap U$ in $\bar{U}$.

We may now use the second residue sequence in Proposition 2.3.7, stating that

$$0 \to \Omega^{n,\log}_{\bar{U}}(\log(D+\bar{Z})) \to \Omega^{n,\log}_{\bar{U}}(\log(D)) \to \Omega^{n-1,\log}_{\bar{Z}}(D|_{\bar{Z}}) \to 0.$$

The global sections of these terms are identified with the sections of $\Omega^{+,n}_{\mathcal{X}}(\mathcal{X})$, $\Omega^{+,n}_{\mathcal{U}}(\mathcal{U})$, and $\Omega^{+,n-1}_{\mathcal{Z}}(\mathcal{Z})$, respectively. Therefore, we obtain an exact sequence

$$0 \to \Omega^{+,n}_{\mathcal{X}} \to j_*\Omega^{+,n}_{\mathcal{U}} \to i_*\Omega^{+,n-1}_{\mathcal{Z}} \to 0, \tag{5.2}$$

which completes the proof. □

Remark 5.2.3.
- In the case $n=0$, the claim is equivalent to $R^1 i^!\mathcal{O}^+_{\mathcal{X}} \cong 0$. This is true since $j_*\mathcal{O}^+_{\mathcal{U}} \cong \mathcal{O}^+_{\mathcal{X}}$, even without assuming resolution of singularities.
- In Chapter 6, we give a proof of Theorem 5.2.2 without the assumption of Resolution of singularities (cf. Corollary 6.2.6)

Corollary 5.2.4. Let k be a perfect field of positive characteristic p. Let (X,Z) be a smooth pair of pure codimension r. Assume that resolution of singularities holds (Conjecture B.2.1 and B.2.3). Define $\mathcal{X} := \mathrm{Spa}(X,k)$ and $\mathcal{Z} := \mathrm{Spa}(Z,k)$. Then we have an isomorphism in $D(\mathcal{Z})$:

$$\phi_{\mathcal{Z}/\mathcal{X}} : \Omega^{+,n-r}_{\mathcal{Z}}[-r] \to Ri^!\Omega^{+,n}_{\mathcal{X}}$$

Proof. This is done by induction on the codimension of Z. When $\mathrm{codim}_Z(X) = 1$ this is proven by combining Theorems 5.1.4 and 5.2.2. For general codimension r, assume that Z is a complete intersection of smooth schemes. Then there exists a smooth subscheme Z', which is also a complete intersection in X, such that $Z \xrightarrow{i_1} Z' \xrightarrow{i_2} X$, and $\mathrm{codim}_{Z'}(X) = r - 1$. We obtain

$$\begin{aligned} Ri^!\Omega^{+,n}_{\mathcal{X}} \cong & R(i_2 \circ i_1)^!\Omega^{+,n}_{\mathcal{X}} \cong Ri_1^!\big(Ri_2^!\Omega^{+,n}_{\mathcal{X}}\big) \\ \cong & Ri_1^!\Omega^{+,n-r+1}_{\mathrm{Spa}(Z',k)}[-r+1] \cong \Omega^{+,n-r}_{\mathcal{Z}}[-r]. \end{aligned}$$

5.2.4 when Z is a complete intersection $\square$

Milne's proof of semi-purity for log de Rham-Witt sheaves [Mil86, Proposition 2.1] uses the following proposition

Proposition 5.2.5. Let k be a field. There is a family of homomorphisms

$$\phi_{Z/X} : \Omega^{s-r}_{Z/k} \to R^r i^!\Omega^s_{X/k},$$

defined for every smooth pair (X, Z) of codimension r, where $i : Z \to X$ is the inclusion. Furthermore, Φ is defined by with the following properties

- $\phi_{Z/X}$ commutes with the differentials d.
- If $Z \xrightarrow{i_{Z/Y}} Y \xrightarrow{i_{Y/X}} X$ are closed immersions of smooth schemes over k, such that (X, Z) is of codimension r and (Y, Z) is of codimension t, then
$$R^{r-t}i^!_{Z/Y}(\phi_{Y/X}) \circ \phi_{Z/Y} = \phi_{Z/X}.$$
- Assume that $r = 1$. Let $z \in Z$ be a closed point and let $A = \mathrm{Spec}(\mathcal{O}_{X,z})$. Let $f = 0$ be a local equation of Z around z. Then $(R^1 i^!\Omega^s_{X/k})_z$ is associated to the A-module $\Omega^s_{A[f^{-1}]} = \Omega^s_A$ and $(\phi_{Z/X})_z$ is associated to the residue map
$$\omega \mapsto \omega \wedge \mathrm{dlog}(f) : \ \Omega^{s-1}_{A/(f)} \to \Omega^s_{A[f^{-1}]}/\Omega^s_A.$$

However Milne gives the reference [Har06, p. III.8], which only states the proposition in characteristic 0. The construction does however work in any characteristic (see the first part of the proof of [Har06, Theorem III.8.1]). We obtain a morphism of sheaves over adic spaces

$$\phi_{\mathcal{Z}/\mathcal{X}} : \Omega^{s-r}_{\mathcal{Z}/k} \to R^r i^!\Omega^s_{\mathcal{X}/k},$$

where $\mathcal{Z} = \mathrm{Spa}(Z, k)$ and $\mathcal{X} = \mathrm{Spa}(X, k)$.

Proposition 5.2.6. Under the assumptions of Corollary 5.2.4, the diagram

$$\begin{tikzcd} \Omega^{+,s-r}_{\mathcal{Z}/k} \arrow[r] \arrow[d, "\phi_{\mathcal{Z}/\mathcal{X}}"'] & \Omega^{s-r}_{\mathcal{Z}/k} \arrow[d, "\phi_{\mathcal{Z}/\mathcal{X}}"] \\ R^r i^!\Omega^{+,s}_{\mathcal{X}/k} \arrow[r] & R^r i^!\Omega^s_{\mathcal{X}/k}. \end{tikzcd} \tag{5.3}$$

commutes.

Proof. We start by assuming $r = 1$. Consider a point $z \in \mathcal{Z}$. Let (A, A^+) the local Huber pair associated to $\mathcal{X}_z$. There exists $f \in A$ such that $\mathcal{Z}_z = \mathrm{Spa}(A/(f), A^+/(f))$. The diagram (5.3) corresponds locally to the diagram

$$\begin{tikzcd} \Omega^{+,s-1}_{(A/(f),A^+/(f))} \arrow[r] \arrow[d, "\omega\mapsto\omega\wedge\mathrm{dlog} f"'] & \Omega^{s-1}_{A/(f)} \arrow[d, "\omega\mapsto\omega\wedge\mathrm{dlog} f"] \\ \Omega^{+,s}(\mathrm{Spa}(A[f^{-1}], A^+))/\Omega^{+,s}_{(A,A^+)} \arrow[r] & \Omega^s_{A[f^{-1}]}/\Omega^s_A. \end{tikzcd} \tag{5.4}$$

Since (5.4) commutes, the diagram (5.3) also commutes for $r = 1$. The claim for $r > 1$ follows by induction. □

End of the proof of Theorem 5.2.4. In the proof of [Gro85, Théorème 3.5.8], Gros proves that the morphism

$$\phi_{\mathcal{Z}/\mathcal{X}} : \Omega^{s-r}_{\mathcal{Z}/k} \to R^r i^! \Omega^s_{\mathcal{X}/k},$$

is an injection. Since $\Omega^{+,s-r}_{\mathcal{Z}/k} \to \Omega^{s-r}_{\mathcal{Z}/k}$ is also injective, the morphism

$$\Omega^{+,s-r}_{\mathcal{Z}/k} \to \Omega^{s-r}_{\mathcal{Z}/k} \xrightarrow{\Phi_{\mathcal{Z}/\mathcal{X}}} R^r i^! \Omega^s_{\mathcal{X}/k}$$

is also injective and factors uniquely over $R^r i^! \Omega^{+,s}_{\mathcal{X}/k}$ locally (due to (5.4)). We conclude that

$$\phi_{\mathcal{Z}/\mathcal{X}} : \Omega^{+,s-r}_{\mathcal{Z}/k} \to R^r i^! \Omega^{+,s}_{\mathcal{X}/k}$$

is defined globally. □

Remark 5.2.7. Note that, since the fact that diagram (5.3) commutes can be checked locally, Proposition 5.2.6 holds even when Z is not a complete intersection.

Corollary 5.2.8. Assume the Conjectures B.2.1 and B.2.3.

Let $i : Z \to X$ be a smooth subscheme of pure codimension r. Let $i : \mathcal{Z} \to \mathcal{X}$ the associated inclusion of discretely ringed adic spaces. Then we have an isomorphism

$$\phi_{\mathcal{Z}/\mathcal{X}} : Z\Omega^{n-r,+}_{\mathcal{Z}}[-r] \to Ri^! Z\Omega^{n,+}.$$

Proof. We first prove the claim when $\underline{r = 1}$.

Step 1: Proof that $Ri^! Z\Omega^{+,n}$ and $Ri^! B\Omega^{+,n}$ are concentrated in degree 1:

This is done by induction on the degree n. For degree $n = 0$, $Z\Omega^{+,0} = (\mathcal{O}^+)^p$ consists of sections of $\mathcal{O}^+$ which are p-powers. This is isomorphic to $\mathcal{O}^+$ by the Frobenius morphism. The claim follows for $Z\Omega^{+,0}$. Since $B\Omega^{+,0} = 0$, the case $n = 0$ is complete.

Note that since $Z\Omega^{+,n} \cong Z\Omega^{n,\log}_{\lim}$ and $B\Omega^{+,n} = B\Omega^{n,\log}_{\lim}$, we have $i^! Z\Omega^{+,n} = i^! B\Omega^{+,n} = 0$. This follows for any sheaf $\mathcal{F}_{\lim}$ when $\mathcal{F}$ is unramified. Indeed [Hüb24, Lemma 8.8] states that for $\mathcal{U} \subset \mathcal{U}'$, one has an inclusion

$$\mathcal{F}_{\lim}(\mathcal{U}') \subset \mathcal{F}_{\lim}(\mathcal{U}).$$

For a given tame morphism $\mathcal{U} \to \mathcal{X}$ the group of section $i^!\mathcal{F}_{\lim}(\mathcal{U})$ is the kernel of $\mathcal{F}_{\lim}(\mathcal{U}\backslash\mathcal{Z}) \to \mathcal{F}_{\lim}(\mathcal{U})$, which is injective. Hence $i^!\mathcal{F}_{\lim}(\mathcal{U}) = 0$.

It is left to check by induction that $R^j i^! Z\Omega^{+,n} = R^j i^! B\Omega^{+,n} = 0$ when $j \geq 2$.

Assume the claim holds for $n \in \mathbb{N}$. Consider the short exact sequence in Corollary 4.3.7

$$0 \to B\Omega^{+,n+1} \to Z\Omega^{+,n+1} \xrightarrow{C} \Omega^{+,n+1} \to 0,$$

as well as the short exact sequence

$$0 \to Z\Omega^{+,n} \to \Omega^{+,n} \to B\Omega^{+,n+1} \to 0.$$

If we consider the long exact sequence associated to the latter we obtain that $R^j i^! B\Omega^{+,n+1}$ is trivial when $j \geq 2$. Applying the long exact sequence of cohomology to the former implies that $R^j i^! Z\Omega^{+,n+1}$ is trivial when $j \geq 2$.

Step 2: Computing $Ri^! Z\Omega^{+,n}$

Since $Ri^!\Omega^{+,n+1}_{\mathcal{X}}$ is concentrated in degree one when $r = 1$ we may apply the residue sequences in Corollary 2.3.9 together with Conjectures B.2.1 and B.2.3 to conclude that $Ri^! Z\Omega^{+,n+1}_{\mathcal{X}} = Z\Omega^{+,n+1}_{\mathcal{Z}}$ by a similar argument to the one used in 5.2.2.

When $\underline{r \geq 1}$, we find $Z \subset Z' \subset X$ locally, with corresponding immersions $Z \xrightarrow{i_1} Z' \xrightarrow{i_2} X$ and $\operatorname{codim}_{Z'}(X) = r - 1$. We obtain

$$\begin{aligned}\phi_{\mathcal{Z}/\mathcal{X}} : Ri^! Z\Omega^{+,n}_{\mathcal{X}} \cong & R(i_2 \circ i_1)^! Z\Omega^{+,n}_{\mathcal{X}} \cong Ri_1^!\big(Ri_2^!\Omega_{\mathcal{X}}\big) \\ \cong & Ri_1^! Z\Omega^{+,n-r+1}_{\mathrm{Spa}(Z',k)}[-r+1] \cong Z\Omega^{+,n-r}_{\mathcal{Z}}[-r].\end{aligned}$$

Since $\phi_{\mathcal{Z}/\mathcal{X}}$ commutes with differentials in both instances, one also gets a diagram for the closed differentials

$$\begin{tikzcd} Z\Omega^{+,s-r}_{\mathcal{Z}/k} \arrow[r] \arrow[d, "\phi_{\mathcal{Z}/\mathcal{X}}"'] & Z\Omega^{s-r}_{\mathcal{Z}/k} \arrow[d, "\phi_{\mathcal{Z}/\mathcal{X}}"] \\ R^r i^! Z\Omega^{+,s}_{\mathcal{X}/k} \arrow[r] & R^r i^! Z\Omega^{s}_{\mathcal{X}/k}. \end{tikzcd} \tag{5.5}$$

Therefore $\phi_{\mathcal{Z}/\mathcal{X}}$ is independent from the choice of Z', hence may be defined globally. $\square$

5.3. Purity for the tame topology in positive characteristic

Étale cohomology fails to satisfy purity for smooth pairs. Consider for example $i : \{0\} \to \mathbb{A}^1$. We have $\nu_{\mathbb{A}^1}(0) = \mathbb{Z}/p\mathbb{Z}$. Hence there is an exact sequence

$$0 \to R^1 i^! \mathbb{Z}/p\mathbb{Z} \to R^1 i^! \mathbb{G}^p_{a,\mathbb{A}^1} \xrightarrow{F^{-1}-1} R^1 i^! \mathbb{G}_{a,\mathbb{A}^1} \to R^2 i^! \mathbb{Z}/p\mathbb{Z} \to 0.$$

Note that $R^1 i^! \mathbb{G}_{a,\mathbb{A}^1}$ is locally isomorphic to $H^1_Z(X, \mathbb{G}_a)$, which is isomorphic to $\operatorname{coker}(\mathcal{O}_{X,x} \to \mathcal{O}_{X,x}[t^{-1}])$ because of the excision sequence and the fact that étale cohomology of quasi-coherent sheaves is zero on affines.

But $F^{-1}-1$ is not surjective on $\mathcal{O}_{X,x}[t^{-1}]$. Indeed, for $x=0\in\mathbb{A}^1$, finding a preimage of $1/t$ corresponds to solving the equation $T^p-T=1/t$, or equivalently finding roots of the polynomial $f(T)=tT^p-tT-1$. But $f'(T)=-t$ which lies in m_x, thus $k[T]/(tT^p-tT-1)\to k[T]$ is not étale. In conclusion the preimage of $1/t$ under $F^{-1}-1$ does not exist in the étale topology.

The fact that $Z\Omega^n_{X,x}[t^{-1}]\xrightarrow{C-1}\Omega^n_{X,x}[t^{-1}]$ is not surjective seems to be an obstruction to purity. It is worth noting that the Cartier operator and F^{-1} coincide when $n=0$.

To remedy this, an idea is to replace the sheaf Ω with Ω^+ and étale topology with tame topology.

Theorem 5.3.1. [c.f. Theorem 1.3.1] Let k be a perfect field of positive characteristic p. Let (X,Z) be a smooth pair over k of pure codimension r. Assume that Conjectures B.2.1 and B.2.3 hold. Define $\mathcal{X}:=\mathrm{Spa}(X,k)$ and $\mathcal{Z}:=\mathrm{Spa}(Z,k)$. Then we have an isomorphism in $D(\mathcal{Z}_t)$:

$$Gys_m:\nu_{\mathcal{Z},m}(n-r)[-r]\to Ri^!\nu_{\mathcal{X},m}(n),\quad \forall m,n\in\mathbb{N}.$$

Proof. We start by assuming $m=1$. The short exact sequence

$$0\to\nu_{\mathcal{X}}(n)\to Z\Omega^{+,n}_{\mathcal{X}}\xrightarrow{C-1}\Omega^{+,n}_{\mathcal{X}}\to 0$$

induces an exact sequence

$$0\to R^ri^!\nu_{\mathcal{X}}(n)\to R^ri^!Z\Omega^{+,n}_{\mathcal{X}}\xrightarrow{C-1}R^ri^!\Omega^{+,n}_{\mathcal{X}}\to R^{r+1}i^!\nu_{\mathcal{X}}(n)\to 0$$

since $Ri^!\Omega^{+,n}_{\mathcal{X}}$ and $Ri^!Z\Omega^{+,n}_{\mathcal{X}}$ are concentrated in degree r as established in the previous section. It is enough to prove that the square on the right hand side of the diagram

$$\begin{tikzcd}
0 \arrow[r] & \nu_{\mathcal{Z}}(n-r) \arrow[r] \arrow[d, dotted] & Z\Omega^{+,n-r}_{\mathcal{Z}} \arrow[r, "C-1"] \arrow[d, "\cong\text{ (Corollary 5.2.8)}"] & \Omega^{n-r}_{\mathcal{Z}} \arrow[r] \arrow[d, "\cong\text{ (Corollary 5.2.4)}"] & 0 \\
0 \arrow[r] & R^ri^!\nu_{\mathcal{X}}(n) \arrow[r] & R^ri^!Z\Omega^{+,n}_{\mathcal{X}} \arrow[r, "C-1"] & R^ri^!\Omega^{+,n}_{\mathcal{X}} \arrow[r] & R^{r+1}i^!\nu_{\mathcal{X}}(n),
\end{tikzcd} \tag{5.6}$$

commutes. By induction on r, it suffices to prove this for $r=1$.

As in the proof of Theorem 5.2.2 and Corollary 5.2.8 it suffices to consider the commutativity at the level of sections for an open $\mathcal{U}=\mathrm{Spa}(U,\bar{U})\subset\mathcal{X}$ satisfying that $\bar{U}$ is smooth over k and $\bar{U}\backslash U$ as well as $Z\cup(\bar{U}\backslash U)=:\bar{Z}\cup D$ are snc divisors, where $\bar{Z}$ is the schematic image of $Z\cap U$ in $\bar{U}$.

The isomorphisms $R^1i^!\Omega^{+,n+1}_{\mathcal{X}}\to\Omega^{+,n}_{\mathcal{Z}}$ and $R^1i^!Z\Omega^{+,n+1}_{\mathcal{X}}\to Z\Omega^{+,n}_{\mathcal{Z}}$ correspond to the residue morphisms

$$\Omega^{n+1}_{\bar{U}}(\log(D+\bar{Z}))\to\Omega^n_{\bar{Z}}(\log D|_{\bar{Z}})\ \text{ and }\ Z\Omega^{n+1}_{\bar{U}}(\log(D+\bar{Z}))\to Z\Omega^n_{\bar{Z}}(\log D|_{\bar{Z}}).$$

It is therefore enough to check that the Cartier operator commutes with the restriction morphism.

Assume that $\bar{Z}$ is the zero set associated to a section f of $\mathcal{O}_{\bar{U}}$. The Cartier operator $C : Z\Omega^{n+1}_{\bar{U}}(\log(D+\bar{Z})) \to \Omega^{n+1}_{\bar{U}}(\log(D+\bar{Z}))$ sends

$$\eta = \gamma + \eta' \wedge d\log(f); \quad \gamma \in Z\Omega^{n+1}_{\bar{U}}(\log D), \eta' \in \Omega^{n}_{\bar{U}}(\log(D+\bar{Z})) \tag{5.7}$$

to $C(\eta) = C(\gamma) + C(\eta') \wedge d\log(f)$. By Corollary 2.3.9 every $\eta \in Z\Omega^{n+1}_{\bar{U}}(\log(D+\bar{Z}))$ takes the form (5.7).

Since γ is a section of $Z\Omega^{n+1}_{\bar{U}}(\log D) \subset Z\Omega^{n+1}_{\bar{U}}(\log(D+\bar{Z}))$, its image $C(\gamma)$ is a section of $\Omega^{n+1}_{\bar{U}}(\log D)$. Hence the image of η and $C(\eta)$ by the residue map is $\eta'|_{\bar{Z}}$ and $C(\eta')|_{\bar{Z}}$ respectively.

It is only left to check that the Cartier operator commutes with the restriction morphism to $\bar{Z}$, i.e. $\eta'|_{\bar{Z}} = C(\eta')|_{\bar{Z}}$. The inverse Cartier isomorphism

$$\bar{C}^{-1} : \Omega_{\bar{U}}(\log(D+\bar{Z})) \to Z\Omega_{\bar{U}}(\log(D+\bar{Z}_U))/B\Omega_{\bar{U}}(\log(D+\bar{Z}))$$

sends elements of the form hdg to $h^p g^{p-1} dg$, which is compatible with restriction to $\bar{Z}$. Hence the Cartier operator C, which is the composition

$$Z\Omega_{\bar{U}}(\log(D+\bar{Z})) \to Z\Omega_{\bar{U}}(\log(D+\bar{Z}))/B\Omega_{\bar{U}}(\log(D+\bar{Z})) \xrightarrow{\bar{C}} \Omega_{\bar{U}}(\log(D+\bar{Z})),$$

commutes with the restriction to $\bar{Z}$. As wedge products commute with restriction and the Cartier operator, the claim follows. 5.3.1 for m=1 $\square$

The semi-purity isomorphism [Mil86, Proposition 2.1], induces by comparison between the étale cohomology of a scheme and of an adic space [Hüb21, Lemma 8.3] an isomorphism

$$\phi_{\mathcal{Z}/\mathcal{X}} : \nu_{\mathcal{Z},m}(n-r) \to R^r i^!_{\text{ét}} \nu_{\mathcal{X},m}(n). \tag{5.8}$$

Let $j : \mathcal{U} := \mathcal{X} \backslash \mathcal{Z} \to \mathcal{X}$ be the natural inclusion. When $r = 1$, excision induces isomorphisms

$$R^1 i^!_{\text{ét}} \nu_{\mathcal{X},m}(n) \cong \left(j_* \nu_{\mathcal{U},m}(n)/\nu_{\mathcal{X},m}(n)\right)_{\text{ét}} \text{ and } R^1 i^!_t \nu_{\mathcal{X},m}(n) \cong \left(j_* \nu_{\mathcal{U},m}(n)/\nu_{\mathcal{X},m}(n)\right)_t,$$

where the subscript ét and t indicates the topology in which $i^!$ and the quotient are taken. Consequently, $R^1 i^!_{\text{ét}} \nu_{\mathcal{X},m}(n)$ is the étale sheaffification of the tame sheaf $R^1 i^!_t \nu_{\mathcal{X},m}(n)$. In order to make such a comparison, we work on the site $(\text{Ét}/\mathcal{X})_t$ instead of $\mathcal{X}_t$, whose objects are étale morphisms over $\mathcal{X}$, and coverings are tame étale coverings. This is possible due to [Mil80, Proposition 3.1].

Proposition 5.3.2. Assume that we are in the situation of Theorem 5.3.1. Assume that $r = 1$. The diagram

$$\begin{tikzcd}
\nu_{\mathcal{Z},1}(n-1) \arrow[r, "Gys_1"] \arrow[dr, "\Phi_{\mathcal{Z}/\mathcal{X}}"'] & R^1 i^!_t \nu_{\mathcal{X},1}(n) \arrow[d, "\text{sheafification}"] \\
& R^1 i^!_{\text{ét}} \nu_{\mathcal{X},1}(n)
\end{tikzcd}$$

commutes. In particular, the sheafification morphism

$$R^1 i_t^! \nu_{\mathcal{X},1}(n) \to R^1 i_{\text{ét}}^! \nu_{\mathcal{X},1}(n)$$

is an isomorphism.

Proof. Consider the diagram

$$\begin{tikzcd}
Z\Omega_{\mathcal{Z}}^{+,n-1} \arrow[rr] \arrow[dr, "C-1"] \arrow[dd, "\phi_{\mathcal{Z}/\mathcal{X}}"'] & & Z\Omega_{\mathcal{Z}}^{n-1} \arrow[dr, "C-1"] \arrow[dd, "\phi_{\mathcal{Z}/\mathcal{X}}"'] & \\
& \Omega_{\mathcal{Z}}^{+,n-1} \arrow[rr] \arrow[dd, "\phi_{\mathcal{Z}/\mathcal{X}}"'] & & \Omega_{\mathcal{Z}}^{n-1} \arrow[dd, "\phi_{\mathcal{Z}/\mathcal{X}}"] \\
R^1 i^! Z\Omega_{\mathcal{X}}^{+,n} \arrow[rr] \arrow[dr, "C-1"'] & & R^1 i^! Z\Omega_{\mathcal{X}}^{n} \arrow[dr, "C-1"'] & \\
& R^1 i^! \Omega_{\mathcal{X}}^{+,n} \arrow[rr] & & R^1 i^! \Omega_{\mathcal{X}}^{n}.
\end{tikzcd}$$

Taking the kernels of all the morphisms denoted by $C-1$ delivers the commutative diagram

$$\begin{tikzcd}
\nu_{\mathcal{Z}}(n-r) \arrow[r, "="] \arrow[d, "Gys_1"] & \nu_{\mathcal{Z}}(n-r) \arrow[d, "\phi_{\mathcal{Z}/\mathcal{X}}"] \\
R^1 i_t^! \nu_{\mathcal{X}}(n) \arrow[r] & R^1 i_{\text{ét}}^! \nu_{\mathcal{X}}(n),
\end{tikzcd}$$

since $i^! \Omega_{\mathcal{X}}^{+,n}$ and $i^! \Omega_{\mathcal{X}}^{n}$ are trivial. □

Proof of Theorem 5.3.1 for all m. By applying $Ri^!$ on the short exact sequence

$$0 \to \nu_{\mathcal{X},m}(n) \to \nu_{\mathcal{X},m+1}(n) \to \nu_{\mathcal{X},1}(n) \to 0.$$

defined in Lemma 4.3.11, we conclude that $Ri^! \nu_{\mathcal{X},m}(n)$ is concentrated in degree r for all $m, n \in \mathbb{N}$. It is then left to find for each m a morphism

$$Gys_m : \nu_{\mathcal{Z},m}(n-r) \to R^r i^! \nu_{\mathcal{X},m}(n),$$

such that

$$\begin{tikzcd}
0 \arrow[r] & \nu_{\mathcal{Z},m}(n-r) \arrow[r] \arrow[d, "Gys_m"] & \nu_{\mathcal{Z},m+1}(n-r) \arrow[r] \arrow[d, "Gys_{m+1}"] & \nu_{\mathcal{Z},1}(n-r) \arrow[r] \arrow[d, "Gys_1"] & 0 \\
0 \arrow[r] & R^r i^! \nu_{\mathcal{X},m}(n) \arrow[r] & R^r i^! \nu_{\mathcal{X},m+1}(n) \arrow[r] & R^r i^! \nu_{\mathcal{X},1}(n) \arrow[r] & 0,
\end{tikzcd}$$

commutes. We first assume that $\underline{r=1}$. By induction on m, there exists Gys_m fitting in the following diagram

$$\begin{tikzcd}
0 \arrow[r] & \nu_{\mathcal{Z},m}(n-1) \arrow[r] \arrow[d, "Gys_m"] & \nu_{\mathcal{Z},m+1}(n-1) \arrow[r] \arrow[d, dotted, "Gys_{m+1}"] & \nu_{\mathcal{Z},1}(n-1) \arrow[r] \arrow[d, "Gys_1"] & 0 \\
0 \arrow[r] & R^1 i_t^! \nu_{\mathcal{X},m}(n) \arrow[r] \arrow[d, "\cong"] & R^1 i_t^! \nu_{\mathcal{X},m+1}(n) \arrow[r] \arrow[d, "\cong"] & R^1 i_t^! \nu_{\mathcal{X},1}(n) \arrow[r] \arrow[d, "\cong (5.3.2)"] & 0 \\
0 \arrow[r] & R^1 i_{\text{ét}}^! \nu_{\mathcal{X},m}(n) \arrow[r] & R^1 i_{\text{ét}}^! \nu_{\mathcal{X},m+1}(n) \arrow[r] & R^1 i_{\text{ét}}^! \nu_{\mathcal{X},1}(n) \arrow[r] & 0,
\end{tikzcd}$$

where the composition of each of the vertical sequences is $\phi_{\mathcal{Z}/\mathcal{X}}$. Notice that the lower short exact sequence is exact since it is isomorphic to the upper one. Furthermore, the lower vertical morphisms are the sheafification morphisms. These are bijective. Hence Gys_{m+1} can be defined as the composition

$$\nu_{\mathcal{Z},m+1}(n-r) \xrightarrow{\phi_{\mathcal{Z}/\mathcal{X}}} R^1 i_{\text{ét}}^! \nu_{\mathcal{X},m+1}(n) \xrightarrow{\text{sheafification}^{-1}} R^1 i_t^! \nu_{\mathcal{X},m+1}(n).$$

Proof for all r and m: We now assume that Z has codimension $r > 1$. Then, locally in the open topology, there exists a smooth closed subvariety $Y \subset X$ of codimension $r-1$ containing Z. Define $\mathcal{Y} := \mathrm{Spa}(Y, k)$. The closed immersion $i : \mathcal{Z} \to \mathcal{X}$ can be factored as follows

$$\mathcal{Z} \xrightarrow{i_1} \mathcal{Y} \xrightarrow{i_2} \mathcal{X}.$$

Define Gys_m on $\mathcal{X}$ as the composition of isomorphisms

$$\nu_{\mathcal{Z},m}(n-r)[-r] \xrightarrow{Gys_m} R i_{1,t}^! \nu_{\mathcal{Y},m}(n-r+1)[-r+1] \xrightarrow{R i_{1,t}^! Gys_m} R i_t^! \nu_{\mathcal{X},m}(n).$$

In order to construct Gys_m globally, we claim that Gys_m is independent from the choice of Y. It is sufficient to prove that the diagram

$$\begin{tikzcd}
\nu_{\mathcal{Z},m}(n) \arrow[r, "\phi_{\mathcal{Z}/\mathcal{X}}"] \arrow[d, "Gys_m"] & R^r i_{\text{ét}}^! \nu_{\mathcal{X},m}(n) \arrow[d] \\
R^r i_t^! \nu_{\mathcal{X},m}(n) \arrow[r] & R^r i^! W_m \Omega_{\mathcal{X}}^n
\end{tikzcd} \tag{5.9}$$

commutes. Indeed, if this is the case, then $R^r i_t^! \nu_{\mathcal{X},m}(n) \to R^r i^! W_m \Omega_{\mathcal{X}}^n$ is an inclusion. As $\nu_{\mathcal{Z},m}(n) \xrightarrow{\Phi_{\mathcal{Z}/\mathcal{X}}} R^r i^! W_m \Omega_{\mathcal{X}}$ is also an inclusion, the morphism $\nu_{\mathcal{Z},m}(n) \xrightarrow{Gys_m} R^r i_t^! \nu_{\mathcal{X},m}(n)$ is unique and independent from the choice of Y.

When $r = 1$, this is true by the previous discussion. Assume that the claim is true for

some r. The diagram (5.9) for $r+1$ may be written as

$$\begin{tikzcd}
\nu_{\mathcal{Z},m}(n-r-1) \arrow[r] \arrow[d] \arrow[dr, phantom, "①"] & R^1 i^!_{1,\text{ét}} \nu_{\mathcal{Y},m}(n-r) \arrow[r] \arrow[dr, phantom, "②"] & R^{r+1} i^!_{\text{ét}} \nu_{\mathcal{X},m} \arrow[dd] \\
R^1 i^!_{1,t} \nu_{\mathcal{Y},m}(n-r) \arrow[r] \arrow[ur] \arrow[d] \arrow[dr, phantom, "③"] & R^1 i^!_{1,t} R^r i^!_{2,\text{ét}} \nu_{\mathcal{X},m}(n) \arrow[ur] \arrow[dr] \arrow[r, phantom, "④"] & \\
R^{r+1} i^!_t \nu_{\mathcal{X},m}(n) \arrow[rr] & & R^{r+1} i^! W_m \Omega^n_{\mathcal{X}},
\end{tikzcd}$$

where i is factored as $\mathcal{Z} \xrightarrow{i_1} \mathcal{Y} \xrightarrow{i_2} \mathcal{X}$ with Y being of codimension r. The upper diagonal morphisms are étale sheafification morphisms. Each of the diagrams ① to ④ commute:

- The diagram ① commutes by the codimension 1 case. Notice here that $\mathcal{Z}$ has codimension 1 in $\mathcal{Y}$.
- The diagram ② commutes since sheafification is compatible with isomorphisms.
- The diagram and ③ commutes by the induction step on r and applying the functor $R^1 i^!_{1,t}$.
- Lastly, the diagram ④ commutes by the universal property of sheafification since the upper diagonal map is an étale sheafification map.

As Gys_m is independent from the choice of $\mathcal{Y}$, it glues to a global isomorphism. 5.3.1 □

Remark 5.3.3. In his proof of [Mil86, Proposition 2.1], Milne does not construct a Gysin morphism for $\nu_m(n)$ when $m > 1$. The reference [Gro85] is more precise in that aspect.

Corollary 5.3.4. Under the same assumptions as in Theorem 5.3.1, we have

$$H^l_{\mathcal{Z},t}(\mathcal{X}, \nu_m(n)) = H^{l-r}_t(\mathcal{Z}, \nu_m(n-r)).$$

Proof. We consider the spectral sequence

$$R^l\Gamma(\mathcal{Z}, R^s i^! \nu_m(n)) \implies R^{s+l}\Gamma_{\mathcal{Z}}(\mathcal{X}, \nu_m(n)).$$

Since $Ri^!\nu_m(n)$ is concentrated in degree r and $R^r i^! \nu_m(n) = \nu_m(n-r)[-r]$, we have

$$R^l\Gamma(\mathcal{Z}, \nu_m(n-r)[-r]) \cong R^{r+l}\Gamma_{\mathcal{Z}}(\mathcal{X}, \nu_m(n-r))$$ □

By [HS21, Section §14] we conclude:

Corollary 5.3.5. Under the same assumptions, we have

$$H^t_{Z,t}(X, \nu_m(n)) = H^{t-r}_t(Z, \nu_m(n-r)).$$

Remark 5.3.6. Corollary 5.3.4 was proven in [Hüb21] in the case where $n = 0$ (Corollary 14.5 loc. cit.).

A proof when $\dim(X) = 1$ is given in [Hüb23, Theorem 1.2.].

Merici applies 5.3.4 in his article [Mer24] in order to prove that tame cohomology is motivic.

5.4. Purity for regular pairs

In this section, we prove cohomological purity for log de Rham-Witt sheaves in the tame topology for a regular pair over a base S of characteristic p, again under the assumption of resolution of singularities. In our context, a pair of schemes (X, Z) is called regular, if Z is a closed subscheme of X, and X and Z are regular.

Note that the immersion $i : Z \to X$ is part of the structure, we choose however to omit it in the notation.

As usual, we say that (X, Z) is of pure codimension r if Z is of pure codimension r in X.

This part is inspired by the work of Shiho [Shi07], who generalized the result of Milne [Mil86] and Gros [Gro85] in the étale cohomology.

Theorem 5.4.1. Let $i : Z \to X$ be a closed immersion of Noetherian schemes over a base scheme S of characteristic p. Assume that (X, Z) is a regular pair of pure codimension r. Then in $D((Z/S)_t)$ we have the isomorphism

$$Ri^! \nu_{X,m}(n) \cong \nu_{Z,m}(n-r)[-r]$$

under the assumption of resolution of singularities.

A key ingredient in the proof is the following result due to Popescu

Proposition 5.4.2 (Popescu's Theorem [Stacks, Theorem 07GC])**.** Let B be a ring and let A be a regular B-algebra. Then A is a filtered colimit of smooth B-algebras.

An equivalent formulation is that for any diagram

$$\begin{tikzcd} B \arrow[r] \arrow[d] & A. \\ C \arrow[ur] & \end{tikzcd} \tag{5.10}$$

with C is a finitely generated B-algebra, there exists a smooth B-algebra D, fitting into a commutative diagram:

$$\begin{tikzcd} B \arrow[r] \arrow[d] & A \\ C \arrow[r] \arrow[ur] & D. \arrow[u] \end{tikzcd} \tag{5.11}$$

Assume now that the schemes X, Z and S are the affine schemes $\mathrm{Spec}(A)$, $\mathrm{Spec}(A/I)$ and $\mathrm{Spec}(R)$. Furthermore, since $Z \to X$ is regular, we may assume that I is defined by a regular sequence $t_1, \ldots, t_r$.

Claim 5.4.3. Assume furthermore that X is local. The chain of morphisms $Z \to X \to S$ can be expressed as a filtered limit of morphisms of affine schemes

$$Z_i := \mathrm{Spec}(A_i/I_i) \to X_i := \mathrm{Spec}(A_i) \to S_i := \mathrm{Spec}(R_i)$$

where each A_i and R_i is of finite type over $\mathbb{F}_p$, and A_i and A_i/I_i are smooth over $\mathbb{F}_p$.

Proof. Observe first that the morphism $R \to A$ is the filtered colimit of morphisms $(R_i \to C_i)_{i \in J}$, where R_i and C_i are of finite type over $\mathbb{F}_p$. By Proposition 5.4.2, find a smooth $\mathbb{F}_p$ algebra A_i such that the diagram

$$\begin{tikzcd} R \arrow[r] & A & \\ R_i \arrow[u] \arrow[r] & C_i \arrow[u] \arrow[r] & A_i \arrow[ul] \end{tikzcd}$$

commutes. Thus $R \to A$ is the filtered colimit of $(R_i \to A_i)_{i \in J}$.

In order to find adequate I_i's, we first need a Lemma

Lemma 5.4.4. [DFJK21, Lemma C.7.] Let $B \to A$ be a local homomorphism of local rings. Let $J \subsetneq B$ be an ideal such that A/JA is regular. Set $n = \dim A - \dim A/JA$. If there exists elements $f_1, \ldots, f_n \in B$ generating J, then f_i form a subset of regular parameters of B, and the ring B/J is regular.

Proof of the Lemma. Let $\mathbf{n}$ and $\mathfrak{m}$ be the respective maximal ideals of B and A. Let $g_1, \ldots, g_n$ be the images of $f_1, \ldots, f_n$ along the morphism $B \to A$. Since A/JA is regular, $g_1, \ldots, g_n$ can be extended to a regular sequence generating $\mathfrak{m}$ according to [Stacks, Lemma 00NR]. This implies that $g_1, \ldots, g_n$ are $A/\mathfrak{m}$-linearly independent elements of $\mathfrak{m}/\mathfrak{m}^2$, due to Nakyama's lemma [Stacks, Lemma 00DV]. Thus the f_i'S are $B/\mathbf{n}$-linearly independent elements of $\mathbf{n}/\mathbf{n}^2$. By applying Nakayama's lemma again, we conclude the claim. □

The ideal $I \subseteq A$ is defined by the regular sequence $t_1, \ldots, t_r$, which extends to a regular sequence $(t_1, \ldots, t_n)$, where $n = \dim X$. By restricting J and localizing each A_i, we may lift, according to Lemma 5.4.4, the regular sequence $t_1, \ldots, t_n$ to compatible regular sequences $t_{1,i}, \ldots, t_{n,i}$, for which each $A_i/(t_{1,i}, \ldots, t_{r,i})$ is smooth over $\mathbb{F}_p$. Set $I_i := (t_{1,i}, \ldots, t_{r,i})$, yielding the desired presentation. □

Remark 5.4.5.

- In Claim 5.4.3 one may drop the condition that X is local. This has been done in [DFJK21, Proposition C.8.].
- Claim 5.4.3 implies that the closed immersion $i : \mathcal{Z} := \mathrm{Spa}(Z, S) \hookrightarrow \mathrm{Spa}(X, S) =: \mathcal{X}$ may also be written as a limit of closed immersions $\mathcal{Z}_i := \mathrm{Spa}(Z_i, S_i) \hookrightarrow \mathrm{Spa}(X_i, S_i) =: \mathcal{X}_i$, under the assumption that X is local. Applying [Hüb21, Corollary 5.4] and excision we conclude that

$$\operatorname*{colim}_{i \in J} H^s_{t,\mathcal{Z}_i}(\mathcal{X}_i, \nu_m(n)) = H^s_{t,\mathcal{Z}}(\mathcal{X}, \nu_m(n)), \quad \text{for all } s \geq 0. \tag{5.12}$$

Since each $\mathcal{X}_i = \mathrm{Spa}(X_i, S_i)$ and $\mathcal{Z}_i = \mathrm{Spa}(Z_i, S_i)$ are open subspaces of $\mathrm{Spa}(X_i, \mathbb{F}_p)$ and $\mathrm{Spa}(Z_i, \mathbb{F}_p)$, respectively, we may apply Theorem 5.3.1 to each pair (X_i, Z_i).

We now remove the assumption that X, Z and S are affine. In order to prove Theorem 5.4.1, we have to define a morphism

$$\Phi_{\mathcal{Z}/\mathcal{X}} : \nu_{\mathcal{Z},m}(n-r)[-r] \to Ri^!\nu_{\mathcal{X},m}(n)$$

and verify that it is an isomorphism.

Proposition 5.4.6. Under the assumptions of Theorem 5.4.1, $Ri^!\nu_{\mathcal{X},m}(n)$ is concentrated in degree r.

Proof. The claim is local on the tame site of $\mathcal{X}$. We can thus assume that $\mathcal{X} = \mathrm{Spa}(A, A^+)$ is a tamely henselian adic space. By assumption, a regular ideal $I \subseteq A$ exists, such that $\mathcal{Z} = \mathrm{Spa}(A/I, A^+) = \mathrm{Spa}(A/I, A^+/I)$. According to Remark 5.4.5, we may describe $\mathrm{Spa}(A, A^+)$ and $\mathrm{Spa}(A/I, A^+)$ as filtered limits of adic spaces $\mathrm{Spa}(A_i, R_i)$ and $\mathrm{Spa}(A_i/I_i, R_i)$, where A_i and R_i are of finite type over $\mathbb{F}_p$, and A_i and A_i/I_i are smooth $\mathbb{F}_p$-algebras.

For $s \neq r$, the sheaf $R^s i^!\nu_m(n)$ corresponds to the cohomology group $H^s_{t,\mathcal{Z}}(\mathcal{X}, \nu_m(n))$. By Remark 5.4.5, the latter is isomorphic to

$$\operatorname*{colim}_i H^s_{t,\mathrm{Spa}(A_i/I_i,R_i)}(\mathrm{Spa}(A_i, R_i), \nu_m(n)) = \operatorname*{colim}_i H^{s-r}_t(\mathrm{Spa}(A_i/I_i, R_i), \nu_m(n-r)) \tag{5.13}$$

By [Hüb21, Corollary 5.4], (5.13) is isomorphic to $H^{s-r}_t(\mathrm{Spa}(A/I, A^+), \nu_m(n-r))$, which is trivial, since $\mathrm{Spa}(A/I, A^+) = \mathrm{Spa}(A/I, A^+/I)$ is tamely henselian. □

Proof of Theorem 5.4.1. It is only left to construct an isomorphism of sheaves in $D((Z/S)_t)$.

$$\Phi_{\mathcal{Z}/\mathcal{X}} : \nu_{\mathcal{Z},m}(n-r)[-r] \to R^r i^!\nu_{\mathcal{X},m}(n).$$

We use the following construction from [Shi07, Section 3] on $W_m\Omega^n_X$.

Definition/Lemma 5.4.7. For each $n > 0$ There exists a morphism

$$\Phi_{Z/X} : W_m\Omega^{n-r}_Z \to R^r i^! W_m\Omega^n_X.$$

Locally, when Z and X are the colimit of affine schemes Z_i and X_i which are smooth over $\mathbb{F}_p$, the morphism $\Phi_{Z/X}$ is the colimit of the morphisms

$$\Phi_{Z_i/X_i} : W_m\Omega^{n-r}_{Z_i} \to R^r i^! W_m\Omega^n_{X_i}$$

defined by Gros in [Gro85, Subsection 3.4].

By [Hüb21, Lemma 8.3], we obtain a morphism

$$\Phi_{\mathcal{Z}/\mathcal{X}} : W_m\Omega^{n-r}_{\mathcal{Z}} \to R^r i^! W_m\Omega^n_{\mathcal{X}}. \tag{5.14}$$

We have established at the end of the proof of Theorem 5.3.1 that the natural morphism $R^r i^! \nu_{\mathcal{X}_i,m}(n) \to R^r i^! W_m \Omega^n_{\mathcal{X}_i}$ is injective. By Definition/Lemma 5.4.7, we conclude that $R^r i^! \nu_{\mathcal{X},m}(n) \to R^r i^! W_m \Omega^n_{\mathcal{X}}$ is also injective: this is checked on local Huber pairs, where the morphism is a filtered colimit of injective maps.

It suffices to check that the composition of morphisms

$$\nu_m(n-r) \hookrightarrow W_m \Omega^{n-r}_Z \xrightarrow{\Phi_{\mathcal{Z}/\mathcal{X}}} R^r i^! W_m \Omega^n_Z.$$

factors through $R^r i^! \nu_m(n)$ and that $\Phi_{\mathcal{Z}/\mathcal{X}} : \nu_m(n-r) \to R^r i^! \nu_m(n)$ is an isomorphism. This claim may be checked on local adic spaces. By expressing $\mathcal{X}$ and $\mathcal{Z}$ as colimits of smooth adic spaces as in Remark 5.4.5, the claim follows by Theorem 5.3.1. □

Remark 5.4.8. In [Shi07, Theorem 3.2], Shiho required X to be part of *the category* $\mathcal{C}$ whose objects are excellent schemes X satisfying $[\kappa(x) : \kappa(x)^p] < \infty$ for all $x \in X$. These conditions are not required in our context, but are necessary in the étale cohomology case [Shi07, Remark 3.10].

6. Avoiding resolution of singularities

For the rest of this chapter fix the following notation: let (X, Z) be a smooth pair of codimension r over a perfect field k of characteristic p, i.e. X and Z are smooth over k and $Z \hookrightarrow X$ is a closed immersion. Set $\mathcal{X} := \mathrm{Spa}(X, k)$ and $\mathcal{Z} := \mathrm{Spa}(Z, k)$ and denote the natural inclusion by $i : \mathcal{Z} \to \mathcal{X}$. Furthermore, set $\mathcal{U} := \mathcal{Z} \backslash \mathcal{X}$ and denote the associated open immersion by $j : \mathcal{U} \to \mathcal{Z}$.

Two main subjects where resolution of singularities played a role in the proof of Theorem 5.3.1 are

- the construction of the Cartier operator and,
- Computing $R^r i^! \Omega^{+,n}$.
- the proof of the vanishing of $R^j i^! \Omega^{+,n}$, $j \neq r$

In this chapter, we apply techniques from the theory of valuation rings in order to avoid the use of resolution of singularities for the first two problems: we construct the Cartier morphism (Corollary 6.1.3) and prove a version of semi-purity when Z has codimension one (Theorem 6.2.9).

6.1. The Cartier operator II: local theory

The aim of this section is to construct the Cartier operator

$$C : Z\Omega^{+,n}_{\mathcal{X}} \to \Omega^{+,n}_{\mathcal{X}}$$

using valuation theoretic techniques. We furthermore prove that its kernel is $B\Omega^{+,n}_{\mathcal{X}}$.

6.1.1. Reduction to valuation rings

The Cartier operator on $\Omega^{+,n}$, if it exists, is the restriction of the Cartier operator on Ω^n. It is therefore enough to prove that $C(Z\Omega^+) \subseteq \Omega^+$, which may be checked on stalks. Consequently, we may replace $\mathcal{X}$ with its localization at a point $\mathrm{Spa}(A, A^+)$. Let $(K, K^+) = (A/\mathfrak{m}, A^+/\mathfrak{m})$ be its residue field. As usual, for a Huber pair (A, A^+), we endow A^+ with the logarithmic structure associated to the inclusion

$$A^\times \cap A^+ \to A^+.$$

For example, when $(A, A^+) = (K, K^+)$ this is simply the usual log structure

$$K^+ \backslash \{0\} \to K^+$$

associated to the valuation ring K^+.

Since we are working over a local Huber pair, $\Omega^{+,n}$ and $Z\Omega^{+,n}$ are isomorphic to $\Omega^{n,\log}$ and $Z\Omega^{n,\log}$, respectively ([Hüb24, Proposition 7.11 and Lemma 7.3]). One may therefore construct a morphism of groups

$$C^{-1} : \Omega^{n,\log}_{(A,A^+)} \to Z\Omega^{n,\log}_{(A,A^+)}/B\Omega^{n,\log}_{(A,A^+)}$$

which acts as follows when $n = 1$:

$$C^{-1}(fdg) = f^p g^{p-1} dg, \quad C^{-1}(f\mathrm{dlog}g) = f^p \mathrm{dlog}g.$$

For $n > 1$ set

$$C^{-1}(\omega \wedge \omega') = C^{-1}(\omega) \wedge C^{-1}(\omega'), \quad \forall \omega \in \Omega^{m,\log}_{(A,A^+)}, \omega' \in \Omega^{n-m,\log}_{(A,A^+)} \tag{6.1}$$

It will suffice to show that C^{-1} is well defined when $n = 1$, as the relation (6.1) will determine it uniquely for $n > 1$. Recall that $\Omega^{\log}_{(A,A^+)}$ is given by

$$\Omega^{\log}_{(A,A^+)} = \left[\Omega^n_{A^+} \oplus A^+ \otimes_{\mathbb{Z}} \left(A^\times \cap A^+\right)^{\mathrm{gp}}\right]/\mathcal{R}$$

where $\mathcal{R}$ is generated as an A^+-submodule by the elements $(-dg) \oplus (g \otimes g)$ ([Ogu18, Proof of Proposition IV.1.1.2]). One defines as usual the notation

$$f\mathrm{dlog}g := 0 \oplus f \otimes g, \quad dg := dg \oplus 0.$$

In particular, $\mathcal{R}$ is generated by $g\mathrm{dlog}g - dg$.

In total, the A^+-module $\Omega^{\log}_{(A,A^+)}$ is generated by the elements dg and $\mathrm{dlog}g$ for $g \in A^+$ modulo the equivalence relations

- $d(g + h) = dg + dh$
- $d(gh) = gdh + hdg$
- $\mathrm{dlog}(g^{n_1} h^{n_2}) = n_1 \mathrm{dlog}g + n_2 \mathrm{dlog}h$ and
- $g\mathrm{dlog}g = dg$.

for all $g, h \in A^+$ and $n_1, n_2 \in \mathbb{Z}$. In order to check that C^{-1} is well defined, we have to check that these relations still hold modulo $B\Omega^{\log}_{(A,A^+)}$ after applying C^{-1}:

- $C^{-1}d(g + h) = (g + h)^{p-1} d(g + h) = g^{p-1}dg + h^{p-1}dh + d(\sum_{0<i<p} \frac{(p-1)!}{i!(p-i)!} g^i h^{p-i})$,
- $C^{-1}d(gh) = (gh)^{p-1} d(gh) = h^p g^{p-1} dg + g^p h^{p-1} dh$,
- $C^{-1}(n_1 \mathrm{dlog}g + n_2 \mathrm{dlog}h) = n_1^p \mathrm{dlog}g + n_2^p \mathrm{dlog}h = n_1 \mathrm{dlog}g + n_2 \mathrm{dlog}h$ and
- $C^{-1}(g\mathrm{dlog}g) = g^p \mathrm{dlog}g = g^{p-1} g\mathrm{dlog}g = g^{p-1}dg$.

Lemma 6.1.1. The diagram

$$\begin{tikzcd} \Omega^{n,\log}_{(A,A^+)} \arrow[r, "C^{-1}"] \arrow[d] & Z\Omega^{n,\log}_{(A,A^+)}/B\Omega^{n,\log}_{(A,A^+)} \arrow[d] \\ \Omega^n_A \arrow[r, "C^{-1}"] & Z\Omega^n_A/B\Omega^n_A \end{tikzcd}$$

commutes. Consequently, C^{-1} is injective on $\Omega^{n,\log}_{(A,A^+)}$.

Proof. The diagram commutes by definition of C^{-1}. We have to show that C^{-1} is injective on $\Omega^{n,\log}_{(A,A^+)}$. The lower horizontal morphism is an isomoprhism since A is regular. The left morphism is injective by [Hüb24, Corollary 4.2]. □

In order to prove the existence of the Cartier operator on smooth discretely ringed adic spaces, it is enough, due to Lemma 6.1.1 and [Hüb24, Proposition 7.11], to prove the following theorem.

Theorem 6.1.2. Let $\mathcal{X} = \mathrm{Spa}(A, A^+)$ be a local adic space which is the localization of a smooth discretely ringed adic space. The morphism

$$C^{-1} : \Omega^{+,n}_{(A,A^+)} \to Z\Omega^{+,n}_{(A,A^+)}/B\Omega^{+,n}_{(A,A^+)}$$

is an isomorphism.

A direct application of Theorem 6.1.2 is the following.

Corollary 6.1.3. Let X be a smooth scheme over a perfect field k of positive characteristic p, and let $\mathcal{X}$ be the discretely ringed adic space $\mathrm{Spa}(X, k)$. Then the short exact sequence of sheaves over $\mathcal{X}$

$$0 \to B\Omega^n_{\mathcal{X}} \to Z\Omega^n_{\mathcal{X}} \xrightarrow{C} \Omega^n_{\mathcal{X}} \to 0$$

restricts to a short exact sequence

$$0 \to B\Omega^{+,n}_{\mathcal{X}} \to Z\Omega^{+,n}_{\mathcal{X}} \xrightarrow{C} \Omega^{+,n}_{\mathcal{X}} \to 0.$$

The remainder of this section is devoted to proving Theorem 6.1.2.

Differentials vanishing in Ω^n_K

Recall that (A, A^+) is a local Huber pair with residue field (K, K^+). Consider the map

$$\pi : \Omega^n_A \to \Omega^n_K, \omega \mapsto \pi(\omega) = \bar{\omega}$$

A differential ω, whose image in Ω^n_K is trivial, takes a certain form which we describe in the following definition.

Definition 6.1.4. Let A be a regular local ring over a perfect field k. Let $\mathfrak{m}$ be its maximal ideal. Consider the A-submodule $\Omega^n_{A,0}$ of Ω^n_A whose elements are sums of wedge products of the form

$$g_0 dg_1 \wedge \cdots \wedge dg_n,$$

such that an $i \in \{0, \ldots, n\}$ exists for which $g_i \in \mathfrak{m}$. When $n = 0$, the A-module $\Omega^0_{A,0}$ is the maximal ideal $\mathfrak{m}$.

The differentials $d : \Omega^n_A \to \Omega^{n+1}_A$ restrict to differentials $\Omega^n_{A,0} \to \Omega^{n+1}_{A,0}$. We define the submodules of exact and closed differentials, $B\Omega^{n+1}_{A,0}$ and $Z\Omega^n_{A,0}$, as the image and kernel of $d : \Omega^n_{A,0} \to \Omega^{n+1}_{A,0}$, respectively. These are A^p-submodules of $\Omega^{n+1}_{A,0}$ and $\Omega^n_{A,0}$.

Lemma 6.1.5. We have an equality of submodules of A

$$\{\eta \in \Omega_A | \bar{\eta} = 0 \in \Omega^n_K\} = \Omega^n_{A,0}.$$

Proof. It is clear that the RHS is a submodule of the LHS. Consider η in the LHS. We need to prove that η is in $\Omega^n_{A,0}$.

For $n = 1$ we have the fundamental short exact sequence

$$0 \to \mathfrak{m}/\mathfrak{m}^2 \to A/\mathfrak{m} \otimes_A \Omega_A \to \Omega_{A/\mathfrak{m}} \to 0.$$

Because A is a localization of a smooth scheme, this sequence is exact and splits. Hence

$$A/\mathfrak{m} \otimes_A \Omega_A \cong \mathfrak{m}/\mathfrak{m}^2 \oplus \Omega_{A/\mathfrak{m}}. \tag{6.2}$$

It follows that $A/\mathfrak{m} \otimes_A \Omega^n_A \cong \bigwedge^n_{A/\mathfrak{m}} \left(A/\mathfrak{m} \otimes_A \Omega_A\right) \cong \bigwedge^n_{A/\mathfrak{m}} \left(\mathfrak{m}/\mathfrak{m}^2 \oplus \Omega_{A/\mathfrak{m}}\right)$.

If η maps to 0 in Ω^n_K, then its image $\tilde{\eta}$ in $A/\mathfrak{m} \otimes_A \Omega^n_A$ must be a sum of elements of the form $\bar{f} dg_1 \wedge \cdots \wedge dg_n$, where $\bar{f} \in A/\mathfrak{m}$ and at least one g_i lies in $\mathfrak{m}$. If $f \in A$ is a preimage of $\bar{f}$ then $f dg_1 \wedge \cdots \wedge dg_n \in \Omega_A$ is a preimage of $\bar{f} dg_1 \wedge \cdots \wedge dg_n \in A/\mathfrak{m} \otimes_A \Omega^n_A$. Such an element already lies in $\Omega^n_{A,0}$.

Therefore, we may find a preimage $\eta' \in \Omega^n_A$ of $\tilde{\eta}$ which lies in $\Omega^n_{A,0}$. Since Ω^n_A is free and therefore flat, the sequence

$$0 \to \mathfrak{m}\Omega^n_A \to \Omega^n_A \to A/\mathfrak{m} \otimes_A \Omega^n_A \to 0$$

is exact. We find that

$$\eta - \eta' \in \ker(\Omega^n_A \to A/\mathfrak{m} \otimes_A \Omega^n_A) = \mathfrak{m}\Omega^n_A \subseteq \Omega^n_{A,0}.$$

Thus, $\eta \in \Omega^n_{A,0}$. $\square$

Lemma 6.1.6. Let A be a regular local ring which is the localization of a smooth algebra over a perfect field k. We have an equality

$$B\Omega^n_{A,0} = B\Omega^n_A \cap \Omega^n_{A,0}.$$

Proof. By assumption, A is isomorphic to a ring which is étale over the localization of $\mathbb{A}^d_k = \mathrm{Spec}(k[x_1, \ldots, x_d])$ at the ideal generated by $(x_1, \ldots, x_m)$ with $m \leq d$ [Stacks, Lemma 054L]. Let B be the k-algebra $(k[x_1, \ldots, x_n])_{(x_1,\ldots,x_m)}$ with maximal ideal $\mathfrak{m}_B$. Since A is an étale B-algebra, we have the isomorphisms $\Omega^n_{A/k} \cong A \otimes_B \Omega^n_{B/k}$ and $\Omega^n_{K/k} \cong K \otimes_B \Omega^n_{(B/\mathfrak{m}_B)/k}$. We may therefore replace A with the local ring $(k[x_1, \ldots, x_d])_{(x_1,\ldots,x_m)}$.

The module of Kähler differentials Ω^n_A has an A-basis

$$dx_{i_1} \wedge \cdots \wedge dx_{i_n}, 1 \leq i_1 < \cdots < i_n \leq d.$$

On the other hand, elements of Ω^n_K are sums of the differentials

$$adx_{i_1} \wedge \cdots \wedge dx_{i_n}, \quad a \in A/\mathfrak{m}, m < i_1 < \cdots < i_n < d. \tag{6.3}$$

Let η be an element in $B\Omega^{n+1}_A \cap \Omega^{n+1}_{A,0}$. Then $\eta = d\omega$ for some $\omega \in \Omega^n_A$ and $d\bar{\omega} = 0$ in Ω^{n+1}_K.

We lift $\bar{\omega}$ to $\omega' \in Z\Omega^n_A$. In order to achieve this, observe that any element of the form (6.3) may be lifted to an element in

$$k(x_{m+1}, \ldots, x_d)dx_{i_1} \wedge \ldots dx_{i_n} \subseteq \Omega^n_A, \quad m < i_1 < \cdots < i_n \leq d.$$

This is due to the fact that the ring $K = A/\mathfrak{m}$ is isomorphic to $k(x_{m+1}, \ldots, x_d) \subseteq A$. Now the fact that $d\bar{\omega} = 0$ implies that $d\omega' = 0$: this property may be checked in

$$\bigoplus_{m<i_1<\cdots<i_{n+1}\leq d} k(x_{m+1}, \ldots, x_d)dx_{i_1} \wedge \cdots \wedge dx_{i_{n+1}},$$

which is isomorphic to Ω^{n+1}_K and may also be concidered a submodule of Ω^{n+1}_A.

Lastly, we note that $\gamma := \omega - \omega'$ lies in $\Omega^n_{A,0}$ since ω' and ω have the same image $\bar{\omega}$ in Ω^n_K. Therefore $d\gamma = d(\omega - \omega')$ is in $B\Omega^{n+1}_{A,0}$. □

Remark 6.1.7. The module $\Omega^n_{A,0}$ is an A^+-submodule of $\Omega^{+,n}_{(A,A^+)}$. Note that the definition of $\Omega^n_{A,0}$ is independent of the subring A^+. Indeed, if $\eta \in \Omega^n_{A,0}$ then $|\eta| = 0 \leq 1$ hence $\eta \in \Omega^{+,n}_{(A,A^+)}$ according to Proposition 4.1.6.

Lemma 6.1.8. We have the equality in Ω^n_A

$$\pi^{-1}(\Omega^{+,n}_{(K,K^+)}) = \Omega^{+,n}_{(A,A^+)}.$$

Proof. This Lemma is a consequence of Proposition 4.1.6: Let $\eta \in \pi^{-1}(\Omega^{+,n}_{(K,K^+)})$. Then its image in Ω^n_K is of the form

$$\overline{\eta} = \sum \overline{g_{0,i}} d\overline{g_{1,i}} \wedge \ldots d\overline{g_{n,i}}$$

where $|\overline{g_{0,i}} \cdots \overline{g_{n,i}}| \leq 1$ for every i. Consider lifts $g_{0,i}, \ldots, g_{n,i}$. Then $|g_{0,i} \cdots g_{n,i}| \leq 1$, which implies that

$$\eta' := \sum_i g_{0,i} dg_{1,i} \wedge \cdots \wedge g_{n,i}$$

is in $\Omega^{+,n}_{(A,A^+)}$ according to Proposition 4.1.6. The difference $\eta - \eta'$ is in $\Omega^n_{A,0}$, hence in $\Omega^{+,n}_{(A,A^+)}$ according to Remark 6.1.7.

On the other hand, a differential $\omega \in \Omega^{+,n}_{(A,A^+)}$ satisfies $|\omega| = |\pi(\omega)| \leq 1$. Thus $\omega \in \pi^{-1}(\Omega^{+,n}_{(K,K^+)})$. □

Proposition 6.1.9. Let $\mathcal{X} = \mathrm{Spa}(A, A^+)$ be a discretely ringed local adic space. Assume that A is the localization of a scheme which is smooth over a perfect field k of positive characteristic p. Suppose that Theorem 6.1.2 holds for $(K, K^+) = (A/\mathfrak{m}, A^+/\mathfrak{m})$. Then the Cartier operator

$$C : Z\Omega^n_A \to \Omega^n_A$$

restricts to a morphism

$$C : Z\Omega^{+,n}_{(A,A^+)} \to \Omega^{+,n}_{(A,A^+)}.$$

Furthermore the kernel of C is $B\Omega^{+,n}_{(A,A^+)}$.

Proof. Consider $\omega \in Z\Omega^{+,n}_{(A,A^+)}$. To prove the first part, we must show that its preimage η under C^{-1} lies in $\Omega^{+,n}_{(A,A^+)}$.

Since $\bar{\omega} = \pi(\omega)$ is in Ω^n_K and satisfies $d\bar{\omega} = \pi(d\omega) = 0$, it follows that $\bar{\omega} \in Z\Omega^n_K$. By Lemma 6.1.8, we conclude that $\bar{\omega} \in Z\Omega^{+,n}_{(K,K^+)}$.

By assumption, the Cartier operator on (K, K^+) has an inverse, so there exists a unique $\bar{\eta} \in \Omega^{+,n}_{(K,K^+)}$ such that $C^{-1}(\bar{\eta}) = \bar{\omega}$. Note that by definition of the inverse Cartier operator, $\eta \in Z\Omega^n_A$ is a lift of $\bar{\eta}$. Due to Lemma 6.1.8, we have $\eta \in \Omega^{+,n}_{(A,A^+)}$, completing the proof of the first statement.

For the second statement, Lemma 6.1.1 ensures that the morphism

$$C^{-1} : \Omega^{+,n}_{(A,A^+)} \to Z\Omega^{+,n}_{(A,A^+)}/(B\Omega^n_A \cap \Omega^{+,n}_{(A,A^+)})$$

is an isomorphism. Therefore, it suffices to show that

$$B\Omega^n_A \cap \Omega^{+,n}_{(A,A^+)} = B\Omega^{+,n}_{(A,A^+)}.$$

By our assumption on (K, K^+), we already know that

$$B\Omega^n_K \cap \Omega^{+,n}_{(K,K^+)} = B\Omega^{+,n}_{(K,K^+)}.$$

If $d\eta \in B\Omega^n_A \cap \Omega^{+,n}_{(A,A^+)}$, then its image $d\bar{\eta}$ in Ω^n_K lies in $B\Omega^n_K \cap \Omega^{+,n}_{(K,K^+)} = B\Omega^{+,n}_{(K,K^+)}$, meaning that there exists $\bar{\gamma} \in \Omega^{+,n-1}_{(K,K^+)}$ such that $d\bar{\eta} = d\bar{\gamma}$.

Lifting $\bar{\gamma}$ to $\gamma \in \Omega^{n-1}_A$, we use Lemma 6.1.8 to conclude $\gamma \in \Omega^{+,n-1}_{(A,A^+)}$. Then,

$$d\eta = d\gamma + \delta,$$

where δ has a trivial image in Ω^n_K and thus lies in $B\Omega^n_{A,0} \subseteq B\Omega^{+,n}_{(A,A^+)}$ according to Lemma 6.1.6. Consequently, $d\eta$ is in $B\Omega^{+,n}_{(A,A^+)}$, completing the proof.

□

Corollary 6.1.10. Let $\mathcal{X} = \mathrm{Spa}(A, A^+)$ be a discretely ringed local adic space. Assume that A is the localization of a scheme which is smooth over a perfect field k of positive characteristic p. Then

$$\left(B\Omega^n_A \cap \Omega^{+,n}_{(A,A^+)}\right) \cong B\Omega^{+,n}_{(A,A^+)}.$$

6.1.2. The log-Cartier isomorphism on valuation ring

From now on, assume that (A, A^+) is the valued field (K, K^+) of positive characteristic p. Gabber proved that the inverse Cartier operator

$$\Omega^n_{K^+} \xrightarrow{C^{-1}} Z\Omega^n_{K^+}/B\Omega^n_{K^+} \tag{6.4}$$

is an isomorphism. The claim was proven in the Appendix of [KST21] and was employed in the second section of [KM21] to compare $K_n(A)/p^r$ and $\nu_{A,r}(n)$ for a valuation ring A. Using similar techniques, we will prove Theorem 6.1.2.

To proceed, we introduce the following concept from logarithmic geometry:

Definition 6.1.11. A pre-logarithmic ring $A^{\log} = (N \to A)$ is *integral* if A is integral and the map $N \to A$ is an inclusion.

Example 6.1.12. The valuation ring K^+, equipped with the log structure $K^+ \setminus \{0\} \to K^+$, is integral. We denote this log ring by $K^{+,\log}$.

Now, consider an extension of integral pre-logarithmic rings of characteristic p,

$$(N \to A) \subseteq (M \to B),$$

meaning that $A \to B$ and $N \to M$ are inclusions. Assume that $M^p \subseteq N$ and $B^p \subseteq A$, which we write as $B^{\log,p} \subseteq A^{\log}$.

We define the *relative inverse Cartier operator*:

$$\begin{aligned} C^{-1}_{B^{\log}/A^{\log}} : A \otimes_{F,B} \Omega^{i,\log}_{B/A} &\longrightarrow Z\Omega^{i,\log}_{B/A}/B\Omega^{i,\log}_{B/A} \\ \begin{pmatrix} a \otimes b db_1 \wedge \cdots \wedge db_m \wedge \\ \mathrm{dlog} b_{m+1} \wedge \cdots \wedge \mathrm{dlog} b_i \end{pmatrix} &\longmapsto \begin{pmatrix} ab^p (b_1 \cdots b_m)^{p-1} db_1 \wedge \cdots \wedge db_m \wedge \\ \mathrm{dlog} b_{m+1} \wedge \cdots \wedge \mathrm{dlog} b_i. \end{pmatrix} \end{aligned} \tag{6.5}$$

Here A is a B-module via the Frobenius map $F : A \to A, a \mapsto a^p$.

Example 6.1.13. Let $A^{\log} = (N \to A)$ be an integral pre-log ring of characteristic p. Define the ring $A^{1/p} := \{\sqrt[p]{a} | a \in A\}$ by extending A by the p-th root of all its elements. Note that the Frobenius on $A^{1/p}$ induces an isomorphism $F : A^{1/p} \to A$.

Define the monoid $N^{1/p} := \{\sqrt[p]{a} | a \in N\}$. Consider the pre-log ring $B^{\log} = A^{\log,1/p} = (N^{1/p} \to A^{1/p})$. The relative inverse Cartier operator $C^{-1}_{B^{\log}/A^{\log}}$ coincides with the usual inverse Cartier operator:

$$C^{-1} : \Omega^{i,\log}_{A/\mathbb{F}_p} \to Z\Omega^{i,\log}_{B/\mathbb{F}_p}/B\Omega^{i,\log}_{B/\mathbb{F}_p}.$$

Proof. Identify the map $A^{\log} \to A^{\log,1/p}$ with the inclusion $A^{\log,p} = (N^p \to A^p) \subseteq A^{\log}$. The module $\Omega^{i,\log}_{A^{\log,1/p}/A^{\log}}$ is then isomorphic to $\Omega^{i,\log}_{A^{\log}/A^{\log,p}}$. Note however, that for any element $\alpha^p \in A^p$ and any $\gamma^p \in N^p$, the differentials $d\alpha^p$ and $\mathrm{dlog}\gamma^p$ are trivial in $\Omega^{\log}_{A^{\log}/\mathbb{F}_p}$. The result follows from the fundamental exact sequence in Lemma 2.2.4,

$$0 \cong \Omega^{\log}_{A^{\log,p}/\mathbb{F}_p} \to \Omega^{\log}_{A^{\log}/\mathbb{F}_p} \xrightarrow{\cong} \Omega^{\log}_{A^{\log}/A^{\log,p}} \to 0. \qquad \square$$

Notation 6.1.14. When given a pre-log ring $A^{\log}$, denote its underlying ring by A and its log structure by $\alpha_A : N_A \to A$.

Elementary extensions of log rings

Elementary extensions of rings in characteristic p, introduced in [KST21, Abstract A], behave analogously to smooth extensions. We now define a logarithmic version of such extensions.

Definition 6.1.15. Let $A \to B$ be an extension of rings in characteristic p. A family $(x_i)_{i\in I}$ in B is a *p-basis* (resp. *p-independent*) over A if the monomials

$$\prod_{i\in I} x_i^{n_i}; \quad 0 \le n_i < p, n_i = 0 \text{ for all but finitely many } i \in I, \tag{6.6}$$

form a basis of (resp. are linearly independent in) of B as an $A[B^p]$-module.

Note that given a ring extension $A \to B$ with a p-basis $(x_i)_{i\in I}$, then $(dx_i)_{i\in I}$ forms a B-basis of $\Omega_{B/A}$ [EGA4.2, Corollaire 21.2.4].

A similar concept exists for logarithmic rings.

Definition 6.1.16. Let $A^{\log} = (N \to A) \xrightarrow{f} B^{\log} = (M \to B)$ be an extension of integral log rings in characteristic p. A family $((x_i)_{i\in I}, (y_j)_{j\in J})$, where $x_i \in B$ and $y_j \in M$, is a *logarithmic p-basis* if:

- The union of $(x_i)_{i\in I}$ and $(y_j)_{j\in J}$ forms a p-basis of B over A.
- The family $(y_j)_{j\in J}$ forms an $\mathbb{F}_p$ basis of $M^{\mathrm{gp}}/(f_*N)^{\mathrm{gp}} \mod p$.

Similarly, the family $((x_i)_{i\in I}, (y_j)_{j\in J})$ is *p-independent* over $A^{\log}$ if

- The union of $(x_i)_{i\in I}$ and $(y_j)_{j\in J}$ is p-independent over A.
- The elements $(y_j)_{j\in J}$ are $\mathbb{F}_p$-linearly independent elements of $M^{\mathrm{gp}}/(f_*N)^{\mathrm{gp}} \mod p$.

Proposition 6.1.17. Let $((x_i)_{i\in I}, (y_j)_{j\in J})$ be a log-p-basis of $B^{\log}$ over $A^{\log}$. Then the set

$$\{dx_i | i \in I\} \cup \{\mathrm{dlog} y_j | j \in J\}$$

forms a B-basis of the module of logarithmic differentials $\Omega^{\log}_{B^{\log}/A^{\log}}$.

Proof. We proceed in two steps: generation and linear independence.

Step 1: Generation. Every element of $\Omega^{\log}_{B^{\log}/A^{\log}}$ can be written as

$$\eta + \sum_i h_i \mathrm{dlog}\alpha_i,$$

where $\eta \in \Omega_{B/A}$ is an ordinary differential, $h_i \in B$, and $\alpha_i \in M$. Each α_i can be expressed in the form

$$\alpha_i = \prod_j y_{t_j}^{s_j} \cdot \beta_i \gamma_i \delta_i^p,$$

where $\beta_i \in N^{\mathrm{gp}}$, $\gamma_i \in B^\times$, $\delta_i \in M^{\mathrm{gp}}$ and $0 \leq s_j < p$.

By differentiating, we obtain

$$h_i \mathrm{dlog}\alpha_i = \sum_j s_j h_i \mathrm{dlog} y_{t_j} + \eta',$$

where $\eta' = d\gamma_i/\gamma_i \in \Omega_{B/A}$. Since dx_i and $dy_j = y_j \mathrm{dlog} y_j$ generate $\Omega_{B/A}$, this shows that every element of $\Omega^{\log}_{B^{\log}/A^{\log}}$ can be written as a B-linear combination of $\{dx_i \mid i \in I\} \cup \{\mathrm{dlog} y_j \mid j \in J\}$.

Step 2: Linear independence. Suppose there is a relation

$$\sum a_i dx_i + \sum h_j \mathrm{dlog} y_j = 0. \tag{6.7}$$

We must show that all coefficients vanish. Consider the commutative diagram:

$$\begin{tikzcd} \Omega_{B/A} \arrow[r] \arrow[dr] & \Omega^{\log}_{B^{\log}/A^{\log}} \arrow[d] \\ & \Omega_{\mathrm{Frac}(B)/\mathrm{Frac}(A)}, \end{tikzcd}$$

where $\Omega_{B/A} \to \Omega_{\mathrm{Frac}(B)/\mathrm{Frac}(A)}$ is injective because B is integral and $\Omega_{B/A}$ is a free B-module (due to the p-basis assumption). This forces the map $\Omega_{B/A} \to \Omega^{\log}_{B^{\log}/A^{\log}}$ to be injective as well.

Rewriting (6.7) as

$$\left(\prod_{j:h_j \neq 0} y_j \right) \sum_i a_i dx_i + \sum_i \left(\prod_{j:h_j \neq 0, j \neq i} y_j \right) h_i dy_i = 0,$$

we see that this equation holds in $\Omega_{B/A}$. Since $(x_i)_i \cup (y_j)_j$ is a log-p-basis and B is integral, it follows that all coefficients must vanish: $a_i = 0$ and $h_j = 0$.

This completes the proof. $\square$

We call an extension $(N \to A) = A^{\log} \subseteq B^{\log}$ of integral log rings of characteristic p *elementary* if there exists a finite logarithmic p-basis $\{x_1, \ldots, x_r, y_1, \ldots, y_{r'}\}$. If, in addition, we assume that $B^{\log,p} \subseteq A^{\log}$, then the elementary extension $A^{\log} \subseteq B^{\log}$ is the logification (Definition 2.1.1) of a prelog ring of the form

$$\begin{aligned} B^{\log} : N \oplus \mathbb{N}^{r'} &\to A[X_1, \ldots, X_r, Y_1, \ldots, Y_{r'}]/(X_i^p - x_i, Y_j^p - y_j) \\ a \oplus (n_1, \ldots, n_{r'}) &\mapsto a \cdot Y_1^{n_1} \cdots Y_{r'}^{n_{r'}}. \end{aligned} \tag{6.8}$$

We write $A^{\log}[((x_i)_i, (y_j)_j)] := B^{\log}$.

Corollary 6.1.18. Let $A^{\log} \to B^{\log}$ be an elementary extension of integral logarithmic rings such that $B^{\log} \subseteq A^{\log,p}$. Then the morphisms in the chain

$$\Omega^i_{B/A} \to \Omega^{i,\log}_{B/A} \to \mathrm{Frac}B \otimes_B \Omega^{i,\log}_{B/A} \cong \Omega^i_{\mathrm{Frac}B/\mathrm{Frac}A}$$

are injective.

Proof. It suffices to prove the claim for $i = 1$.

The injectivity of the leftmost inclusion follows from the proof of Proposition 6.1.17.

For the isomorphism on the right, we apply [Ogu18, Proposition 1.1.3] to the co-Cartesian diagram of logarithmic rings:

$$\begin{tikzcd} A^{\log} \arrow[r] \arrow[d] & \mathrm{Frac}A \arrow[d] \\ B^{\log} \arrow[r] & \mathrm{Frac}B, \end{tikzcd}$$

where $\mathrm{Frac}A$ and $\mathrm{Frac}B$ are endowed with the trivial logarithmic structure. The diagram is co-Cartesian because $A \to B$ is an integral extension of integral rings, implying that the generators of B as an A-algebra also generate $\mathrm{Frac}B$ as a $\mathrm{Frac}A$-algebra.

The middle morphism is injective since $\Omega^{\log}_{B/A}$ is a free B-module and hence torsion-free. □

Proposition 6.1.19.

- If the extension $A^{\log} \subseteq B^{\log}$ is elementary, then the relative inverse Cartier operator $C^{-1}_{B^{\log}/A^{\log}}$ is an isomorphism.
- If the extension $A^{\log} \subseteq A^{\log,1/p}$ is a filtered colimit of elementary extensions $A^{\log} \subseteq B^{\log}$, then $C^{-1}_{A^{\log}/\mathbb{F}_p}$ is an isomorphism.

Proof. For the first claim, we observe that an elementary extension admits a presentation as in (6.8). This decomposes into a product of extensions of the form:

$$\begin{aligned} A^{\log}[x] &:= A[X]/(X^p - x), \quad \text{with trivial extension of log structures,} \\ A^{\log}[\log y] &:= (\mathbb{N} \to A[Y]/(Y^p - y), n \mapsto Y^n)^{\log}. \end{aligned}$$

By the Künneth theorem (cf. [Kat70, Theorem 7.2.]), it suffices to prove that

$$C^{-1}_{B^{\log}/A^{\log}} : A \otimes_{F,B} \Omega^{1,\log}_{B/A} \to Z\Omega^{1,\log}_{B/A}/B\Omega^{1,\log}_{B/A}$$

is an isomorphism in the two cases $B = A^{\log}[x]$ and $B^{\log} = A^{\log}[\log y]$.

The first case is the classical case as $\Omega_{A^{\log}[x]/A^{\log}} = \Omega_{B/A}$. This is due to [KST21, Proposition A.2].

In the second case, $\mathrm{dlog}y$ forms a B-basis of $\Omega^{\log}_{B/A}$, and therefore an A-basis of $A \otimes_{F,B} \Omega^{\log}_{B/A}$. On the other hand, $Z\Omega^{\log}_{B/A}$ has an A^p-basis given by $\{\mathrm{dlog}y, \dots, y^{p-1}\mathrm{dlog}y\}$, while $B\Omega^{\log}_{B/A}$ has the A^p-basis $\{dy = y\mathrm{dlog}y, \dots, y^{p-2}dy\}$. Consequently, we find that

$$Z\Omega^{\log}_{B/A}/B\Omega^{\log}_{B/A}$$

is generated by $\mathrm{dlog}y$, which is precisely $C^{-1}_{B^{\log}/A^{\log}}(\mathrm{dlog}y)$.

For the second claim, consider an elementary subextension $A^{\log} \subseteq B^{\log} \subseteq A^{\log,1/p}$. We claim that the colimit of the relative inverse Cartier operators over all elementary subextensions is equal to the Cartier operator of $A^{\log,1/p}/A^{\log}$.

By Corollary 6.1.18, we have a chain of inclusions

$$A \otimes_{F,B} \Omega^{i,\log}_{B/A} \subseteq A \otimes_{F,B} \Omega^{i}_{\mathrm{Frac}B/\mathrm{Frac}A} \subseteq A \otimes_{F,A^{1/p}} \Omega^{i}_{\mathrm{Frac}A^{1/p}/\mathrm{Frac}A} \cong \Omega^{i}_{\mathrm{Frac}A/\mathbb{F}_p}.$$

This inclusion factors through

$$A \otimes_{F,B} \Omega^{i,\log}_{B/A} \to A \otimes_{F,B} \Omega^{i,\log}_{A^{1/p}/A} \cong \Omega^{i,\log}_{A/\mathbb{F}_p}.$$

If one assumes that $A^{\log} \subseteq A^{\log,1/p}$ is a filtered union of elementary subextensions, $A^{\log} \subseteq B^{\log} \subseteq A^{\log,1/p}$, then we have a filtered colimit of inclusions

$$\Omega^{i,\log}_{A/\mathbb{F}_p} = \operatorname*{colim}_{A^{\log} \subseteq B^{\log} \subseteq A^{\log,1/p}} A \otimes_{F,B} \Omega^{i,\log}_{B/A}.$$

Similarly, $Z\Omega^{i,\log}_{A/\mathbb{F}_p}$ and $B\Omega^{i,\log}_{A/\mathbb{F}_p}$ are equal to the filtered colimits $Z\Omega^{i,\log}_{B/A}$ and $B\Omega^{i,\log}_{B/A}$ respectively. Since colimits are right exact, we also conclude that

$$Z\Omega^{i,\log}_{A/\mathbb{F}_p}/B\Omega^{i,\log}_{A/\mathbb{F}_p} = \operatorname*{colim}_{A^{\log} \subseteq B^{\log} \subseteq A^{\log,1/p}} Z\Omega^{i,\log}_{A/\mathbb{F}_p}/B\Omega^{i,\log}_{A/\mathbb{F}_p}.$$

The inverse Cartier operator on $\Omega^{i,\log}_{A/\mathbb{F}_p}$ is thus the filtered colimit of relative inverse Cartier operators on elementary extensions, which are isomorphisms by the first claim. □

Purely inseparable extensions of valued fields

Let $K \subset K'$ be an extension of valued fields over $\mathbb{F}_p$ with $(K')^p \subseteq K$. Let $K^+ \subseteq K'^+$ the corresponding (unique) extension of valuation rings. A subextension of logarithmic rings $K^{+,\log} \subseteq B^{\log} \subseteq K'^{+,\log}$ is called *elementary* if $K^{+,\log} \subseteq B^{\log}$ is elementary.

Theorem 6.1.20. Let $K^+ \subseteq K'^+$ be an extension of valuation rings satisfying $K'^{+,p} \subseteq K^+$. Then $K'^{+,\log}$ is the *directed* union of elementary extensions $K^{+,\log} \subseteq B^{\log} \subseteq K'^{+,\log}$.

While it is evident that $K'^{\log}$ can be expressed as the union of its elementary subextensions over $K^{\log}$, the directedness of this system requires proof.

We call extensions satisfying the property of Theorem 6.1.20 *good*. An equivalent formulation states that every subextension $A^{\log}$ of $K'^{+,\log}$ which is of finite type over $K^{+,\log}$ is contained in some elementary extension. Here, an extension $A^{\log} = (N \to A) \to (M \to B) = B^{\log}$ of integral logarithmic rings is *of finite type* (or finitely generated) if the ring map $A \to B$ and the induced monoid map $f_*N \to M$ are of finite type.

Corollary 6.1.21. Let K be any valuation ring. Then the Cartier operator $C^{-1}_{K^{+,\log}/\mathbb{F}_p}$ is an isomorphism.

Proof. Applying Theorem 6.1.20 to $K \subseteq K^{1/p}$, the result follows from Proposition 6.1.19. □

Following the approach in the appendix of [KST21], we reduce the proof to the case where $[K' : K] = p$ and K^+ has height one.

Lemma 6.1.22. Consider extensions $K^+ \subseteq K'^+ \subseteq K''^+$ of valuation rings with $(K''^+)^p \subseteq K^+$. If both $K^{+,\log} \subseteq K'^{+,\log}$ and $K'^{+,\log} \subseteq K''^{+,\log}$ are good extensions, then $K^{+,\log} \subseteq K''^{+,\log}$ is also good.

Proof. Let $x = ((x_1, \dots, x_r), (y_1, \dots, y_s))$ and $x' = ((x'_1, \dots, x'_{r'}), (y'_1, \dots, y'_{s'}))$ be p-independent sets for $K'^{+,\log}/K^{+,\log}$ and $K''^{+,\log}/K'^{+,\log}$ respectively. Their union $x \cup x'$ forms a p-independent set for $K''^{+,\log}/K^{+,\log}$. We equip $K^+[x, x']$ with the logarithmic structure induced by the prelogarithmic map

$$\mathbb{N}^{s+s'} \to K^+[x, x'], \quad (n_1, \dots, n_s, m_1, \dots, m_{s'}) \mapsto y_1^{n_1} \cdots y_s^{n_s} (y'_1)^{m_1} \cdots (y'_{s'})^{m_{s'}},$$

making $K^{+,\log}[x, x']$ an elementary extension over $K^{+,\log}$ with log p-basis $x \cup x'$.

Now consider a finite type subextension $A^{\log}$ of $K''^{+,\log}$ over $K^{+,\log}$. Its extension $K'^{+,\log} \cdot A^{\log}$ (the ring $K'^+ \cdot A$ with log structure from $(K'^+ \setminus \{0\}) \cdot N_A \to K'^+ \cdot A$) remains finitely generated over $K'^{+,\log}$. By goodness, there exists an elementary extension $K'^{+,\log}[x']$ containing $K'^{+,\log} \cdot A^{\log}$.

We have $A^{\log} \subseteq B^{\log}[x']$ for some finitely generated $K^{+,\log}$-algebra $B^{\log}$. Since $K^{+,\log} \subseteq K'^{+,\log}$ is good, $B^{\log}$ embeds into some elementary extension $K^{+,\log}[x]$. Thus, $A^{\log} \subseteq K^{+,\log}[x, x']$. □

Notation 6.1.23. Given an extension $K \subseteq K'$ of valued fields and $x \in K'^+$, we define $K^{+,\log}[\log x] := K^{+,\log}[(\emptyset, x)]$ and $K^{+,\log}[x] := K^{+,\log}[(x, \emptyset)]$.

The next claim will help us reduce the proof of Theorem 6.1.20 to rank 1 valuations. This is done by induction on the number of primes in $\mathrm{Spec}(K^+)$ and the next lemma.

Lemma 6.1.24. (cf. [KST21, Lemma A.8] for the non-logarithmic version) Let $K \subseteq K'$ be an extension of valued fields with $K' \subseteq K$ and $[K' : K] = p$. Let $\mathfrak{p}' \subseteq K'^+$ be a prime ideal lying over $\mathfrak{p}$ in $\mathrm{Spec}(K^+)$. Assume either:

(1) The residue field extension $\kappa(\mathfrak{p}')/\kappa(\mathfrak{p})$ is trivial and $K^{+,\log}_{\mathfrak{p}} \subseteq K'^+_{\mathfrak{p}'}$ is good; or

(2) $[\kappa(\mathfrak{p}') : \kappa(\mathfrak{p})] = p$, $K^+/\mathfrak{p} \subseteq K'^+/\mathfrak{p}'$ is good, such that a sequence $(\bar{x}_n)_n$ in $(K'^+/\mathfrak{p})$, for which $K'^+/\mathfrak{p}' = \cup_{n\in\mathbb{N}} K^+/\mathfrak{p}[\bar{x}_n]$, exists. Furthermore if

$$e = [|(K'/\mathfrak{p}')^\times| : |(K/\mathfrak{p})^\times|] = p$$

then $\bar{x}_n$ satisfies the following properties:

- $|\bar{x}_n| < 1$ is non-decreasing,
- $\bar{w}_n \cdot \bar{x}_n = \bar{x}_1$ with $\bar{w}_n \in K^+/\mathfrak{p}$ for all n and
- $|\bar{x}_n|$ generates $|\kappa(\mathfrak{p}')^\times|/|\kappa(\mathfrak{p})^\times| \cong \mathbb{Z}/p$.

Then $K^+ \subseteq K'^+$ is good.

Proof. Since $[K' : K] = p$, all p-independent sets are singletons. For any $x \in K'^+$, both $K^{+,\log}[x]$ and $K^{+,\log}[\log x]$ are elementary (and equal if $x \in (K^+[x])^\times$). Thus, $K'^{+,\log} = \bigcup_{x\in K'^+} K^{+,\log}[\log x]$, though directedness requires verification.

Case 1. We first claim the following:

Claim 6.1.25. Under condition (1), we have:

$$K'^{+,\log}_{\mathfrak{p}'} = \bigcup_{x\in\mathfrak{p}'} K^{+,\log}_{\mathfrak{p}}[\log x],$$

and this union is directed.

Proof. Indeed, any $x \in K'^+_{\mathfrak{p}'}\backslash\{0\}$ is either in $\mathfrak{p}'K'^+_{\mathfrak{p}'}$ or in $K'^+_{\mathfrak{p}'}\backslash\mathfrak{p}'K'^+_{\mathfrak{p}'}$. The ideal $\mathfrak{p}'K'^+_{\mathfrak{p}'}$ is equal to $\mathfrak{p}'K'^+_{\mathfrak{p}'} \cap K'^+ = \mathfrak{p}'$ by [KST21, Claim A.9.]. If x is in $\mathfrak{p}'K'^+_{\mathfrak{p}'} = \mathfrak{p}'$, then obviously $x \in N_{K^+_{\mathfrak{p}}[\log x]} \subseteq K^+_{\mathfrak{p}}[\log x]$. Furthermore, if $x \in K'^+_{\mathfrak{p}'}\backslash\mathfrak{p}'$, then it may be written as a sum

$$x = y + h, y \in K^+_{\mathfrak{p}}\backslash\mathfrak{p}, h \in \mathfrak{p}'K'^+_{\mathfrak{p}'} = \mathfrak{p}'$$

since $\kappa(\mathfrak{p}') = K'^+_{\mathfrak{p}'}/\mathfrak{p}' \cong K^+_{\mathfrak{p}}/\mathfrak{p} = \kappa(\mathfrak{p})$. Therefore x lies in the ring $K^+_{\mathfrak{p}}[h]$. Since $x^{-1} = x^{-p}x^{p-1}$ and $x^{-p} \in K'^+_{\mathfrak{p}'} \cap K = K^+_{\mathfrak{p}}$, we conclude that x^{-1} is also in $K^+_{\mathfrak{p}}[h]$. Hence x is in the monoid $N_{K^{+,\log}_{\mathfrak{p}}[\log h]}$. Consequently, we have the inclusion $K^{+,\log}_{\mathfrak{p}}[\log x] \subseteq K^{+,\log}_{\mathfrak{p}}[\log h]$.

Since by assumption the union $\bigcup_{x\in K'^+} K^{+,\log}_{\mathfrak{p}}[\log x]$ is directed, the claim follows by replacing $x \in K'^+$ by $h \in \mathfrak{p}'$ whenever $x \notin \mathfrak{p}'$. □

We now show that the $K^{+,\log}$-algebras $K^{+,\log}[\log(x/t)]$ for $x \in \mathfrak{p}'$ and $t \in K^+ \setminus \mathfrak{p}$ form a directed system whose union equals $K'^{+,\log}$. For fixed $x \in \mathfrak{p}'$, define:

$$A^{\log}_x = \bigcup_{t\in K^+\backslash\mathfrak{p}} K^{+,\log}[\log(x/t)],$$

in the sense that $A_x := \bigcup_{t\in K^+\backslash\mathfrak{p}} K^+_{\mathfrak{p}}[x]$ and $N_{A^{\log}_x} = \bigcup_{t\in K^+\backslash\mathfrak{p}} N_{K^{+,\log}[\log(x/t)]}$.

Claim 6.1.26.

$$A_x^{\log} = K_{\mathfrak{p}}^{+,\log}[\log x] \cap K'^{+,\log}.$$

Proof of Claim 6.1.26. We first check that $A_x = K_{\mathfrak{p}}^+[x] \cap K'^+$ (cf [KST21, Proof of Lemma A.8]). Note that x/t is in K'^+ for every $x \in \mathfrak{p}'$ and $t \in K^+ \setminus \mathfrak{p}$ [KST21, Claim A.9], thus $A_x^{\log} \subseteq K_{\mathfrak{p}}^+[x] \cap K'^+$. On the other hand, every element in $K_{\mathfrak{p}}^+[x]$ is a sum

$$f = \sum_{i \geq 0} (a_i/t_i)x^i; \text{ s.t. } a_i \in K^+, t_i \in K^+ \setminus \mathfrak{p} \quad \forall i \in \mathbb{N}. \tag{6.9}$$

If $i \geq 1$, then $(a_i/t_i)x^i$ already lies in K'^+ due to [KST21, Claim A.9]. Assume that f is in $K_{\mathfrak{p}}^+[x] \cap K'^+$, then $a_0/t_0 = f - \sum_{i \geq 1}(a_i/t_i)x_i \in K'^+ \cap K_{\mathfrak{p}}^+ = K^+$. Furthermore, if $i \geq 1$ we have $a_i(x^i/t_i) \in K^+[x^i/t_i] \subseteq A_x$. In conclusion f is in A_x.

The logarithmic structure on $K^{+,\log}[\log(x/t)]$ is generated over $N_{K^{+,\log}}$ by x, since $t \in N_{K^{+,\log}} = K^+ \setminus \{0\}$. Thus the log structure on $A_x^{\log}$ is generated by x over $N_{K^{+,\log}}$, finishing the proof of Claim 6.1.26. □

Due to Claim 6.1.25, the log rings $A_x^{\log}$ form a filtered system with $\bigcup_{x \in \mathfrak{p}'} A_x^{\log} = K'^{+,\log}$. Since each $x/t \in K'^+$ by [KST21, Claim A.9], the claim follows for Case 1.

Case 2: As shown in [KST21, Claim A.9(b)], the directed system $K^+/\mathfrak{p}[\bar{x}_n]$ lifts to $K^+[x_n]$ covering K'^+ by lifting $\bar{x}_n$ to x_n. In the above mentioned reference, it was proven that $K^+[x_n]$ is independent from the choice of the lift of $\bar{x}_n$.

We choose lifts x_n of $\bar{x}_n$ such that:

- $K'^{+,\log} = \bigcup_{n \in \mathbb{N}} K^{+,\log}[\log x_n]$
- The union remains filtered

The choice depends on the ramification index $e = [|K'^\times| : |K^\times|]$, as we show after proving the following claim.

Claim 6.1.27. The ramification indices $[|K'^\times| : |K^\times|]$ and $e = [|(\kappa(\mathfrak{p}'))^\times| : |(\kappa(\mathfrak{p}))^\times|]$ are equal.

Proof of Claim 6.1.27. Since $[\kappa(\mathfrak{p}') : \kappa(\mathfrak{p})] = p$, the ramification index $e(K_{\mathfrak{p}'}'^+/K_{\mathfrak{p}}^+)$ equals 1. The value group of $K_{\mathfrak{p}}^+$ and $K_{\mathfrak{p}'}'^+$ is the quotient of $|K^\times|$ and $|K'^\times|$ by the convex subgroups

$$H := \{|a|; a \in K^\times, a \notin \mathfrak{p} \text{ and } a^{-1} \notin \mathfrak{p}\} \text{ and } H' := \{|a| : a \in K'^\times, a \notin \mathfrak{p}' \text{ and } a^{-1} \notin \mathfrak{p}'\},$$

respectively.

On the other hand, the value groups associated to $K^+/\mathfrak{p}$ and $K'^+/\mathfrak{p}'$ are H and H', respectively. The claim then follows by group theoretical abstract nonesense. □

Subcase 2.1 ($e = 1$): Pick any lift x_n of $\bar{x}_n$. We claim that $K'^{+,\log} = \cup K^{+,\log}[\log x_n]$ and that the union is filtered. Let $y \in K'^{+}\backslash\{0\}$. It is enough to prove that

$$y \in N_{K^{+,\log}[\log x_l]}, l \gg 0.$$

Since $e = 1$, there exists $\tilde{y} \in K^+$ and $u \in (K'^{+})^\times$ such that $y = u\tilde{y}$. Then $u, u^{-1} \in K^+[x_l]$ for sufficiently large l, so $y \in N_{K^{+,\log}[\log x_l]}$.

Subcase 2.2 ($e = p$): We lift $\bar{x}_n$ as follows

- Fix a lift $x_1 \in K'^{+}$ of $\bar{x}_1$
- Lift each $\bar{w}_n$ to $w_n \in K^+ \setminus \{0\}$.
- Set $x_n := w_n^{-1} \cdot x_1$.

These satisfy $x_n \mod \mathfrak{p}' = \bar{x}_n$ since in $\kappa(\mathfrak{p}')$:

$$w_n^{-1} \cdot x_1 \mod \mathfrak{p}' \equiv \bar{w}_n^{-1} \cdot \bar{x}_1 = \bar{x}_n$$

Therefore, x_n is indeed a lift of $\bar{x}_n$ in K'^{+}.

By construction we have $x_n = w_n/w_{n+1}x_{n+1}$, with $w_n/w_{n+1} \in K^+$, which implies that $K^{+,\log}[\log(x_n)] \subseteq K^{+,\log}[\log(x_{n+1})]$. Consequently, the system $K^{+,\log}[\log x_n]$ is a directed system of logarithmic rings. The proof of [KST21, Lemma A.8(b)] implies that $K'^{+} = \bigcup_{n\in\mathbb{N}} K^+[x_n]$ (independently from the choice of lift x_n). It is left to prove that every $y \in K'^{+}\backslash\{0\}$ is in $N_{K^{+,\log}[\log x_n]}$ for some $n \in \mathbb{N}$.

Let $y \in K'^{+}\backslash\{0\}$. If $y \in (K'^{+})^\times$, then there exists $n \in \mathbb{N}$, such that y and y^{-1} are in $K^+[x_n]$. Hence y is contained in $N_{K^{+,\log}[\log x_n]}$.

Assume now that $|y| < 1$ in K'^{+}. For $n \gg 0$, $y \in K^+[x_n]$, i.e. there exist unique $a_0, \ldots, a_{p-1} \in K^+$, such that

$$y = a_0 + \cdots + a_{p-1}x_n^{p-1}.$$

The values $|a_i x_n^i|$ are distinct (otherwise $\exists i \neq j \mid x_n^{i-j} = \frac{a_j}{a_i} \in K^\times$, contradicting that $|x_n| = |1/w_n||x_1|$ generates $|K'^{\times}|/|K^\times| \cong \mathbb{Z}/p\mathbb{Z}$). Thus

$$|y| = \max\{|a_i x_n^i|, 0 \leq i < p-1\}.$$

Let j be the integer for which this maximum is realized. Write $y = ua_j x_n^j$, where $u := y/(a_j x_n^j)$ is a unit in K'^{+}, and therefore lies in $K[x_m]^\times$ for some $m \geq n$. We express y as the product

$$y = ua_j(w_n/w_m)^j x_m^j \in N_{K^{+,\log}[\log x_m]}$$

completing the proof. $\square$

Proof of Theorem 6.1.20. We proceed through several reduction steps.

Reduction 1: Represent K as a colimit of finitely generated fields, allowing us to assume K has finite rank.

Reduction 2: Since K' is a filtered colimit of finite subextensions, we assume $[K' : K] < \infty$ (a p-power). By Lemma 6.1.22, we further reduce to $[K' : K] = p$.

Reduction 3: Reduce to height one valuations, i.e. $\dim \operatorname{Spec}(K^+) = 1$. If $\dim \operatorname{Spec}(K^+) = 0$ then the proof is trivial. Assume that $\dim \operatorname{Spec}(K^+) \geq 1$. Let $\mathfrak{p}$ be the maximal ideal for which $[\kappa(\mathfrak{p}') : \kappa(\mathfrak{p})] = p$, where $\mathfrak{p}'$ is the image of $\mathfrak{p}$ in $\operatorname{Spec}(K'^+)$. If $\mathfrak{p}$ is equal to the maximal ideal of K^+ then replace it by the next prime generalizing it. Set $\mathfrak{p} \subseteq \mathfrak{q}$ to be the next specialization of $\mathfrak{p}$. Let $\mathfrak{q}'$ be the image of $\mathfrak{q}$ in $\operatorname{Spec}(K'^+)$. Note that $K^+_{\mathfrak{q}}/\mathfrak{p}$ is a rank one valuation ring. Assume that the extension $K^+_{\mathfrak{q}}/\mathfrak{p} \subseteq K'^+_{\mathfrak{q}'}/\mathfrak{p}'$ satisfies the conditions specified in Case 2 of Lemma 6.1.24. Then $K^+_{\mathfrak{q}} \subseteq K'^+_{\mathfrak{q}'}$ is good by Lemma 6.1.24(2). Consequently, $K^+ \subseteq K'^+$ is good, either by Lemma 6.1.24(1), or because $\mathfrak{q}$ is the maximal ideal of K^+.

The rank one valuation on K' can be expressed as an absolute value:

$$|\cdot| : K' \to \mathbb{R}.$$

Following [KST21, Theorem A.3], we distinguish three cases:

Case 1: $f = [\kappa(K'^+) : \kappa(K^+)] = p$.
Choose $x \in K'^+$ lifting a generator of $\kappa(K'^+)/\kappa(K^+)$. Then $K'^+ = K^+[x]$ by construction. Let $\alpha \in K'^+ \backslash \{0\}$. Since the ramification index $e = [|K'^\times| : |K^\times|]$ is 1, there exists $\beta \in K^\times$ such that $|\beta| = |\alpha| \leq 1$. Thus $\frac{\alpha}{\beta} \in (K^+)^\times$ and $\alpha = \beta \cdot \frac{\alpha}{\beta} \in N_{K^{+,\log}[x]}$.

In conclusion, $K'^{+,\log} = K^{+,\log}[x]$.

Case 2: Discrete valuation with ramification index $e = p$.
Let $x \in K'^+$ be a uniformizer. Then $K'^+ = K^+[x]$ due to the proof of Case 2 of [KST21, Theorem A.3]. Any $y \in K'^+ \setminus \{0\}$ may be written as αx^i, with $\alpha \in (K'^+)^\times$. Therefore, $K'^{+,\log} = K^{+,\log}[\log x]$.

In the Cases 1 and 2, set $x_n = x$ for all $n \in \mathbb{N}$ in order to satisfy the conditions of Case 2 of Lemma 6.1.24.

Case 3: Remaining cases.
When K^+ doesn't fit Cases 1–2, express K'^+ as the directed union $\bigcup_{n\in\mathbb{N}} K^+[x_n]$ with $|x_n| \to 1$ using [KST21, Theorem A.3]. After restricting to a subsequence, one may assume that $|x_n|$ is a non-decreasing sequence.

<u>Subcase 3.1:</u> Nontrivial defect: $f = 1$ and $e = 1$.
We claim that $K^{+,\log}[\log x_n]$ is filtered and its union is $K'^{+,\log}$. It is enough to prove that every $y \in K'^+\backslash\{0\}$ satisfies $y \in N_{K^{+,\log}[\log x_l]}, \forall l \gg 0$. Indeed, this would imply that for every $n \in \mathbb{N}, K^{+,\log}[\log x_n] \subseteq K^{+,\log}[\log x_l], \forall l \gg n$.

Let $y \in K'^+\backslash\{0\}$. Since $e = 1$, there exists $\tilde{y} \in K^+$ and $u \in (K'^+)^\times$ satisfying $y = u\tilde{y}$. Notice that u lies in $(K^+[x_l])^\times$ for all $l \gg 0$. Therefore $y \in N_{K^{+,\log}[\log x_l]}$ for all $l \gg 0$.

<u>Subcase 3.2:</u> $|\cdot|$ is not discrete, $f = 1$ and $e = p$.
We first replace the sequence x_n by a sequence $\tilde{x}_n$, such that $K'^{+,\log}$ becomes the increasing union of $K^{+,\log}[\log \tilde{x}_n]$.We first study the properties of the sequence $(x_n)_n$ defined in [KST21, Theorem A.3 Case 3].

Step 1: $|x_n| \in |K'^\times| \backslash |K^\times| \forall n \gg 1$.
Let $y \in \mathfrak{p}'$, i.e. $y \in K'^+$ and $|y| < 1$. According to Lemma A.11 of [KST21], there exists n, for which $y = a_0^{(n)} + \cdots + a_{p-1}^{(n)} x_n^{p-1}$, $|a_i^{(n)}| \leq 1$ and

$$|y| = \max\{|a_0^{(n)}|, \ldots, |a_{p-1}^{(n)} x_n^{p-1}|\}.$$

Let $i = i(n) \in \{0, \ldots, p-1\}$ be an integer satisfying $|y| = |a_i^{(n)} x_n^i|$, then this i is unique: if $|a_i^{(n)} x_n^i| = |a_j^{(n)} x_n^j|$ then $|x_n^{j-i}| = |a_i^{(n)}/a_j^{(n)}| \in |K^\times|$ and therefore $|x_n| \in |K^\times|$. However $[|K'^\times| : |K^\times|] = p$, which contradicts the fact that any value in $|K'^\times|$ may be expressed in $|K^\times||x_n^i|$ for some i and $n \gg 0$.

Step 2: Constructing $(\tilde{x}_n)_n$ out of $(x_n)_n$.
By restricting to a subsequence of $(x_n)_n$, we may assume that $|x_n|$ describes the same (non-trivial) class in $|K'^\times|/|K^\times|$ for all n. Consider an integer $n \in \mathbb{N}$. Then there exists $u_n \in (K'^+)^\times$ and $w_n \in K^+ \backslash \{0\}$ such that

$$x_1 = w_n u_n x_n.$$

Define for $n \in \mathbb{N}$, $\tilde{x}_n := u_n x_n$. Then $|x_n| = |\tilde{x}_n| \xrightarrow{n \to \infty} 1$. We claim that

$$K'^{+,\log} = \cup_{n \in \mathbb{N}} K^{+,\log}[\log \tilde{x}_n],$$

and that the union is increasing.

Step 3: Proof that $K^{+,\log}[\log \tilde{x}_n]$ are directed.
The fact that the logarithmic rings are increasing is a consequence of the equality $\tilde{x}_n = w_n/w_m \tilde{x}_m$ when $m > n$. Note that, since $|\tilde{x}_n| = |x_n| < |x_m| = |\tilde{x}_m|$, we have

$$|w_n/w_m| = |\tilde{x}_n/\tilde{x}_m| < 1.$$

Therefore $w_{nm} := w_n/w_m$ is in $K^+ \backslash \{0\}$, $K^+[\tilde{x}_n] \subseteq K^+[\tilde{x}_m]$ and $N_{K^{+,\log}[\log \tilde{x}_n]} \subseteq N_{K^{+,\log}[\log \tilde{x}_m]}$.

Step 4: $K'^+ = \bigcup_n K^+[\tilde{x}_n]$.
Let $y \in K'^+$. If $|y| = 1$, choose $y' \in K^+$ and $v \in K'^+$ satisfying $|v| < 1$, such that $y = y' + v$. This is possible since $f = 1$, which implies that $\kappa(K'^+) = \kappa(K^+)$. We may therefore replace y with v, achieving that $|y| < 1$.

For each $n \in \mathbb{N}$, we may describe y as the sum

$$y = a_0^{(n)} + \cdots + a_{p-1}^{(n)} \tilde{x}_n^{p-1}, \quad a_0^{(n)}, \ldots, a_{p-1}^{(n)} \in K. \tag{6.10}$$

By applying the same argument as above, there exists a unique $0 \leq i < p$ which depends on the class of $|y|$ in $|K'^\times|/|K^\times|$ and satisfies

$$|y| = |a_i^{(n)} \tilde{x}_m^i| = \max\{|a_0^{(n)}|, \ldots, |a_{p-1}^{(n)} \tilde{x}_n^{p-1}|\}.$$

The coefficients in (6.10) satisfy $|a_j^{(n)}| \leq |y|/|\tilde{x}_n^j| \xrightarrow{n \to \infty} |y| < 1$. Therefore $a_j^{(n)}$ are in K^+ for $n \gg 0$. This implies that $y \in K^+[\tilde{x}_n]$ for $n \gg 0$.

Step 5: $K'^+ \backslash \{0\} = \bigcup_n N_{K^+[\log \tilde{x}_n]}$.
For $y \in K'^+ \backslash \{0\}$, write $y = u a x_1^i$, with $u \in (K'^+)^\times$ and $a \in K^+$. Choose $m \in \mathbb{N}$, such

that $u \in K^+[\tilde{x}_m]$. Since $|u| = 1$ and $K'^p \subseteq K$, $u^{-p} \in K^+$. Therefore $u^{-1} = u^{p-1} \cdot u^{-p}$ is in $K^+[\tilde{x}_m]$. We deduce that $u \in K^+[\tilde{x}_m]^\times \subseteq N_{K^+,\log[\tilde{x}_m]}$. One can now write y as the product

$$y = uaw_m^i \tilde{x}_m^i,$$

which is an element in $N_{K^+,\log[\log \tilde{x}_m]}$. This completes the proof.

Note that, in both subcases, $\tilde{x}_n$ satisfies the desired properties of Case 2 in Lemma 6.1.24. □

To summarize:

Proof of Theorem 6.1.2. Due to Proposition 6.1.9, it is enough to prove that the logarithmic inverse Cartier operator is surjective on (K, K^+), where K is a valuation field and K^+ is its valuation ring. This is exactly Corollary 6.1.21. □

6.2. The residue sequence on adic spaces

We keep the same notation as in the start of the chapter. This section was developed in collaboration with Katharina Hübner.

Theorem 5.2.2 serves as a key ingredient in the proof of the purity theorem (Theorem 5.3.1). We present an alternative proof of Theorem 5.2.2 that avoids resolution of singularities.

Let $\tau \in \{$étale, tame, Zariski$\}$. For any sheaf $\mathcal{F} \in Sh(\mathcal{X}_\tau)$, sheafification of the excision long exact sequence (3.1)gives an isomorphism:

$$R^1 i^! \mathcal{F} = j_* j^* \mathcal{F} / \mathcal{F}. \tag{6.11}$$

Theorem 5.2.2 establishes that for $\mathcal{F} = \Omega_{\mathcal{X}}^{+,n}$ with Z of codimension 1 (assuming resolution of singularities):

$$R^1 i^! \Omega_{\mathcal{X}}^{+,n} \cong \Omega_{\mathcal{Z}}^{+,n}.$$

A crucial component is the exactness of (5.2)

$$0 \to \Omega_{\mathcal{X}}^{+,n} \to j_* \Omega_{\mathcal{U}}^{+,n} \to i_* \Omega_{\mathcal{Z}}^{+,n-1} \to 0.$$

Theorem 6.2.1. Assume that Z has codimension 1 in X. There exists a short exact residue sequence

$$0 \to \Omega_{\mathcal{X}}^{+,n} \to j_* \Omega_{\mathcal{U}}^{+,n} \xrightarrow{\text{res}} i_* \Omega_{\mathcal{Z}}^{+,n-1} \to 0,$$

which satisfies the following property:

Locally on X, when Z is of the form $V(f)$ for $f \in \mathcal{O}_X(X)$, the residue map satisfies $\text{res}(\omega \wedge d \log f) = \omega|_{\mathcal{Z}}$ for every $\omega \in \Omega_{\mathcal{X}}^{+,n-1}(\mathcal{X})$.

The local version of Theorem 6.2.1 is the following

Proposition 6.2.2. Assume that $Z = V(f)$ has codimension one in X. Let x be a point in $\mathcal{Z}$. Let (A, A^+) be the local Huber pair associated to x. We have a short exact sequence

$$0 \to \Omega_{(A,A^+)}^{+,n} \to \Omega_{(A[f^{-1}],A^+)}^{+,n} \xrightarrow{\text{res}} \Omega_{(A/f,A^+/f)}^{+,n} \to 0.$$

The second residue sequence from (2.3.7) localizes to

$$0 \to \Omega^n_A \to \Omega^{n,\log}_{(A[f^{-1}],A)} \xrightarrow{\text{res}} \Omega^{n-1}_{A/f} \to 0. \tag{6.12}$$

Note that $\Omega^{n,\log}_{(A[f^{-1}],A)}$ is isomorphic to $\Omega^{+,n}_{(A[f^{-1}],A)}$: This is due to [Hüb24, Theorem 8.12] and the fact that $\mathrm{Spa}(A[f^{-1}], A)$ is the limit of the adic spaces $\mathrm{Spa}(V \setminus Z, V)$, with $V \subseteq X$ open.

The sequence (6.12) fits into the diagram

$$\begin{tikzcd}
0 \arrow[r] & \Omega^{+,n}_{(A,A^+)} \arrow[r] \arrow[d] & \Omega^{+,n}_{(A[f^{-1}],A^+)} \arrow[r, dashed, "\text{res}"] \arrow[d] & \Omega^{+,n-1}_{(A/f,A^+/f)} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \Omega^n_A \arrow[r] & \Omega^{n,\log}_{(A[f^{-1}],A)} \arrow[r, "res"] & \Omega^{n-1}_{A/f} \arrow[r] & 0.
\end{tikzcd} \tag{6.13}$$

It is therefore enough to prove that $\mathrm{res} : \Omega^{n,\log}_{(A[f^{-1}],A)} \to \Omega^{n-1}_{A/f}$ restricts to a morphism $\mathrm{res} : \Omega^{+,n}_{(A[f^{-1}],A^+)} \to \Omega^{+,n-1}_{(A/f,A^+/f)}$. The next proposition provides a description of the module $\Omega^{+,n}_{(A[f^{-1}],A^+)}$.

Proposition 6.2.3. Any $\omega \in \Omega^{+,n}_{(A[f^{-1}],A^+)}$ decomposes as:

$$\omega = \omega_1 + \omega_0 \wedge d\log f,$$

where $\omega_1 \in \Omega^{+,n}_{(A,A^+)}$ and $\omega_0 \in \Omega^{+,n-1}_{(A,A^+)}$.

We first need a construction and a lemma.

Construction 6.2.4. Recall the following general construction: Let v be a valuation on a field K with valuation ring $\mathcal{O}$. Let v' be a valuation on the residue field $\kappa(\mathcal{O})$ with valuation ring $\mathcal{O}'$. Then the preimage of $\mathcal{O}'$ in $\mathcal{O}$ is a valuation ring, say $\mathcal{O}''$, with fraction field K. Denote by $v' \circ v$ the valuation on K associated to $\mathcal{O}''$.

We will construct a valuation in $\mathrm{Spa}(A[f^{-1}], A^+)$ as follows:
First consider a regular system of parameters $(f_1, \dots, f_n)$ with $f_1 = f$, and $n = \dim A$. We used here that A is regular. Then f_i defines a prime ideal in $\mathrm{Spec}(A/(f_1, \dots, f_{i-1}))$. Let v_i be the valuation associated to f_i on $\mathrm{Spec}(A/(f_1, \dots, f_{i-1}))$.

We define the valuation

$$v = x \circ v_n \circ \cdots \circ v_1$$

in $\mathrm{Spa}(A[f^{-1}], A^+)$. The valuation v is a horizontal generalization of x and a vertical specialization of v_f.

Lemma 6.2.5. Let $\omega \in \Omega^+_{(A[f^{-1}],A)}$ and $\delta \in \mathcal{R}_{\Gamma_v}$ satisfying $\delta > |\omega(v)|$, then ω has a presentation of the form

$$\omega = \omega_1 + a\mathrm{dlog} f,$$

with $\omega_1 \in \Omega_A$ and $a \in A$ satisfying

$$|\omega_1(v)| < \delta \quad \text{and} \quad |a(v)| < \delta.$$

If $|\omega(v)|$ is in the value group Γ_x, then

$$|\omega_1(v)| \leq |\omega(v)| \quad \text{and} \quad |a(v)| \leq |\omega(v)|.$$

Proof. Let $\omega = \sum g_i dh_i$ with $g_i, h_i \in A[f^{-1}]$ and $|g_i h_i(v)| < \delta$ (resp. $\leq |\omega(v)|$ if $|\omega(v)| \in \Gamma_v$ due to [Hüb24, Lemma 7.2]).

Since A is regular, we may rewrite g_i and h_i as the products

$$h_i = h_i' f^{n_i} \quad \text{and} \quad g_i = g_i' f^{m_i},$$

such that $|h_i'(v_f)| = |g_i'(v_f)| = 1$. Since $|h_i g_i(v)| \leq 1$, it follows, by definition of v, that $|h_i g_i(v_f)| \leq 1$ and therefore $n_i + m_i \geq 0$ for all i.

We have

$$g_i dh_i = g_i' f^{m_i} d(h_i' f^{n_i}) = (g_i' f^{m_i+n_i}) dh_i' + (n_i g_i' h_i' f^{m_i+n_i}) \mathrm{dlog} f$$

and

$$|g_i' f^{m_i+n_i}(v)| \cdot |h_i'(v)| = |g_i h_i(v)| \leq 1, \quad |n_i g_i' h_i' f^{m_i+n_i}(v)| \leq |g_i h_i(v)| \leq 1.$$

We can therefore set $\omega_1 := \sum_i (g_i' f^{m_i+n_i}) dh_i'$ and $a = \sum_i n_i g_i' h_i' f^{m_i+n_i}$. □

Proof of Proposition 6.2.3. Let $\omega \in \Omega^+_{(A[f^{-1}],A^+)}$. Consider a presentation of ω of the form

$$\omega = \omega_1 + a \mathrm{dlog} f, \tag{6.14}$$

with $\omega_1 \in \Omega_A$ and $a \in A$ satisfying

$$|\omega_1(v)| \leq 1 \text{ and } |a(v)| \leq 1.$$

This is possible since either $|\omega(v)| = 1$ or $\delta := 1 > |\omega(v)|$.

Since v is a horizontal generalization of x we also have inequalities

$$|\omega_1(x)| \leq 1 \text{ and } |a(x)| \leq 1,$$

hence $\omega_1(x) \in \Omega^+_{(A,A^+)}$ and $a \in A^+$.

By taking wedge products of differentials of the form (6.14), we obtain the desired result. □

Proof of 6.2.2. Consider a differential $\omega \in \Omega^{+,n}_{(A[f^{-1}],A^+)}$. Proposition 6.2.3 describes ω as a sum $\omega_1 + \omega_0 \wedge \mathrm{dlog} f$, with $\omega_1 \in \Omega^{+,n}_{(A,A^+)}$ and $\omega_0 \in \Omega^{+,n-1}_{(A,A^+)}$. Hence the image of ω by $\mathrm{res} : \Omega^{+,n}_{(A[f^{-1}],A)} \to \Omega^{n-1}_{A/f}$ in (6.12) is $\omega_0|_{A/f}$, which lies in $\Omega^{+,n-1}_{(A/f,A^+/f)}$. Thus the map res restricts to a map

$$\mathrm{res} : \Omega^{+,n}_{(A[f^{-1}],A^+)} \to \Omega^{+,n}_{(A/f,A^+/f)},$$

with kernel $\Omega^{+,n}_{(A[f^{-1}],A^+)} \cap \Omega^n_A$. Again by Proposition 6.2.3, the latter is equal to $\Omega^{+,n}_{(A,A^+)}$ □

In order to prove the Theorem 6.2.1 one needs to glue the residue morphism defined in Proposition 6.12 into a global morphism.

Proof of 6.2.1. Let $\mathrm{Spa}(V,\bar{V})$ be an open in $\mathcal{X}$, where $V \to \bar{V}$ is an open immersion. Then we have a diagram

$$\begin{tikzcd}
0 \arrow[r] & \Omega^{+,n}_{\mathcal{X}}(\mathrm{Spa}(V,\bar{V})) \arrow[r] \arrow[d, hook] & \Omega^{+,n}_{\mathcal{X}}(\mathrm{Spa}(V\cap Z,\bar{V})) \arrow[r, dashed] \arrow[d, hook] & \Omega^{+,n-1}_{\mathcal{Z}}(\mathrm{Spa}(Z\cap V,\bar{V})) \arrow[r] \arrow[d, hook] & 0 \\
0 \arrow[r] & \Omega^{n}_{V}(V) \arrow[r] & \Omega^{n}_{V}(\log Z) \arrow[r, "\mathrm{res}"] & \Omega^{n-1}_{Z}(Z\cap V) \arrow[r] & 0.
\end{tikzcd} \tag{6.15}$$

The dashed morphism exists for the following reason: an element $\omega \in \Omega^{+,n}_{\mathcal{X}}(\mathrm{Spa}(V \cap U,\bar{V}))$ is an element in $\Omega^n_V(\log Z)$, such that for every $x \in \mathrm{Spa}(V\cap U,\bar{V})$, $|w(x)| \leq 1$. It follows by Proposition 6.2.2 that

$$|\mathrm{res}(w)(x)| \leq 1,$$

for every $x \in \mathcal{Z}$ and therefore $\mathrm{res}(\omega)$ is in $\Omega^{+,n-1}_{\mathcal{Z}}(\mathrm{Spa}(Z\cap V,\overline{V}))$.

Since $\mathrm{res} : \Omega^n_{\mathcal{X}} \to i_*\Omega^{n-1}_{\mathcal{Z}}$ is well defined (globally), it restricts globally to a morphism

$$\mathrm{res} : \Omega^{+,n}_{\mathcal{X}} \to i_*\Omega^{+,n-1}_{\mathcal{Z}}.$$

□

In view of (6.11) and Theorem 6.2.1, we immediately obtain the following.

Corollary 6.2.6. Assume that Z has codimension 1, then we have a canonical isomorphism

$$R^1 i^! \Omega^{+,n}_{\mathcal{X}} \cong \Omega^{+,n-1}_{\mathcal{Z}}.$$

Corollary 6.2.6 may be used in order to prove a semi-purity result for the tame topology when Z has codimension 1, analogous to -though weaker than- the semi-purity theorem established by Milne in [Mil86] and Gros in [Gro85] for étale cohomology. We have to prove the analogue of Corollary 6.2.6 for closed differentials.

Proposition 6.2.7. Assume that Z has codimension 1 in X. By restricting the residue morphism in Theorem 6.2.1 to closed differentials, we obtain a short exact residue sequence

$$0 \to Z\Omega^{+,n}_{\mathcal{X}} \to j_* Z\Omega^{+,n}_{\mathcal{U}} \xrightarrow{\mathrm{res}} i_* Z\Omega^{+,n-1}_{\mathcal{Z}} \to 0.$$

Proof. This claim may be checked locally. Let $x \in \mathcal{X}$ be a point in $\mathcal{Z} \subseteq \mathcal{X}$ and let (A, A^+) be its local Huber pair. Assume that $Z = V(f)$ and let ω be an element in $Z\Omega_{(A[f^{-1}],A^+)}$. The image of ω by $\mathrm{res} : \Omega^{+,n}_{(A[f^{-1}],A)} \to \Omega^{n-1}_{A/f}$ is in $\Omega^{+,n-1}_{(A/f,A^+/f)}$ due to proposition 6.2.2 and in $Z\Omega^{n-1}_{A/f}$ due to Corollary 2.3.9. The claim follows from the equality $Z\Omega^{n-1}_{A/f} \cap \Omega^{+,n-1}_{(A/f,A^+/f)} = Z\Omega^{+,n-1}_{(A/f,A^+/f)}$. □

Corollary 6.2.8. Assume that Z has codimension 1, then we have a canonical isomorphism

$$R^1 i^! Z\Omega^{+,n}_{\mathcal{X}} \cong Z\Omega^{+,n-1}_{\mathcal{Z}}$$

Proof. This is a direct consequence of Proposition 6.2.7. □

Theorem 6.2.9. [Semi-purity in codimension 1] Assume that (X, Z) is a smooth pair of pure codimension 1 over a perfect field k of characteristic p. Then

$$R^1 i^! \nu_{\mathcal{X},m}(n) \cong \nu_{\mathcal{Z},m}(n-1),$$

where $\mathcal{X} = \mathrm{Spa}(X, k)$ and $\mathcal{Z} = \mathrm{Spa}(Z, k)$.

Proof. Due to Corollary 6.1.3, the Cartier operator $C : Z\Omega^n_{\mathcal{X}} \to \Omega^n_{\mathcal{X}}$ restricts to a morphism $C : Z\Omega^{+,n}_{\mathcal{X}} \to \Omega^{+,n}_{\mathcal{X}}$. In particular, we have a short exact sequence

$$0 \to \nu_{\mathcal{X},1}(n) \to Z\Omega^{+,n}_{\mathcal{X}} \xrightarrow{C-1} \Omega^{+,n}_{\mathcal{X}} \to 0. \tag{6.16}$$

After applying $R^1 i_t^!$ to (6.16), we obtain the claim for $m = 1$. We use induction on m and the short exact sequence

$$0 \to \nu_{\mathcal{X},m}(n) \to \nu_{\mathcal{X},m+1}(n) \to \nu_{\mathcal{X},1}(n) \to 0$$

to deduce the rest of the claim, as in the proof of Theorem 5.3.1. □

A. De Rham-Witt sheaves

One way to define logarithmic de Rham-Witt sheaves $\nu_m(n)$ for general m is the use of de Rham-Witt sheaves introduced by Illusie in [Ill79]. A good english reference is the article of Langer and Zink [LZ04]

A.1. Definition of de Rham-Witt sheaves

We begin by recalling the Witt vectors associated to the structure sheaf. Fix a prime number p.

The ring of Witt vectors

Let A be a ring. The ring of Witt vectors $W(A)$ of A relative to p is defined as follows. As a set, we have $W(A) = A^{\mathbb{N}}$. For $a \in W(A)$, write $a = (a_0, a_1, a_2, \dots)$, where each $a_n \in A$. The ring structure on $W(A)$ is uniquely determined by requiring that the map

$$w : W(A) \to A^{\mathbb{N}}, a \mapsto \begin{cases} w_0 & = a_0 \\ w_1 & = a_0^p + pa_1 \\ & \dots \\ w_n & = \sum_{i=0}^n p^i a_i^{p^{n-i}}, \\ & \dots \end{cases}$$

is a natural transformation, for variable A, of functors of rings. For example

$$(a+b)_0 = a_0 + b_0, \quad (a+b)_1 = a_1 + b_1 + \sum_{0<i<p} p^{-1} \binom{p}{i} a_0^i b_0^{p-i}$$

and

$$(ab)_0 = a_0 b_0, \quad (ab)_1 = b_0^p a_1 + b_1 a_0^p + pa_1 b_1.$$

The components $w_n(a)$ are called the *ghost components* of a.

Example A.1.1. The ring of Witt vectors associated to $\mathbb{Z}/p$ is $W(\mathbb{Z}/p) \cong \mathbb{Z}_p$.

In general the ring of Witt vectors $W(A)$ is a $\mathbb{Z}_p = W(\mathbb{Z}/p)$-algebra. We define the ring of Witt vecotrs of length n, denoted by $W_n(A)$, which is isomorphic to A^n as a set. Multiplication and additions are defined such that the ghost coordinates $(w_0, \dots, w_{n-1})$ define a morphism of functors of rings $W_n(-) \to (-)^n$.

Example A.1.2. The ring of Witt vecotrs of length n associated to $\mathbb{Z}/p$ is

$$W_n(\mathbb{Z}/p) \cong \mathbb{Z}/p^n.$$

Each $W_n(A)$ is a $\mathbb{Z}/p^n$-module and a quotient of $W(A)$ via the truncation map

$$(a_0, a_1, \dots) \mapsto (a_0, \dots, a_{n-1}).$$

Moreover, the full ring $W(A)$ is the projective limit of the system $(W_n(A))_n$ with respect to the *restriction* maps

$$R : W_{n+1}(A) \to W_n(A), \quad (a_0, \dots, a_n) \mapsto (a_0, \dots, a_{n-1}).$$

There is a natural additive map

$$V : W(A) \to W(A), \quad (a_0, a_1, \dots) \mapsto (0, a_0, a_1, \dots),$$

which induces morphisms $V : W_n(A) \to W_{n+1}(A)$. This gives rise to short exact sequences

$$0 \to W_n(A) \xrightarrow{V^r} W_{n+r}(A) \xrightarrow{R^n} W_r(A) \to 0.$$

Passing to the colimit yields:

$$0 \to W(A) \xrightarrow{V^r} W(A) \to W_r(A) \to 0.$$

Another key morphism is the *Frobenius*:

$$F : W_{n+1}(A) \to W_n(A),$$

which satisfies the relation $F \circ V = p$.

For an $x \in A$, the element

$$\underline{x} = (x, 0, \dots) \in W(A),$$

is called the *multiplicative representative* of x. It is invertible with inverse $\underline{x^{-1}} = (x^{-1}, 0, \dots)$. By abuse of notation we also write $\underline{x}_{\leq n} = (x, 0, \dots, 0) \in W_n(A)$. We sometimes drop the subscript $\leq n$ if the situation is clear.

Every Witt vector $a = (a_0, a_1, \dots)$ can be uniquely written as

$$a = \sum_{i=0}^{\infty} V^i \underline{a_i}.$$

LLet $X = (X, \mathcal{O}_X)$ be a ringed topos. If $\mathcal{O}_X$ is a $\mathbb{Z}_{(p)}$-algebra, then the presheaf $U \mapsto W(\mathcal{O}_X(U))$ (resp. $W_n(\mathcal{O}_X(U))$) is a sheaf. We denote $(X, W(\mathcal{O}_X))$ by $W(X)$ and $(X, W_n(\mathcal{O}_X))$ by $W_n(X)$. The maps w, R, V, F extend to sheaf morphisms on $W(X)$.

If X is of characteristic p, then $X = W_1(X)$. Given a morphism $f : X \to Y$ of ringed topoi, we get induced morphisms

$$W(f) : W(X) \to W(Y), \quad W_n(f) : W_n(X) \to W_n(Y).$$

The de Rham-Witt pro-complex

We now recall the construction of the de Rham-Witt pro-complex $W_\bullet\Omega^\bullet_X$, introduced in [Ill79, Ch. 2]. It satisfies a universal property in the category of *DR-V pro-complexes*, also called *F-V pro-complexes* in [LZ04].

Definition A.1.3. Let X be a topos. A *DR-V pro-complex* over X is a projective system $M_\bullet = ((M_n)_{n\in\mathbb{Z}}, R : M_{n+1} \to M_n)$ of differential graded $\mathbb{Z}$-algebras, equipped with additive maps $V : M^i_n \to M^i_{n+1}$ for all $n, i \in \mathbb{Z}$, satisfying $RV = VR$, and the following properties:

(V1) $M_n = 0$ for $n \leq 0$. Moreover, M^0_1 is an $\mathbb{F}_p$-algebra, and for $n \geq 1$, we have $M^0_n = W_n(M^0_1)$ with the usual restriction and Verschiebung maps.

(V2) For all $n, i, j \in \mathbb{N}$, $x \in M^i_n$ and $y \in M^j_n$ we have

$$V(xdy) = (Vx)dVy.$$

(V3) For all $n \geq 1$, $x \in M^0_1$ and $y \in M^0_n$, we have

$$(Vy)d\underline{x} = V(\underline{x}^{p-1}y)dV\underline{x},$$

where $\underline{x} = (x, 0, \ldots, 0) \in M^0_n = W_n(M^0_1)$.

The DR-V pro-complexes over X form a category $\mathrm{VDR}(X)$.

We have a forgetful functor

$$\mathrm{VDR}(X) \to \mathbb{F}_p - \mathrm{alg}(X), \quad M_\bullet \mapsto M^0_1. \tag{A.1}$$

Definition/Lemma A.1.4. The forgetful functor (A.1) has a left adjoint

$$\mathbb{F}_p - \mathrm{alg}(X) \to VDR(X), \quad A \mapsto W_\bullet\Omega^\bullet_A.$$

In particular, we have an isomorphism

$$\mathrm{Hom}_{\mathrm{VDR}(X)}(W_\bullet\Omega^\bullet, M_\bullet) \to \mathrm{Hom}_{\mathbb{F}_p-\mathrm{alg}(X)}(A, M^0_1). \tag{A.2}$$

By adjunction, we have a natural isomorphism $A \cong W_1\Omega^0_A$. Consequently we automatically have $W_n\Omega^0_A \cong W_nA$. We thus get natural morphisms of $\mathbb{F}_p$-dg-algebras $\pi_n\Omega^\bullet_{W_nA}W_n\Omega^\bullet$. Note that π_n is surjective for all n and bijective for $n = 1$.

The DR-V pro-complex associated to $\mathcal{O}_X$ is denoted by $W_\bullet\Omega^\bullet_X$. It is called the de *Rham-Witt pro-complex*. By passing to the limit, we obtain the *de Rham Witt complex*

$$W\Omega^\bullet_X = \lim_{n\in\mathbb{Z}} W_n\Omega^\bullet_X.$$

Assume from now on that X is an $\mathbb{F}_p$-scheme. Fir every $n \geq 1, i \geq 0$, $W_n\Omega^i_X$ is a quasi-coherent sheaf over X. Furthermore, for every affine $U = \mathrm{Spec}(A)$, $\Gamma(U, W_n\Omega^i_X) = W_n\Omega^i_A$.

The de Rham-Witt pro-complex is furthermore an étale sheaf: if $f : Y \to X$ is an étale morphism, then

$$f^*W_n\Omega^i_X \cong W_n\Omega^i_Y.$$

Proposition A.1.5. Let X be a ringed topos over $\mathbb{F}_p$. The homomorphism of projective systems of rings $RF = FR : W_\bullet \mathcal{O}_X \to W_{\cdot-1}\mathcal{O}_X$ can be prolonged uniquely to a homomorphism of projective system of dg-algebras

$$F : W_\bullet \Omega_X^\bullet \to W_{\cdot-1}\Omega_X^\bullet$$

such that

(a) For all $n \geq 2$ and $x \in \mathcal{O}_X$, we have

$$F d\underline{x}_{\leq n} = \underline{x}_{\leq n-1}^{p-1} d\underline{x}_{\leq n-1}.$$

(b) For all $n \geq 1$, $FdV = d : W_n\mathcal{O}_X \to W_n\Omega_X^1$.

Let us describe the Frobenius morphism F on $W_n\Omega^\bullet$ for $n \geq 2$. Let $x = (x_0, \ldots, x_{n-1}) \in W_n(A)$ then

$$x = (\underline{x}_0)_{\leq n} + V(\underline{x}_1)_{\leq n-1} + \cdots + V^{n-1}(\underline{x}_{n-1})_{\leq 1}.$$

Consequently,

$$Fdx = (\underline{x}_0)_{\leq n-1}^{p-1} d(\underline{x}_0)_{\leq n-1} + d(\underline{x}_1)_{\leq n-1} + \cdots + dV^{n-1}(\underline{x}_{n-1})_{\leq 1}.$$

A.2. Logarithmic de Rham-Witt sheaves

In this section we define the logarithmic de Rham-Witt sheaf and state its relevant properties.

Description using the de Rham-Witt pro-complex

Let X be a regular scheme over $\mathbb{F}_p$. The logarithmic de Rham-Witt sheaf $\nu_{m,X}(n)$ is the étale subsheaf of $W_m\Omega_X^n$ which is étale locally generated (as a sheaf of groups) by wedge products of logarithmic differentials

$$\mathrm{dlog}(\underline{x}_1) \wedge \ldots \mathrm{dlog}(\underline{x}_n), x \in \mathcal{O}_X^\times.$$

Recall that $\mathrm{dlog}(\underline{x}) = d\underline{x}/x$.

In the literature (e.g. [Gro85]), one writes $W_m\Omega_X^n$ instead of $\nu_{m,X}(n)$. We sometimes drop the subscript X if the situation is clear. Furthermore, we may write $\nu(n)$ instead of $\nu_1(n)$ or $\nu_{1,X}(n)$.

According to [Ill79, Theorem I.5.7.2], we have a sequence of pro-sheaves in the étale topology

$$0 \to \nu_{\bullet,X}(n) \to W_\bullet\Omega_X^n \xrightarrow{F-1} W_\bullet\Omega_X^n \to 0.$$

In particular, we have a short exact sequence of sheaves in the étale topology

$$0 \to \nu_{n,X}(n) \to W_m\Omega_X^n \xrightarrow{F-1} W_m\Omega_X^n / dV^{m-1}\Omega_X^{n-1} \to 0, \tag{A.3}$$

see for example [Shi07, Proposition 2.8.]. For $m = 1$, (A.3) corresponds to the short exact sequence

$$0 \to \nu_X(n) \to \Omega^n_X \xrightarrow{F-1} \Omega^n_X/B\Omega^n_X \to 0. \tag{A.4}$$

The short exact sequence (A.4) fits into the commutative diagram

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
 & & B\Omega^n_X \arrow[r, "\mathrm{id}"] \arrow[d] & B\Omega^n_X \arrow[d] & \\
0 \arrow[r] & \nu_X(n) \arrow[r] \arrow[d, "\mathrm{id}"] & Z\Omega^n_X \arrow[r, "C-1"] \arrow[d, "C"] & \Omega^n_X \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \nu_X(n) \arrow[r] & \Omega^n_X \arrow[r, "F-1"] \arrow[d] & \Omega^n_X/B\Omega^n_X \arrow[r] \arrow[d] & 0 \\
 & & 0 & 0 &
\end{tikzcd} \tag{A.5}$$

where $C : Z\Omega^n_X \to \Omega^n_X$ is the Cartier operator. See for example [Mil76, Lemma 1.5.] or [Shi07, Remark 2.11.].

As it is easier to work with the Cartier operator than the Frobenius, we generally work with $\nu_{1,X}(n)$, then use induction for $m > 1$. For that we need the following short exact sequence

$$0 \to \nu_{m,X}(n) \to \nu_{m+m',X}(n) \to \nu_{m',X}(n) \to 0.$$

The proof may be found for example in [Mil86, Lemma 1.7].

K-theoretic description

Let X be a regular scheme over $\mathbb{F}_p$. We will revise in this paragraph the chain of isomorphisms introduced in [Mor15, Theorem 5.1]

$$K^M_n(X)/p^m \xrightarrow{\mathrm{dlog}[\cdot]} \nu_{m,X}(n).$$

Let $\tau \in \{Zar, \text{ét}\}$. Let dlog be the τ-sheafification of the map of presheaves

$$\mathrm{dlog}[\cdot] : (\mathcal{O}^\times_X)^{\otimes n} \to W_m\Omega^n_X, \quad \alpha_1 \otimes \cdots \otimes \alpha_n \mapsto \mathrm{dlog}\underline{\alpha}_1 \cdots \mathrm{dlog}\underline{\alpha}_n.$$

Denote by $\nu_{m,X,\tau}$ the sheaffification of the image of dlog$[\cdot]$ in the τ-topology. The morphism dlog$[\cdot]$ induces a surjective morphism

$$\mathrm{dlog}[\cdot] : K^M_n(X)/p^r \to \nu_{m,X,\tau}(n),$$

since $\mathrm{dlog}a \wedge \mathrm{dlog}(1-a) = 0$. This is an isomorphism by [Mor15, Theorem 5.1]. Since $K^M_n(X)/p^r$ is a Zariski sheaf, one concludes that $\nu_{m,X,\tau}(n) = \nu_{m,X,\text{ét}}(n)$ [Mor15, Theorem 1.2].

B. Resolution of singularities

Let $\mathcal{X} := \mathrm{Spa}(X, S)$ be a discretely ringed adic space, where X and S are qcqs schemes, and $X \to S$ is separated. In our applications, we often seek an étale cover of $\mathcal{X}$ by affinoids of the form $\mathrm{Spa}(U, \bar{U})$, where $U \subset \bar{U}$ is an open immersion. Such opens form a basis for the topology on $\mathcal{X}$. This construction relies on Nagata's compactification theorem and its refinement by Conrad [Con07]. This is the subject of Appendix B.1.

Our goal is to replace pairs $(U, \bar{U})$ by log regular log schemes, since logarithmic differentials admit a simpler and more explicit description in this context (see [Hüb24]). However, this approach runs into the classical problem of resolution of singularities, which remains open in dimension > 3 in positive characteristic.

B.1. Compactifications

We recall the following version of Nagata's compactification theorem:

Proposition B.1.1. [Con07, Theorem 4.3.] Let S be a qcqs scheme, and let X be a separated S-scheme of finite type. Then there exists an open immersion $j : X \hookrightarrow \bar{X}$ over S such that $\bar{X} \to S$ is proper. We call such an immersion a *compactification* of X over S.

This implies that the topology on a discretely ringed adic space $\mathcal{X} = \mathrm{Spa}(X, S)$, where $X \to S$ is separated and of finite type, is generated by affinoids $\mathrm{Spa}(U, \bar{U})$, where $U \subset \bar{U}$ is an open immersion. After replacing $\bar{U}$ by a blow-up along $\bar{U} \setminus U$, we may assume that the morphism $U \to \bar{U}$ is affine. Indeed, the complement $Z := \bar{U} \setminus U$ is a Cartier divisor, and one can cover $\bar{U}$ by affine opens $\mathrm{Spec}(A)$ where $Z \cap \mathrm{Spec}(A) = V(f)$ for some $f \in A$, so $U \cap \mathrm{Spec}(A) = \mathrm{Spec}(A[f^{-1}])$.

Given a separated morphism $X \to S$ of qcqs schemes, the category of compactifications forms a filtered system. In particular, if $X \hookrightarrow \bar{X}_1$ and $X \hookrightarrow \bar{X}_2$ are two compactifications, then the induced map $X \to \bar{X}_1 \times_S \bar{X}_2$ is also a compactification.

Proposition B.1.2. [Con07, Theorem 2.11] Let $f : X' \to X$ be a proper morphism of qcqs schemes that restricts to an isomorphism on a quasi-compact dense open subscheme $U \subseteq X, X'$. Then there exist U-admissible blow-ups $\bar{X} \to X$ and $\bar{X}' \to X'$ and an isomorphism $\bar{X}' \cong \bar{X}$ over f.

In particular, the category of compactifications is dominated by those in which all morphisms are blow-ups.

B.2. Resolutions

Let $X \to S$ be a smooth separated morphism of finite type, and assume $S = \mathrm{Spec}(k)$ for a perfect field k of characteristic p. Then X is quasi-excellent. We would like to find a smooth compactification $\bar{X}$ of X such that $\bar{X} \setminus X$ is an snc divisor. Ideally, we also want every such compactification to be dominated by a sequence of blow-ups in smooth centers lying within the boundary. This allows us to apply Theorem 2.4.5 to compute cohomology of logarithmic differentials. These goals remain conjectural in dimensions > 3.

Conjecture B.2.1. Let $\bar{X}$ be a reduced, separated, Noetherian, quasi-excellent scheme. Then there exists a proper birational morphism $\pi : \bar{X}' \to \bar{X}$ such that:

- $\bar{X}'$ is regular;
- π is an isomorphism over the regular locus of $\bar{X}$;
- The exceptional divisor $\pi^{-1}(\bar{X}_{\mathrm{sing}})$ is an snc divisor in $\bar{X}'$.

Conjecture B.2.1 holds when $\bar{X}$ is of dimension ≤ 3 [CP19]. Now back to our situation, where $X \to S$ is separated and smooth and assume that $S = \mathrm{Spec}(k)$, where k is a perfect field of characteristic p. Then X is quasi-excellent. Under the assumption of Conjecture B.2.1, one can find a smooth compactification $\bar{X}$ of X such that $\bar{X} \setminus X$ is an snc divisor, and hence $(X, \bar{X})$ is a log smooth pair.

Conjecture B.2.2 (Embedded resolution)**.** Let Z be a reduced, excellent, Noetherian scheme, and X a regular excellent scheme. Suppose $Z \hookrightarrow X$ is a closed immersion. Then there exists a projective, surjective morphism $\pi : X' \to X$, which is an isomorphism over $X \setminus Z$, such that $\pi^{-1}(Z)$ is an snc divisor in X'.

This is known when $\dim(Z) \leq 2$ [CJS20, Corollary 1.5]. Together, Conjectures B.2.1 and B.2.2 imply that every smooth discretely ringed adic space $\mathrm{Spa}(X, S)$ over a perfect field k of characteristic p has a cover by affinoids $\mathrm{Spa}(U, \bar{U})$ such that $\bar{U}$ is smooth and $\bar{U} \setminus U$ is an snc divisor. This holds up to dimension 3 since Conjecture B.2.2 holds up to dimension 2 and Conjecture B.2.1 holds up to dimension 3.

We present a stronger version of the conjecture, which implies the Conjecture of embedded resolution when X is a defined over a perfect field k.

Conjecture B.2.3. [Embedded resolution of singularities with boundary] Consider a smooth scheme $\bar{X}$ over a perfect field k. Let $D \subset \bar{X}$ be an snc divisor. Let $\bar{Z} \subset \bar{X}$ be a reduced closed subscheme, which is smooth over $\bar{X} \setminus D$, s.t. $\bar{Z} \setminus D$ is smooth over k. Then the triple $(\bar{X}, D, \bar{Z})$ admits a resolution $\pi : \bar{Y} \to \bar{X}$ by blowups in smooth centers

$$\bar{Y} := \bar{X}_m \to \bar{X}_{m-1} \to \cdots \to \bar{X}_2 \to \bar{X}_1 := \bar{X},$$

satisfying the following conditions

- Each morphism $\bar{X}_{i+1} \xrightarrow{\pi_i} \bar{X}_i$ is a blowup in a smooth center W_i with exceptional divisor E_{i+1}. Set $D_1 := D$, and let D_{i+1} be the strict transform of D_i in $\bar{X}_{i+1}$.

- D_i is an snc divisor and $W_i \subset D_i$.
- $\pi^{-1}(\bar{Z})$ is a smooth k-scheme.

Conjecture B.2.3 is proven when $\bar{Z}$ has dimension ≤ 2 [CJS20, Theorem 6.9].

Remark B.2.4. Assuming Conjectures B.2.1 and B.2.2, we conclude that opens of the form $\mathrm{Spa}(U, \bar{U})$, where $(U, \bar{U})$ is a log smooth pair, form a basis for the topology on $\mathrm{Spa}(X, S)$. By further covering $\bar{U}$ by affine opens, we may assume that these are affinoids. Assuming Conjecture B.2.3, any U-modification of $\bar{U}$ is dominated by a sequence of blow-ups in smooth centers.

Indeed, modifications are dominated by blow-ups [Tem11, Corollary 3.4.8]. Moreover, in our setting, U-blow-ups and U-admissible blow-ups coincide [Tem11, Remark 3.4.5(i)]. Applying Conjecture B.2.3 to the center $Z \subset \bar{U} \setminus U$ allows us to resolve Z within the boundary divisor, ultimately yielding a smooth blow-up dominating any given U-modification.

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